My Humble Solution to

Volume 1 of Course of Theoretical Physics

Mechanics

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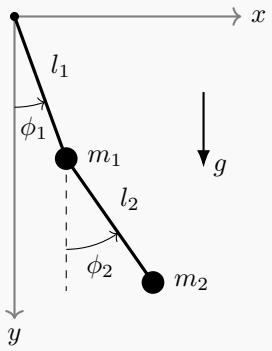
Chapter 1.

The Equations of Motion

§5 The Lagrangian for a system of particles

Problem 1.

Find the Lagrangian of a coplanar double pendulum when placed in a uniform gravitational field (acceleration g).



Solution: The generalized co-ordinates of the system are the two angles ϕ_1 and ϕ_2. We need to express the Cartesian co-ordinates in terms of those two angles. First, the Cartesian position of the particle m_1 is

$$\begin{aligned} x_1 &= f_1(\phi_1) = l_1 \sin\phi_1 \\ y_1 &= g_1(\phi_1) = l_1 \cos\phi_1. \end{aligned}$$

By taking the time derivative of those, we obtain

$$\begin{aligned} \dot{x}_1 &= \frac{\partial f_1}{\partial \phi_1}\dot{\phi}_1 = l_1 \cos\phi_1 \dot{\phi}_1 \\ \dot{y}_1 &= \frac{\partial g_1}{\partial \phi_1}\dot{\phi}_1 = -l_1 \sin\phi_1 \dot{\phi}_1. \end{aligned}$$

Substituting these expressions in the Lagrangian, we obtain the desired Lagrangian form, that is

$$\begin{aligned} L_1 &= T_1 - U_1 \\ &= \frac{1}{2} m_1 (\dot{x}_1^2 + \dot{y}_1^2) + m_1 g y_1 \\ &= \frac{1}{2} m_1 l_1^2 \dot{\phi}_1^2 + m_1 g l_1 \cos\phi_1. \end{aligned} \tag{1.1}$$

Second, the Cartesian position of the particle m_2 is

$$\begin{aligned} x_2 &= f_2(\phi_1, \phi_2) = l_1 \sin\phi_1 + l_2 \sin\phi_2 \\ y_2 &= g_2(\phi_1, \phi_2) = l_1 \cos\phi_1 + l_2 \cos\phi_2. \end{aligned}$$

By taking the time derivative of those, we obtain

$$\begin{aligned} \dot{x}_2 &= \sum_k \frac{\partial f_2}{\partial \phi_k}\dot{\phi}_k = l_1 \dot{\phi}_1 \cos\phi_1 + l_2 \dot{\phi}_2 \cos\phi_2 \\ \dot{y}_2 &= \sum_k \frac{\partial g_2}{\partial \phi_k}\dot{\phi}_k = -l_1 \dot{\phi}_1 \sin\phi_1 - l_2 \dot{\phi}_2 \sin\phi_2. \end{aligned}$$

Substituting these expressions in the Lagrangian, we obtain the desired Lagrangian form, that is

$$\begin{aligned} L_2 &= T_2 - U_2 \\ &= \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2) + m_2 g y_2 \\ &= \frac{1}{2} m_2 (l_1^2 \dot{\phi}_1^2 + l_2^2 \dot{\phi}_2^2 + 2 l_1 l_2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_1 - \phi_2)) + m_2 g (l_1 \cos\phi_1 + l_2 \cos\phi_2), \end{aligned} \tag{1.2}$$

where the angle difference identity as been used. Finally, the Lagrangian of the complete system is simply

the sum of Eq. (1.1) and Eq. (1.2), thus

Answer to Problem 1.

$$L = \frac{1}{2}(m_1 + m_2) l_1^2 \dot{\phi}_1^2 + \frac{1}{2} m_2 l_2^2 \dot{\phi}_2^2 + m_2 l_1 l_2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_1 - \phi_2) + (m_1 + m_2) g l_1 \cos\phi_1 + m_2 g l_2 \cos\phi_2$$

Verification: We compare the Lagrangian prediction with an independent simulation of two masses connected by rigid rods under gravity: point masses moving by Newton's laws, with the rod tensions as constraint forces and no angles in the model. Two cases were run for 5 s with $g = 9.81$ m/s^2, both starting with $\dot{\phi}_1 = 0$:

- *tumbling*: $m_1 = m_2 = 1$ kg, $l_1 = l_2 = 1$ m, $\phi_1 = 2$ rad, $\phi_2 = 2.5$ rad, $\dot{\phi}_2 = 4$ rad/s;
- *unequal*: $m_1 = 2$ kg, $m_2 = 0.5$ kg, $l_1 = 1.2$ m, $l_2 = 0.7$ m, $\phi_1 = 0.9$ rad, $\phi_2 = -0.6$ rad, $\dot{\phi}_2 = 1$ rad/s.

Each case was run at *standard* integration tolerances (10^{-9} for the particles, 10^{-11} for the Lagrangian) and *tight* tolerances, 100 times smaller. Fig. 1.1 overlays both pendulum angles of the two models for the tumbling case, a violent motion in which the lower rod goes over the top twice: they are indistinguishable.

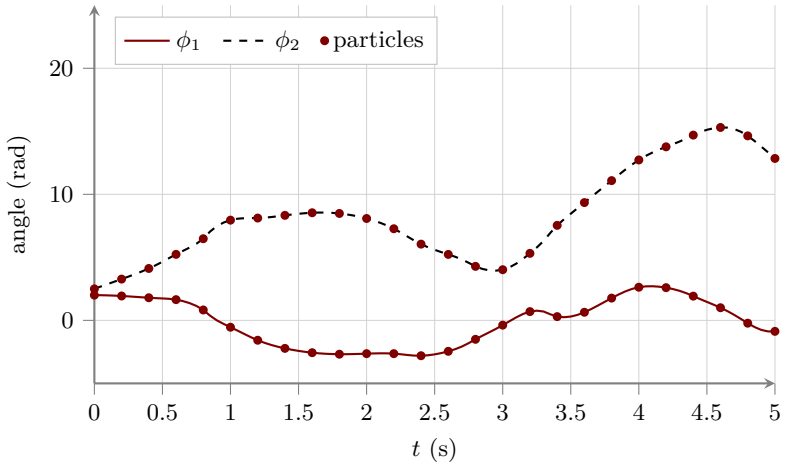


Figure 1.1: Pendulum angles of the double pendulum versus time in the tumbling case, from the Lagrangian (curves) and from the particle simulation (dots).

Fig. 1.2 gives the maximum Cartesian position difference between the models, that is the largest distance between the same mass in the two simulations. With tight tolerances the position difference stays below 3×10^{-8} m, and it drops by a factor of 70 or more from standard to tight, so what remains is numerical error. We also check velocity agreement (difference below 2×10^{-7} m/s with tight tolerances), energy conservation (relative drift below 4×10^{-11} in both models) and constant rod lengths (error below 2×10^{-15} m). The Lagrangian thus gives the same motion as Newton's laws for particles on rods, in the cases tried; this is numerical evidence, not a proof.

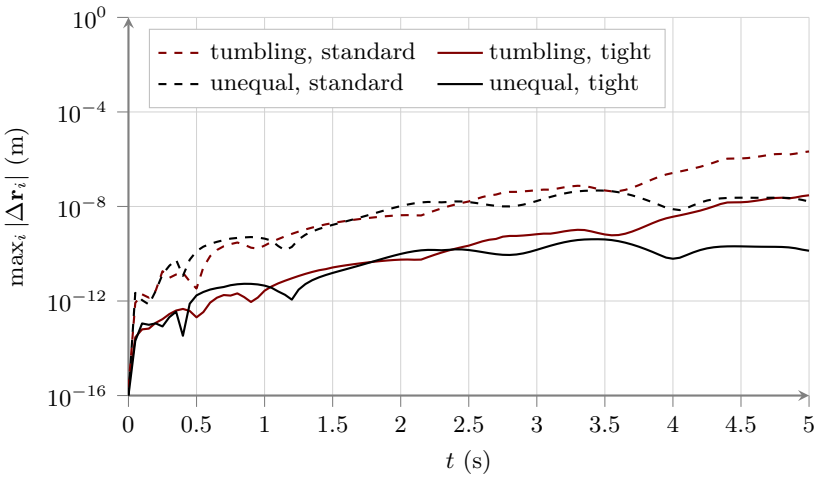
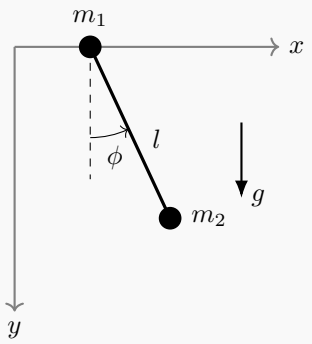


Figure 1.2: Maximum Cartesian position difference between the Lagrangian and particle models of the double pendulum versus time, at standard (dashed) and tight (solid) tolerances, for the tumbling (red) and unequal (black) cases.

Problem 2.

Find the Lagrangian of a simple pendulum of mass m_2, with a mass m_1 at the point of support which can move on a horizontal line lying in the plane in which m_2 moves when placed in a uniform gravitational field (acceleration g).



Solution: The generalized co-ordinates q of the system are the position x and the angle ϕ. Therefore, the Lagrangian of the first particle is simply

$$L_1 = T_1 = \frac{1}{2} m_1 \dot{x}^2. \tag{1.3}$$

The Cartesian position of the particle m_2 can be express in terms of the generalized co-ordinates by

$$\begin{aligned} x_2 &= f_2(x, \phi) = x + l \sin \phi \\ y_2 &= g_2(x, \phi) = l \cos \phi. \end{aligned}$$

The time derivative of the position is then

$$\begin{aligned} \dot{x}_2 &= \sum_k \frac{\partial f_2}{\partial q_k} \dot{q}_k = \dot{x} + l\dot{\phi} \cos \phi \\ \dot{y}_2 &= \sum_k \frac{\partial g_2}{\partial q_k} \dot{q}_k = -l\dot{\phi} \sin \phi. \end{aligned}$$

$$\begin{aligned} L_2 &= T_2 - U_2 \\ &= \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2) + m_2 g y_2 \\ &= \frac{1}{2} m_2 (\dot{x}^2 + l^2 \dot{\phi}^2 + 2l\dot{x}\dot{\phi} \cos \phi) + m_2 g l \cos \phi. \end{aligned} \tag{1.4}$$

Finally, the Lagrangian of the complete system is the sum of the two Lagrangian Eq. (1.3) and Eq. (1.4), thus

Answer to Problem 2.

$$L = \frac{1}{2}(m_1 + m_2)\dot{x}^2 + \frac{1}{2} m_2 (l^2 \dot{\phi}^2 + 2l\dot{x}\dot{\phi} \cos \phi) + m_2 g l \cos \phi$$

Verification: We compare the Lagrangian prediction with an independent simulation of a pendulum attached to a horizontally sliding mass: two point masses moving by Newton's laws, m_1 held on the horizontal line by frictionless guides and joined to m_2 by a rigid rod, with the guide reaction and the rod tension as constraint forces and no angle in the model. Two cases were run for 5 s with $g = 9.81$ m/s^2:

- *looping*: $m_1 = m_2 = 1$ kg, $l = 1$ m, $x = 0$, $\phi = 0.5$ rad, $\dot{x} = -3.5$ m/s, $\dot{\phi} = 8$ rad/s;
- *swinging*: $m_1 = 3$ kg, $m_2 = 0.8$ kg, $l = 1.5$ m, $x = 0.2$ m, $\phi = 1.2$ rad, $\dot{x} = -0.3$ m/s, $\dot{\phi} = 0$.

Each case was run at *standard* integration tolerances (10^{-9} for the particles, 10^{-11} for the Lagrangian) and *tight* tolerances, 100 times smaller. Fig. 1.3 overlays the support position and the pendulum angle of the two models for the looping case, in which m_2 goes over the top more than three times while the support moves back and forth: they are indistinguishable.

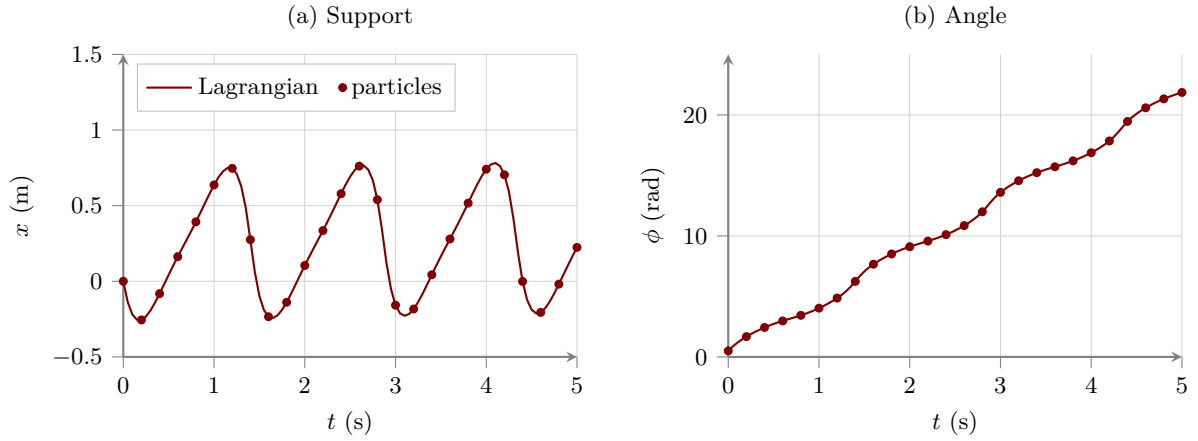


Figure 1.3: (a) Support position and (b) pendulum angle versus time in the looping case, from the Lagrangian (curves) and from the particle simulation (dots).

Fig. 1.4 gives the maximum Cartesian position difference between the models, that is the largest distance between the same mass in the two simulations. With tight tolerances the position difference stays below 10^{-9} m, and it drops by a factor of 20 or more from standard to tight, so what remains is numerical error. We also check velocity agreement (difference below 10^{-8} m/s with tight tolerances), conservation of energy (relative drift below 6×10^{-11} in both models) and of horizontal momentum (relative drift below 10^{-11}, the generalized momentum $\partial L/\partial \dot{x}$ of the cyclic coordinate x), and satisfaction of the rod and guide constraints (error below 2×10^{-15} m). The Lagrangian thus gives the same motion as Newton's laws for these particles, in the cases tried; this is numerical evidence, not a proof.

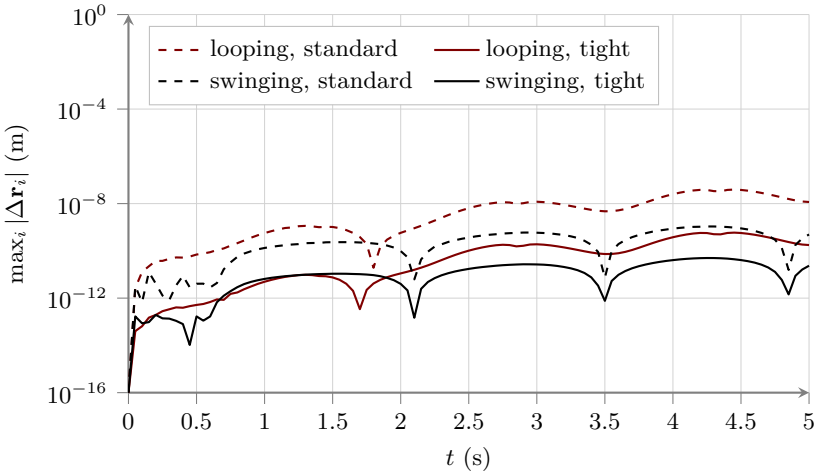
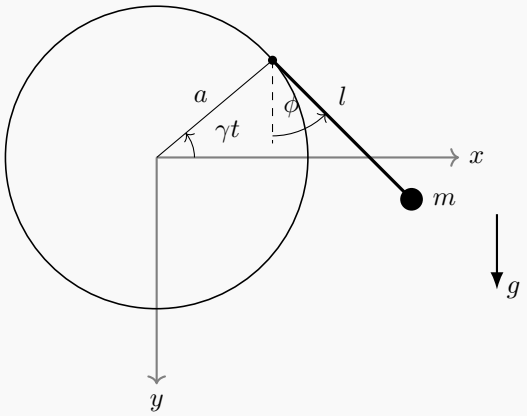


Figure 1.4: Maximum Cartesian position difference between the Lagrangian and particle models of the pendulum with a sliding support versus time, at standard (dashed) and tight (solid) tolerances, for the looping (red) and swinging (black) cases.

Problem 3.

Find the Lagrangian of a simple pendulum of mass m, when placed in a uniform gravitational field (acceleration g), whose point of support ...



(a)

moves uniformly on a vertical circle with constant frequency γ.

Solution: If we set that the rotation of the point of support is counterclockwise, the Cartesian position of the particle m is

$$\begin{aligned} x &= f(\phi_1) = a\cos\gamma t + l\sin\phi \\ y &= g(\phi_1) = -a\sin\gamma t + l\cos\phi. \end{aligned}$$

The time derivative of the position is then

$$\begin{aligned} \dot{x} &= -a\gamma\sin\gamma t + l\dot{\phi}\cos\phi \\ \dot{y} &= -a\gamma\cos\gamma t - l\dot{\phi}\sin\phi. \end{aligned}$$

The potential energy of the system is

$$\begin{aligned} U &= -mgy \\ &= -mg(-a\sin\gamma t + l\cos\phi). \end{aligned}$$

The term $mga\sin\gamma t$ only depend on time and can therefore be ignored (does not contribute to the equations of motion). The kinetic energy of the particle, for its part, is

$$\begin{aligned} T &= \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \\ &= \frac{1}{2}m\left(l^2\dot{\phi}^2 + a^2\gamma^2 + 2a\gamma l\dot{\phi}\sin\left(\phi - \gamma t\right)\right), \end{aligned}$$

using again the angle difference identity (I will stop mentioning it). We can observe that the term $\frac{1}{2}ma^2\gamma^2$ is a constant, thus can be ignored. The last term of the kinetic energy can also be simplified. Indeed,

$$\begin{aligned} ma\gamma l\dot{\phi}\sin\left(\phi - \gamma t\right) &= mal\gamma(\dot{\phi} - \gamma)\sin(\phi - \gamma t) + mal\gamma^2\sin(\phi - \gamma t) \\ &= \frac{d}{dt}\left(-mal\gamma\cos(\phi - \gamma t)\right) + mal\gamma^2\sin(\phi - \gamma t). \end{aligned}$$

After dropping the total time derivative, we can finally get the Lagrangian

Answer to Problem 3 (a).

$$L = \frac{1}{2}ml^2\dot{\phi}^2 + mal\gamma^2\sin(\phi - \gamma t) + mgl\cos\phi$$

Verification: We compare the Lagrangian prediction with an independent simulation of a point mass on a rigid rod whose end is driven around the circle: the particle moves by Newton's laws, with the rod tension as constraint force and no angle in the model. Since the dropped terms do not change the motion, the two must agree. Two cases were run for 5 s with $g = 9.81$ m/s^2:

- *tumbling*: $m = 0.5$ kg, $l = 0.8$ m, $a = 0.4$ m, $\gamma = 7$ rad/s, $\phi = -0.3$ rad, $\dot{\phi} = 1$ rad/s;
- *swinging*: $m = 1$ kg, $l = 1$ m, $a = 0.3$ m, $\gamma = 2$ rad/s, $\phi = 0.5$ rad, $\dot{\phi} = 0$.

Each case was run at *standard* integration tolerances (10^{-9} for the particle, 10^{-11} for the Lagrangian) and *tight* tolerances, 100 times smaller. Fig. 1.5 overlays the pendulum angle of the two models for the tumbling case, in which the fast support drives the pendulum over the top four times: they are indistinguishable.

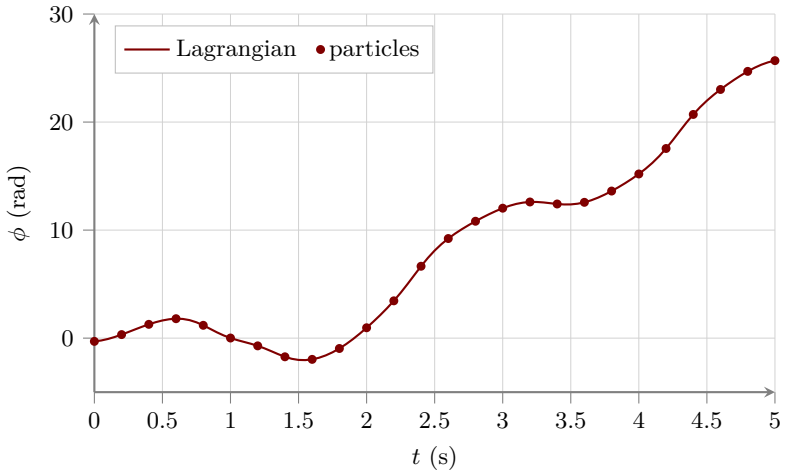


Figure 1.5: Pendulum angle versus time in the tumbling case, from the Lagrangian (curve) and from the particle simulation (dots).

Fig. 1.6 gives the Cartesian position difference between the models, that is the distance between the mass in the two simulations. With tight tolerances it stays below 4×10^{-9} m, and in the tumbling case it drops by a factor of 60 from standard to tight, so what remains is numerical error. We also check velocity agreement (difference below 4×10^{-8} m/s with tight tolerances) and constant rod length (error below 2×10^{-15} m). The energy is not conserved, since the moving support does work on the pendulum, but the particle simulation balances the energy change with this work to a relative error below 2×10^{-9}. The Lagrangian thus gives the same motion as Newton's laws for a pendulum with a moving support, in the cases tried; this is numerical evidence, not a proof.

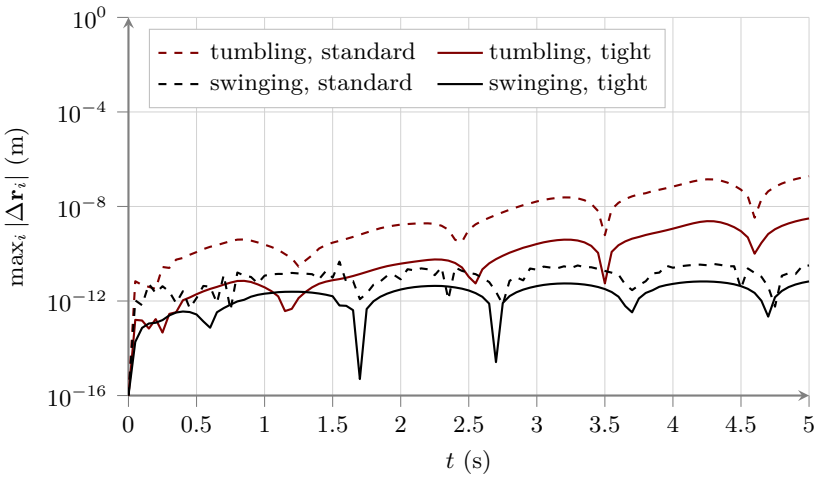


Figure 1.6: Cartesian position difference between the Lagrangian and particle models of the pendulum with a support on a circle versus time, at standard (dashed) and tight (solid) tolerances, for the tumbling (red) and swinging (black) cases.

(b)

oscillates horizontally in the plane of motion of the pendulum according to the law $x = a\cos\gamma t$.

Solution: The Cartesian position of the particle m is

$$x = f(\phi_1) = a\cos\gamma t + l\sin\phi$$
$$y = g(\phi_1) = l\cos\phi.$$

The time derivative of the position is then

$$\dot{x} = -a\gamma\sin\gamma t + l\dot{\phi}\cos\phi$$
$$\dot{y} = -l\dot{\phi}\sin\phi.$$

The potential energy of the system is

$$U = -mgy = -mgl\cos\phi,$$

and the kinetic energy

$$\begin{aligned} T &= \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \\ &= \frac{1}{2}m\left(l^2\dot{\phi}^2 + a^2\gamma^2\sin^2\gamma t - 2a\gamma l\dot{\phi}\sin\gamma t\cos\phi\right). \end{aligned}$$

We first see that the term $\frac{1}{2}ma^2\gamma^2\sin^2\gamma t$ only depend on time. The last term of the kinetic energy can also be simplified. Indeed,

$$\begin{aligned} -ma\gamma l\dot{\phi}\sin\gamma t\cos\phi &= -mla\gamma(\gamma\cos\gamma t\sin\phi + \dot{\phi}\sin\gamma t\cos\phi) + mla\gamma^2\cos\gamma t\sin\phi \\ &= \frac{d}{dt}\left(-mla\gamma\sin\gamma t\sin\phi\right) + mla\gamma^2\cos\gamma t\sin\phi. \end{aligned}$$

After dropping the total time derivative, we can finally get the Lagrangian

Answer to Problem 3 (b).

$$L = \frac{1}{2}ml^2\dot{\phi}^2 + mla\gamma^2\cos\gamma t\sin\phi + mgl\cos\phi$$

Verification: We compare the Lagrangian prediction with an independent simulation of a point mass on a rigid rod whose end is driven back and forth along the horizontal line: the particle moves by Newton's laws, with the rod tension as constraint force and no angle in the model. Two cases were run for 5 s with $g = 9.81$ m/s^2:

- *resonant*: $m = 1$ kg, $l = 1$ m, $a = 0.1$ m, $\gamma = 3$ rad/s, close to $\sqrt{g/l} = 3.13$ rad/s, $\phi = 0$, $\dot{\phi} = 0$;
- *fast*: $m = 0.5$ kg, $l = 0.6$ m, $a = 0.05$ m, $\gamma = 15$ rad/s, $\phi = 0.8$ rad, $\dot{\phi} = 0$.

Each case was run at *standard* integration tolerances (10^{-9} for the particle, 10^{-11} for the Lagrangian) and *tight* tolerances, 100 times smaller. Fig. 1.7 overlays the pendulum angle of the two models for the resonant case, in which the drive builds up a swing from rest: they are indistinguishable.

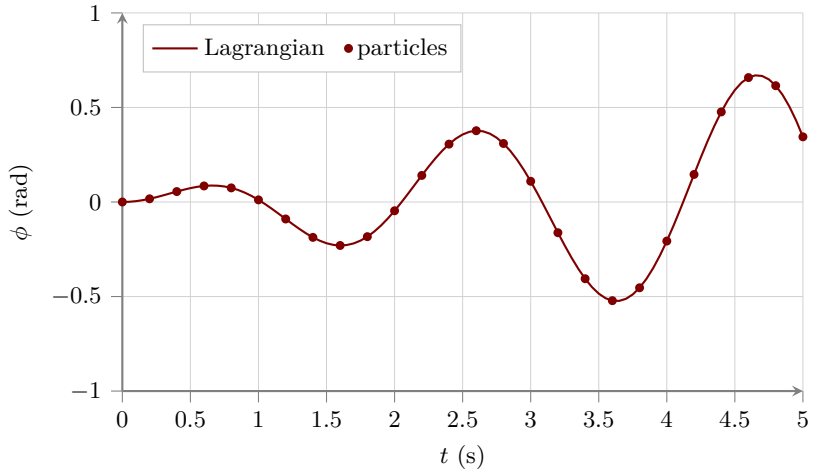


Figure 1.7: Pendulum angle versus time in the resonant case, from the Lagrangian (curve) and from the particle simulation (dots).

Fig. 1.8 gives the Cartesian position difference between the models, that is the distance between the mass in the two simulations. With tight tolerances it stays below 4×10^{-11} m, and in the fast case it drops by a factor of 100 from standard to tight, so what remains is numerical error. We also check velocity agreement (difference below 10^{-10} m/s with tight tolerances) and constant rod length (error below 10^{-15} m). The energy is not conserved, since the moving support does work on the pendulum, but the particle simulation balances the energy change with this work to a relative error below 2×10^{-11}. The Lagrangian thus gives the same motion as Newton's laws for a pendulum with a moving support, in the cases tried; this is numerical evidence, not a proof.

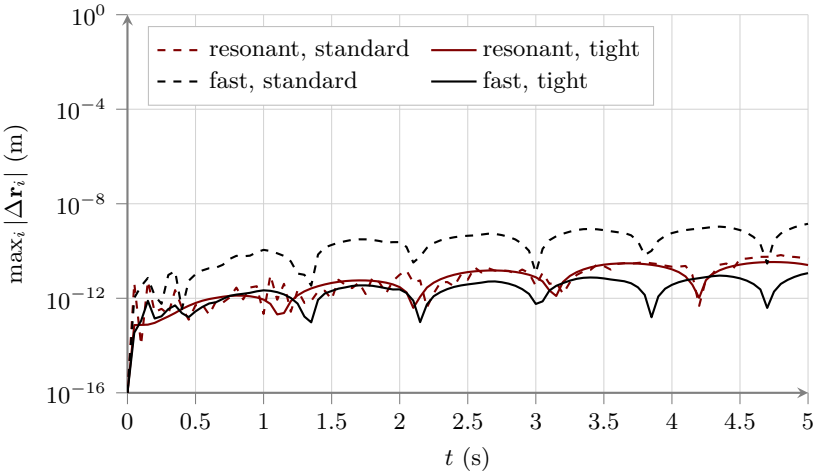


Figure 1.8: Cartesian position difference between the Lagrangian and particle models of the pendulum with a horizontally oscillating support versus time, at standard (dashed) and tight (solid) tolerances, for the resonant (red) and fast (black) cases.

(c)

oscillates vertically according to the law $y = a\cos\gamma t$.

Solution: The Cartesian position of the particle m is

$$\begin{aligned} x &= f(\phi_1) = l\sin\phi \\ y &= g(\phi_1) = a\cos\gamma t + l\cos\phi. \end{aligned}$$

The time derivative of the position is then

$$\begin{aligned} \dot{x} &= l\dot{\phi}\cos\phi \\ \dot{y} &= -a\gamma\sin\gamma t - l\dot{\phi}\sin\phi. \end{aligned}$$

The potential energy of the system is

$$U = -mgy = -mg\left(a\cos\gamma t + l\cos\phi\right),$$

The term $mga\cos\gamma t$ only depend on time and can therefore be ignored. The kinetic energy is

$$\begin{aligned} T &= \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \\ &= \frac{1}{2}m\left(l^2\dot{\phi}^2 + a^2\gamma^2\sin^2\gamma t + 2a\gamma l\dot{\phi}\sin\gamma t\sin\phi\right). \end{aligned}$$

We first see that the term $\frac{1}{2}ma^2\gamma^2\sin^2\gamma t$ only depend on time. The last term of the kinetic energy can also be simplified. Indeed,

$$\begin{aligned} ma\gamma l\dot{\phi}\sin\gamma t\sin\phi &= mla\gamma(-\gamma\cos\gamma t\cos\phi + \dot{\phi}\sin\gamma t\sin\phi) + mla\gamma^2\cos\gamma t\cos\phi \\ &= \frac{d}{dt}\left(-mla\gamma\sin\gamma t\cos\phi\right) + mla\gamma^2\cos\gamma t\cos\phi. \end{aligned}$$

After dropping the total time derivative, we can finally get the Lagrangian

Answer to Problem 3 (c).

$$L = \frac{1}{2}ml^2\dot{\phi}^2 + mla\gamma^2\cos\gamma t\cos\phi + mgl\cos\phi$$

Verification: We compare the Lagrangian prediction with an independent simulation of a point mass on a rigid rod whose end is driven up and down: the particle moves by Newton's laws, with the rod tension as constraint force and no angle in the model. Two cases were run for 5 s with $g = 9.81$ m/s^2 and $m = 1$ kg:

- *Kapitza*: $l = 0.3$ m, $a = 0.05$ m, $\gamma = 60$ rad/s, $\phi = 2.8$ rad, $\dot{\phi} = 0$, a drive fast enough ($a^2\gamma^2 > 2gl$) to hold the pendulum upside down;
- *parametric*: $l = 1$ m, $a = 0.1$ m, $\gamma = 6.26$ rad/s, close to $2\sqrt{g/l}$, $\phi = 0.2$ rad, $\dot{\phi} = 0$, a parametric resonance in which the swing grows.

Each case was run at *standard* integration tolerances (10^{-9} for the particle, 10^{-11} for the Lagrangian) and *tight* tolerances, 100 times smaller. Fig. 1.9 overlays the pendulum angle of the two models for the Kapitza case, in which the pendulum swings slowly about the upward vertical $\phi = \pi$ while jittering at the drive frequency: they are indistinguishable.

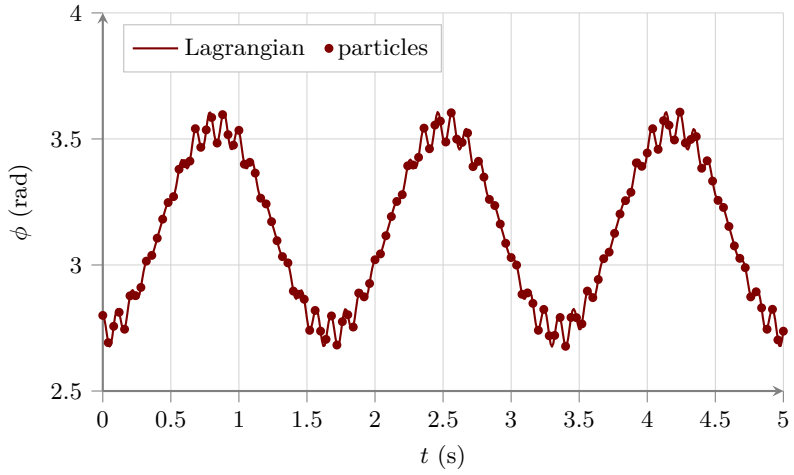


Figure 1.9: Pendulum angle versus time in the Kapitza case, from the Lagrangian (curve) and from the particle simulation (dots).

Fig. 1.10 gives the Cartesian position difference between the models, that is the distance between the mass in the two simulations. With tight tolerances it stays below 2×10^{-11} m, and it drops by a factor of 50 or more from standard to tight, so what remains is numerical error. We also check velocity agreement (difference below 2×10^{-10} m/s with tight tolerances) and constant rod length (error below 10^{-15} m). The energy is not conserved, since the moving support does work on the pendulum, but the particle simulation balances the energy change with this work to a relative error below 3×10^{-9}. The Lagrangian thus gives the same motion as Newton's laws for a pendulum with a moving support, in the cases tried; this is numerical evidence, not a proof.

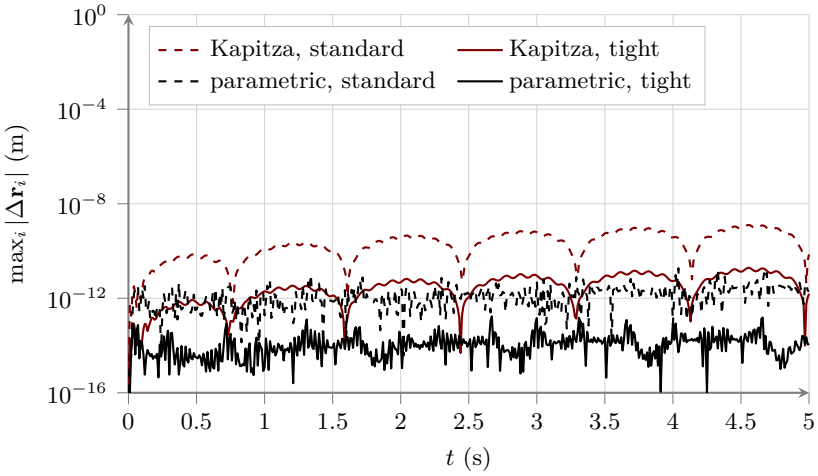
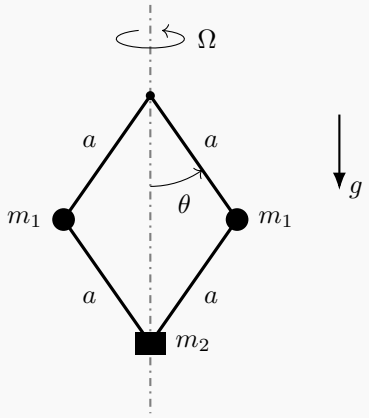


Figure 1.10: Cartesian position difference between the Lagrangian and particle models of the pendulum with a vertically oscillating support versus time, at standard (dashed) and tight (solid) tolerances, for the Kapitza (red) and parametric (black) cases.

Problem 4.

Find the Lagrangian of a simple pendulum of the system below when placed in a uniform gravitational field (acceleration g). The particle m_2 moves on a vertical axis and the whole system rotates about this axis with a constant angular velocity Ω.



Solution: The position of each particle m_1 is best described in cylindrical coordinates, which is

$$\begin{aligned} r_1 &= a\sin\theta \\ \phi_1 &= \phi \\ z_1 &= a\cos\theta, \end{aligned}$$

where ϕ is the angle of rotation of the system about the axis; $\dot{\phi} = \Omega$. The kinetic energy of each particle m_1 is thus

$$\begin{aligned} T_1 &= \frac{1}{2}m_1(\dot{r}_1^2 + r_1^2\dot{\phi}_1^2 + \dot{z}_1^2) \\ &= \frac{1}{2}m_1(a^2\dot{\theta}^2\cos^2\theta + a^2\Omega^2\sin^2\theta + a^2\dot{\theta}^2\sin^2\theta) \\ &= \frac{1}{2}m_1a^2(\dot{\theta}^2 + \Omega^2\sin^2\theta). \end{aligned} \tag{1.5}$$

The potential energy of this particle can be found by using the z component of his position, namely

$$\begin{aligned} V_1 &= -m_1gz_1 \\ &= -m_1ga\cos\theta. \end{aligned} \tag{1.6}$$

The particle m_2, in its case, can only move up and down, thus its position can be completely defined by

$$z_2 = 2a\cos\theta.$$

Then, since $\dot{z}_2 = -2a\dot{\theta}\sin\theta$, the kinetic energy of the single particle m_2 is

$$\begin{aligned} T_2 &= \frac{1}{2}m_2\dot{z}_2^2 \\ &= 2m_2a^2\dot{\theta}^2\sin^2\theta \end{aligned} \tag{1.7}$$

and its potential energy is

$$\begin{aligned} V_2 &= -m_2gz_2 \\ &= -2m_2ga\cos\theta. \end{aligned} \tag{1.8}$$

Using Eq. (1.5), Eq. (1.6), Eq. (1.7) and Eq. (1.8), and remembering that there are two particles m_1 but

only one particle m_2, the Lagrangian of the system is

$$L = 2T_1 + T_2 - 2V_1 - V_2,$$

therefore

Answer to Problem 4.

$$L = m_1 a^2(\dot\theta^2 + \Omega^2 \sin^2\theta) + 2m_2 a^2 \dot\theta^2 \sin^2\theta + 2(m_1 + m_2) g a \cos\theta$$

Verification: We compare the Lagrangian prediction with an independent simulation of three point masses moving by Newton's laws: each m_1 hangs on a rigid rod from the fixed pivot and is joined to m_2 by another rod, m_2 is held on the axis by frictionless guides, and the two m_1 are held in a vertical plane through the axis that turns at the constant rate Ω, which imposes the rotation. The rod tensions and the guide and plane reactions are constraint forces, with no angle in the model. Two cases were run for 5 s with $g = 9.81$ m/s^2:

- *oscillating*: $m_1 = m_2 = 1$ kg, $a = 0.5$ m, $\Omega = 8$ rad/s, $\theta = 0.5$ rad, $\dot\theta = 0$;
- *fast*: $m_1 = 0.5$ kg, $m_2 = 1$ kg, $a = 0.3$ m, $\Omega = 15$ rad/s, $\theta = 0.7$ rad, $\dot\theta = 1$ rad/s.

In both cases the system oscillates about the angle $\cos\theta = (m_1 + m_2)g/(m_1 a\Omega^2)$ at which the rotation balances gravity. Each case was run at *standard* integration tolerances (10^{-9} for the particles, 10^{-11} for the Lagrangian) and *tight* tolerances, 100 times smaller. Fig. 1.11 overlays the angle θ of the two models for the oscillating case: they are indistinguishable.

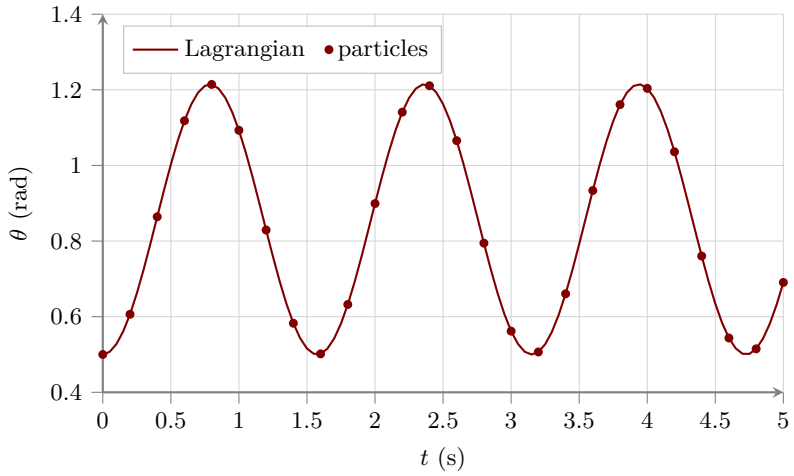


Figure 1.11: Angle of the rods versus time in the oscillating case, from the Lagrangian (curve) and from the particle simulation (dots).

Fig. 1.12 gives the maximum Cartesian position difference between the models, that is the largest distance between the same mass in the two simulations. With tight tolerances the position difference stays below 10^{-11} m, and it drops by a factor of 60 or more from standard to tight, so what remains is numerical error. We also check velocity agreement (difference below 4×10^{-11} m/s with tight tolerances) and satisfaction of the rod, guide and plane constraints (error below 10^{-15} m). The energy is not conserved, since the turning plane does work on the masses, but L does not depend on time, so the energy function $\dot\theta\,\partial L/\partial\dot\theta - L$ is; it equals $E - \Omega M_z$, with M_z the angular momentum about the axis, and its relative drift stays below 2×10^{-12} in both models. The Lagrangian thus gives the same motion as Newton's laws for these particles, in the cases tried; this is numerical evidence, not a proof.

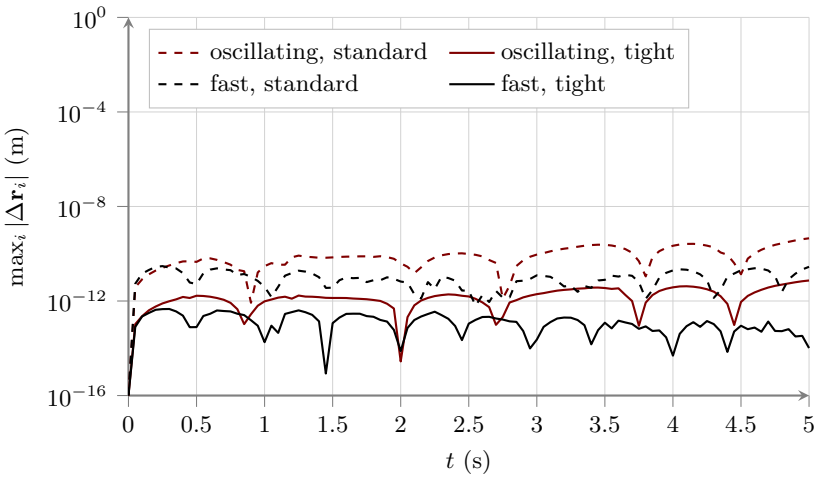


Figure 1.12: Maximum Cartesian position difference between the Lagrangian and particle models of the rotating system versus time, at standard (dashed) and tight (solid) tolerances, for the oscillating (red) and fast (black) cases.

Chapter 2.

Conservation Laws

§7 Momentum

Problem 1.

A particle of mass m moving with velocity $\mathbf{v}_1$ leaves a half-space in which its potential energy is a constant U_1 and enters another in which its potential energy is a different constant U_2. Determine the change in the direction of motion of the particle.

Solution: The potential energy only depend on the co-ordinate perpendicular to the plane separating the half-space. Thus, the component of the momentum in that plane is conserved. Denoting by θ_1 and θ_2 the angles between the normal to the plane and the velocities v_1 and v_2 of the particle before and after passing the plane, we have

$$\begin{aligned} & P_1 \sin\theta_1 = P_2 \sin\theta_2 \\ \Rightarrow & v_1 \sin\theta_1 = v_2 \sin\theta_2 \\ \Rightarrow & \frac{\sin\theta_1}{\sin\theta_2} = \frac{v_2}{v_1}. \end{aligned} \tag{2.1}$$

The potential energy of the system is also independent of time, therefore the energy of the particle is conserved. Posing $E_1 = T_1 + U_1$ and $E_2 = T_2 + U_2$ as the energy of the particle before and after passing the plane, the law of conservation of energy requires

$$\begin{aligned} & E_1 = E_2 \\ \Rightarrow & T_1 + U_1 = T_2 + U_2 \\ \Rightarrow & \frac{1}{2}mv_1^2 + U_1 = \frac{1}{2}mv_2^2 + U_2 \\ \Rightarrow & v_2^2 = v_1^2 + \frac{2}{m}(U_1 - U_2). \end{aligned} \tag{2.2}$$

By substituting Eq. (2.2) in the square of Eq. (2.1), we get

$$\left(\frac{\sin\theta_1}{\sin\theta_2}\right)^2 = \frac{v_1^2 + \frac{2}{m}(U_1 - U_2)}{v_1^2}.$$

After taking the square root, the result is

Answer to Problem 1.

$$\frac{\sin\theta_1}{\sin\theta_2} = \sqrt{1 + \frac{2}{mv_1^2}(U_1 - U_2)}$$

§8 Centre of mass

Problem 1.

Find the law of transformation of the action S from one inertial frame to another.

Solution: The Lagrangian L and L' of a mechanical system in two inertial frames of reference K and K' are respectively

$$L = T - U = \frac{1}{2}\sum_a m_a v_a^2 - U$$

and

$$L' = T' - U = \frac{1}{2}\sum_a m_a v_a'^2 - U.$$

If the frame K' moves with velocity $\mathbf{V}$ relative to the frame K, the velocities of the particles of the mechanical system relative to the two frames are related by $\mathbf{v}_a = \mathbf{v}'_a + \mathbf{V}$. We can now express the relation of the Lagrangian of the system in the two frames by

$$\begin{aligned} L &= \frac{1}{2}\sum_a m_a(\mathbf{v}'_a + \mathbf{V})^2 - U \\ &= \frac{1}{2}V^2\sum_a m_a + \mathbf{V}\cdot\sum_a m_a\mathbf{v}'_a + \frac{1}{2}\sum_a m_a v_a'^2 - U \\ &= L' + \mathbf{V}\cdot\mathbf{P}' + \frac{1}{2}\mu V^2. \end{aligned} \tag{2.3}$$

Integrating Eq. (2.3) with respect to time, we obtain

$$\begin{aligned} S &= \int L dt \\ &= S' + \int (\mathbf{V}\cdot\mathbf{P}')\, dt + \frac{1}{2}\mu V^2 t \\ &= S' + \mathbf{V}\cdot\int\sum_a m_a\mathbf{v}'_a dt + \frac{1}{2}\mu V^2 t \\ &= S' + \mathbf{V}\cdot\sum_a m_a\mathbf{r}'_a + \frac{1}{2}\mu V^2 t. \end{aligned}$$

The law of transformation of the action S is then

Answer to Problem 1.

$$S = S' + \mu\mathbf{V}\cdot\mathbf{R}' + \frac{1}{2}\mu V^2 t$$

§9 Angular momentum

Problem 1.

Obtain the expressions for the Cartesian components and the magnitude of the angular momentum of a particle in cylindrical co-ordinates r, ϕ, z.

Solution: The Cartesian components of the angular momentum are simply

$$\mathbf{M} = \mathbf{r} \times \mathbf{p} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \times m \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = m \begin{bmatrix} y\dot{z} - z\dot{y} \\ z\dot{x} - x\dot{z} \\ x\dot{y} - y\dot{x} \end{bmatrix}. \tag{2.4}$$

In cylindrical co-ordinates, the Cartesian components are expressed by

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r\cos\phi \\ r\sin\phi \\ z \end{bmatrix}$$

and

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \dot{r}\cos\phi - r\dot{\phi}\sin\phi \\ \dot{r}\sin\phi + r\dot{\phi}\cos\phi \\ \dot{z} \end{bmatrix}.$$

By substituting those in Eq. (2.4), we get

$$\begin{aligned} \mathbf{M} &= m \begin{bmatrix} y\dot{z} - z\dot{y} \\ z\dot{x} - x\dot{z} \\ x\dot{y} - y\dot{x} \end{bmatrix} \\ &= m \begin{bmatrix} (r\dot{z} - z\dot{r})\sin\phi - rz\dot{\phi}\cos\phi \\ -(r\dot{z} - z\dot{r})\cos\phi - rz\dot{\phi}\sin\phi \\ r^2\dot{\phi} \end{bmatrix}. \end{aligned} \tag{2.5}$$

By taking the magnitude of Eq. (2.5), we finally get

Answer to Problem 1.

$$\begin{aligned} M_x &= m(r\dot{z} - z\dot{r})\sin\phi - mrz\dot{\phi}\cos\phi \\ M_y &= -m(r\dot{z} - z\dot{r})\cos\phi - mrz\dot{\phi}\sin\phi \\ M_z &= mr^2\dot{\phi} \\ M^2 &= m^2r^2\dot{\phi}^2(r^2 + z^2) + m^2(r\dot{z} - z\dot{r})^2 \end{aligned}$$

Problem 2.

Obtain the expressions for the Cartesian components and the magnitude of the angular momentum of a particle in spherical co-ordinates r, θ, ϕ.

Solution: Let θ be the polar angle (measured from the z axis) and ϕ the azimuth (measured in the xy plane from the x axis). In spherical co-ordinates, the Cartesian components are expressed by

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r\sin\theta\cos\phi \\ r\sin\theta\sin\phi \\ r\cos\theta \end{bmatrix}$$

and

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \dot{r}\sin\theta\cos\phi + r\dot{\theta}\cos\theta\cos\phi - r\dot{\phi}\sin\theta\sin\phi \\ \dot{r}\sin\theta\sin\phi + r\dot{\theta}\cos\theta\sin\phi + r\dot{\phi}\sin\theta\cos\phi \\ \dot{r}\cos\theta - r\dot{\theta}\sin\theta \end{bmatrix}.$$

By substituting those in Eq. (2.4) of §9 Prob. 1, we get

$$\begin{aligned} \mathbf{M} &= m \begin{bmatrix} y\dot{z} - z\dot{y} \\ z\dot{x} - x\dot{z} \\ x\dot{y} - y\dot{x} \end{bmatrix} \\ &= m \begin{bmatrix} -r^2(\dot{\theta}\sin\phi + \dot{\phi}\sin\theta\cos\theta\cos\phi) \\ r^2(\dot{\theta}\cos\phi - \dot{\phi}\sin\theta\cos\theta\sin\phi) \\ r^2\dot{\phi}\sin^2\theta \end{bmatrix}. \end{aligned} \tag{2.6}$$

By taking the magnitude of Eq. (2.6), we finally get

Answer to Problem 2.

$$\begin{aligned} M_x &= -mr^2(\dot{\theta}\sin\phi + \dot{\phi}\sin\theta\cos\theta\cos\phi) \\ M_y &= mr^2(\dot{\theta}\cos\phi - \dot{\phi}\sin\theta\cos\theta\sin\phi) \\ M_z &= mr^2\dot{\phi}\sin^2\theta \\ M^2 &= m^2r^4(\dot{\theta}^2 + \dot{\phi}^2\sin^2\theta) \end{aligned}$$

Problem 3.

Which components of momentum **P** and angular momentum **M** are conserved in motion in the following fields?

(a)

The field of an infinite homogeneous plane.

Solution: Let us first see how a symmetry of the field leads to the conservation of a component. For a particle of mass m in a stationary external field U, the Lagrangian in Cartesian co-ordinates is

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - U(x, y, z).$$

Since $P_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x}$, Lagrange's equation for x gives

$$\dot{P}_x = \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x} = -\frac{\partial U}{\partial x}.$$

If the field is unchanged by any translation along x, *i.e.* $U(x + a, y, z) = U(x, y, z)$ for every a, then U does not depend on x and P_x is constant. Any direction can be taken as the x axis, so a component of **P** is conserved when the field is unchanged by arbitrary translations along that direction.

Similarly, in the cylindrical co-ordinates r, ϕ, z, the Lagrangian is

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2 + \dot{z}^2) - U(r, \phi, z),$$

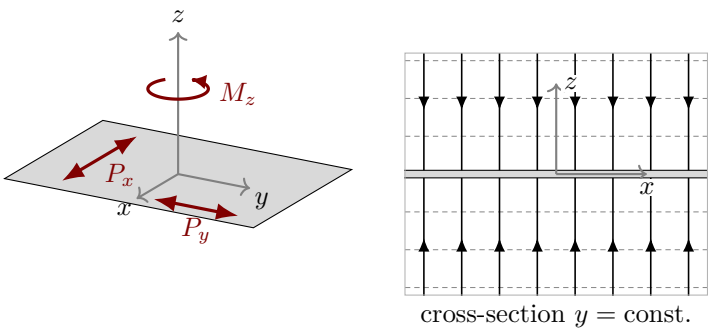
and $\frac{\partial L}{\partial \dot{\phi}} = mr^2\dot{\phi} = M_z$ is the component of the angular momentum along the z axis found in §9 Prob. 1. Lagrange's equation for ϕ gives

$$\dot{M}_z = \frac{d}{dt}\frac{\partial L}{\partial \dot{\phi}} = \frac{\partial L}{\partial \phi} = -\frac{\partial U}{\partial \phi}.$$

If the field is unchanged by any rotation about the z axis, then U does not depend on ϕ and M_z is constant. Here, the angular momentum is measured about the symmetry axis, *i.e.* with the origin on that axis. The symmetries involved are continuous: the field must be unchanged by translations or rotations of any amount, however small, for the corresponding derivative of U to vanish. Invariance under a few isolated rotations is not enough. The same reasoning holds for the total momentum and angular momentum of a system of particles in the field.

In each of the following figures, the body is shown on the left. The red arrows are the transformations leaving its field unchanged (a double arrow for a translation, a circular arrow for a rotation), each labelled with the quantity it conserves. The field is shown on the right in a cross-section. The arrows of the field lines give the direction of the force $\mathbf{F} = -\nabla U$ on the particle, and the dashed curves are equipotentials, along which U is constant.

Let the plane be $z = 0$, as in the figure. Each point of an infinite homogeneous plane is equivalent to any other, so its field depends only on the distance to the plane, *i.e.* $U = U(z)$. It is thus unchanged by any translation along x or y and by any rotation about the z axis: the field lines are perpendicular to the plane and the equipotentials are planes parallel to it.



cross-section $y = \text{const.}$

In terms of the force, this means that

$$F_x = -\frac{\partial U}{\partial x} = 0 \quad \text{and} \quad F_y = -\frac{\partial U}{\partial y} = 0,$$

and the torque about the z axis vanishes:

$$\dot{M}_z = xF_y - yF_x = 0.$$

However, there is generally a force $F_z = -dU/dz$ normal to the plane, so P_z is not conserved, and neither are M_x and M_y, whose torques yF_z and $-xF_z$ do not vanish. The conserved components are then

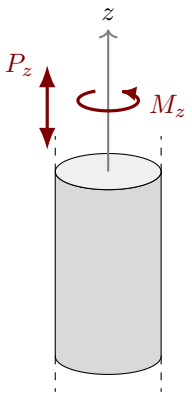
Answer to Problem 3 (a).

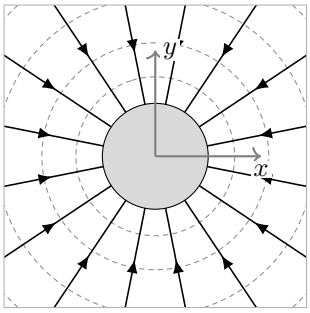
$$P_x, \quad P_y, \quad M_z$$

(b)

That of an infinite homogeneous cylinder.

Solution: Let the axis of the cylinder be the z axis, as in the figure, and r the distance from that axis. The field of an infinite homogeneous cylinder is the same at every height and in every direction around the axis, so it depends only on r, *i.e.* $U = U(r)$. It is thus unchanged by any translation along z and by any rotation about the z axis: in a cross-section, the field lines are radial and the equipotentials are circles.





cross-section $z = \text{const.}$

The force $\mathbf{F} = -U'(r)\,\hat{\mathbf{r}}$ is radial. It has no axial component, so $\dot{P}_z = F_z = 0$, and its line of action meets the axis, so it produces no torque about it. Indeed, (F_x, F_y) is parallel to (x, y) and

$$\dot{M}_z = xF_y - yF_x = 0.$$

The components F_x and F_y do not vanish, so P_x and P_y are not conserved, nor are M_x and M_y. The conserved components are then

Answer to Problem 3 (b).

$$P_z, \quad M_z$$

(c)

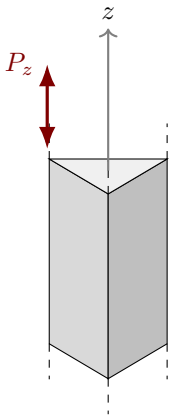
That of an infinite homogeneous prism.

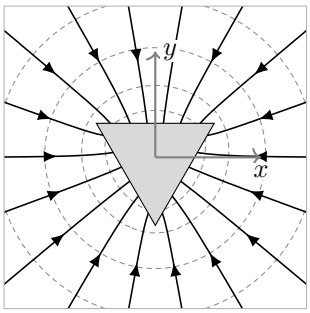
Solution: Let the edges of the prism be parallel to the z axis, as in the figure. The prism is the same in every cross-section perpendicular to its edges, so its field is unchanged by any translation along

z, *i.e.* $U = U(x, y)$, and

$$\dot{P}_z = -\frac{\partial U}{\partial z} = 0.$$

However, its cross-section is a polygon, so the field generally depends on the direction about any axis parallel to the edges. In the figure, the equipotentials near the prism follow the triangular cross-section and only become circular far from it. Even for a regular polygon of n sides, such as the equilateral triangle shown, the field is unchanged only by the rotations by multiples of $2\pi/n$ about its axis. These isolated rotations do not make $\partial U/\partial\phi$ vanish, so they do not imply the conservation of M_z.





cross-section $z = \text{const.}$

The only conserved component is then

Answer to Problem 3 (c).

$$P_z$$

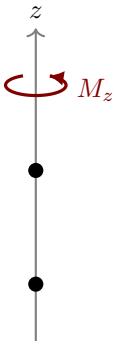
(d)

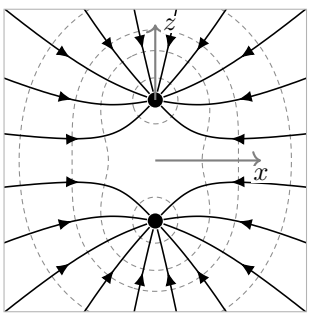
That of two points.

Solution: Let the two points be on the z axis, as in the figure. The field is the sum of the fields of the two points, each depending only on the distance from the particle to that point, *i.e.*

$$U = U_1(\rho_1) + U_2(\rho_2),$$

where ρ_1 and ρ_2 are the distances from the particle to the two points. A rotation of the particle about the line joining the points leaves both distances unchanged, so U does not depend on ϕ. In other words, the force exerted by each point is directed towards that point, so its line of action meets the z axis and it produces no torque about it. This holds whether or not the two points have the same strength. In the figure, the field lines and equipotentials are shown in a plane containing the axis; they are the same in every such plane.





plane $y = 0$

On the other hand, a translation moves the particle with respect to the fixed points and changes these distances, so no component of **P** is generally conserved. The only conserved component is then

Answer to Problem 3 (d).

$$M_z$$

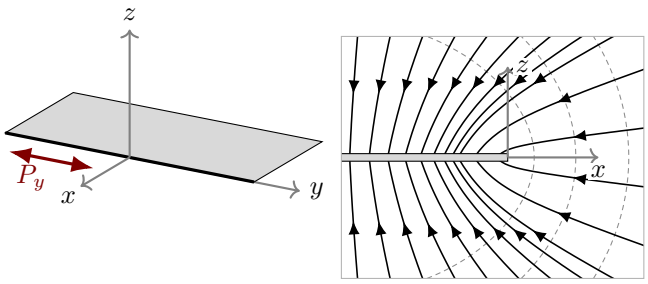
(e)

That of an infinite homogeneous half-plane.

Solution: Let the half-plane be $z = 0$, $x < 0$, as in the figure, so that its edge is the y axis. A translation parallel to the edge maps the half-plane onto itself, so its field is the same in every cross-section $y =$ constant, *i.e.* $U = U(x, z)$, and

$$\dot{P}_y = -\frac{\partial U}{\partial y} = 0.$$

The edge removes the other symmetries of the whole plane. A translation along x moves the edge, and the field depends on the distance to it: in the figure, the field lines bend towards the edge and the equipotentials curve around it. A rotation about the z axis turns the edge, and a rotation about any other axis tilts the half-plane, so no rotation leaves the field unchanged and no component of **M** is generally conserved.



cross-section $y =$ const.

The only conserved component is then

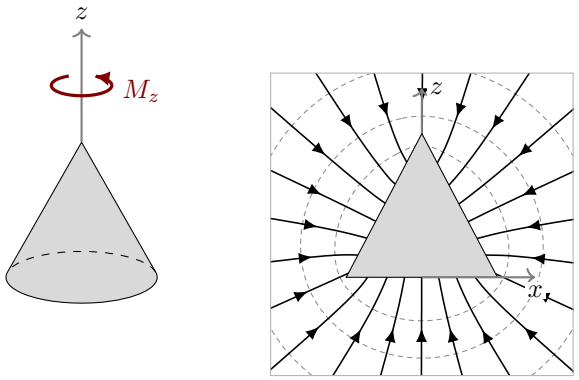
Answer to Problem 3 (e).

$$P_y$$

(f)

That of a homogeneous cone.

Solution: Let the axis of the circular cone be the z axis, as in the figure. A rotation about its axis maps the cone onto itself, so its field can depend on r and z but not on ϕ, *i.e.* $U = U(r, z)$, and M_z is conserved. In the figure, the field lines and equipotentials are shown in a plane containing the axis; they are the same in every such plane.



plane $y = 0$

On the other hand, the apex of the cone is a fixed point. Any translation moves it, even one along the axis, so the translated cone is not the same and the field changes. This would remain true for an infinite cone. Hence, no component of **P** is generally conserved. The only conserved component is then

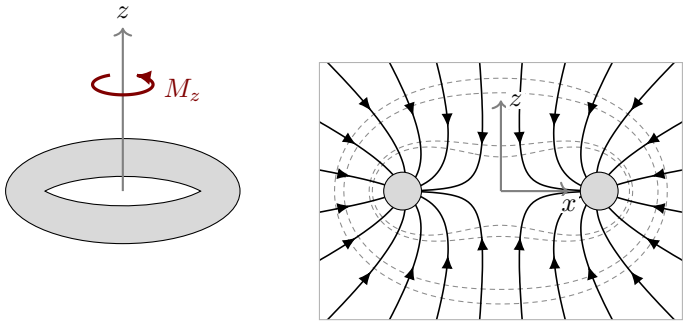
Answer to Problem 3 (f).

M_z

(g)

That of a homogeneous circular torus.

Solution: Let the axis of the torus be the z axis, as in the figure. A circular torus is generated by turning a circle about this axis, so any rotation about it maps the torus onto itself. Its field can then depend on r and z but not on ϕ, *i.e.* $U = U(r, z)$, and M_z is conserved. In the figure, the field lines and equipotentials are shown in a plane containing the axis; they are the same in every such plane.



plane $y = 0$

On the other hand, the torus has a fixed position, with its centre at the origin and its central circle in the plane $z = 0$. Any translation moves it to another position and changes the field, so no component of **P** is generally conserved. The only conserved component is then

Answer to Problem 3 (g).

M_z

(h)

That of an infinite homogeneous cylindrical helix.

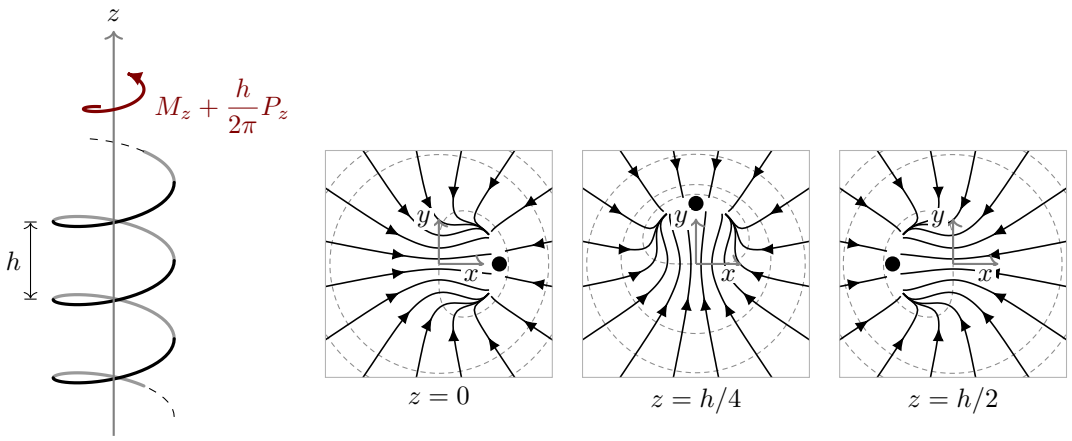
Solution: Let the axis of the helix be the z axis and R its radius. As in the figure, its points are

$$(x, y, z) = \left(R\cos\phi, R\sin\phi, \frac{h\phi}{2\pi} \right),$$

where h is the pitch: the helix advances by h along the axis during one positive rotation of 2π. A rotation by $\delta\phi$ alone does not map the helix onto itself, nor does an arbitrary translation δz alone (the translations by whole multiples of h do, but they are isolated transformations, as the rotations of the prism). However, the combined screw displacement

$$\delta z = \frac{h}{2\pi}\delta\phi$$

maps each point of the helix to the point at $\phi+\delta\phi$, so it does preserve the helix and its field. Consequently, the field in the cross-section at the height z is the one at $z = 0$ turned by the angle $2\pi z/h$. This is what the three cross-sections of the figure show: increasing z by $h/4$ turns the pattern by $\pi/2$. In these cross-sections, the field lines show the components of the field in each plane, and not the complete three-dimensional field lines, since the field also has a component along z.



In cylindrical co-ordinates, the field thus depends on ϕ and z only through the combination

$$\xi = z - \frac{h}{2\pi}\phi, \qquad U = U(r, \xi).$$

Since ϕ and $\phi + 2\pi$ describe the same point while ξ changes by h, the function $U(r, \xi)$ is periodic in ξ with period h. The Lagrangian is

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2 + \dot{z}^2) - U\left(r, z - \frac{h\phi}{2\pi}\right),$$

and its derivatives with respect to ϕ and z are

$$\frac{\partial L}{\partial \phi} = \frac{h}{2\pi}\frac{\partial U}{\partial \xi} \quad \text{and} \quad \frac{\partial L}{\partial z} = -\frac{\partial U}{\partial \xi}.$$

Using Lagrange's equations for ϕ and z, with $M_z = mr^2\dot{\phi}$ and $P_z = m\dot{z}$, we get

$$\dot{M}_z = \frac{h}{2\pi}\frac{\partial U}{\partial \xi} \quad \text{and} \quad \dot{P}_z = -\frac{\partial U}{\partial \xi},$$

so that

$$\frac{d}{dt}\left(M_z + \frac{h}{2\pi}P_z\right) = \frac{h}{2\pi}\frac{\partial U}{\partial \xi} - \frac{h}{2\pi}\frac{\partial U}{\partial \xi} = 0.$$

No individual Cartesian component of $\mathbf{P}$ or $\mathbf{M}$ is generally conserved here. The screw symmetry conserves a combination of the axial angular momentum and the axial momentum, whose two terms have the same dimensions since h is a length. The conserved quantity is then

Answer to Problem 3 (h).

$$M_z + \frac{h}{2\pi}P_z = \text{constant}$$

§10 Mechanical similarity

Problem 1.

Find the ratio of the times in the same path for particles having different masses but the same potential energy.

Solution: If the two particles have the same path, then the ratio of linear dimension is

$$\frac{l'}{l} = \alpha = 1. \tag{2.7}$$

We can define the ratio of time and mass by

$$\frac{t'}{t} = \beta$$

and

$$\frac{m'}{m} = \gamma.$$

Corresponding lengths scale as α and corresponding times as β, so the speeds at corresponding points of the two paths scale as $v'/v = \alpha/\beta$. Then, the ratio of kinetic energy is the ratio of the masses times the ratio of the squared speeds,

$$\frac{T'}{T} = \frac{m'v'^2}{mv^2} = \frac{m'}{m}\left(\frac{\alpha}{\beta}\right)^2 = \frac{\gamma\alpha^2}{\beta^2}.$$

To leave the equation of motion unaltered, the ratio of the kinetic energy and the potential energy must be the same, *i.e.* ,

$$\begin{aligned} \frac{U'}{U} &= \frac{T'}{T} \\ \Rightarrow \alpha^k &= \frac{\gamma\alpha^2}{\beta^2}. \end{aligned}$$

Using Eq. (2.7), we get

$$\begin{aligned} 1 &= \frac{\gamma}{\beta^2} \\ \Rightarrow \beta &= \sqrt{\gamma}. \end{aligned}$$

The ratio of the times is then

Answer to Problem 1.

$$\frac{t'}{t} = \sqrt{\frac{m'}{m}}$$

Problem 2.

Find the ratio of the times in the same path for particles having the same mass but potential energies differing by a constant factor.

Solution: We proceed as in §10 Prob. 1. If the two particles have the same path, then the ratio of linear dimension is

$$\frac{l'}{l} = \alpha = 1. \tag{2.8}$$

We can define the ratio of time by

$$\frac{t'}{t} = \beta.$$

Then, the ratio of kinetic energy is

$$\frac{T'}{T} = \frac{\alpha^2}{\beta^2}.$$

The potential energy of the two particles differ by a constant factor, which mean that

$$\frac{U'}{U} = \gamma.$$

To leave the equation of motion unaltered, the ratio of the kinetic energy and the potential energy must be the same, *i.e.* ,

$$\begin{aligned} \frac{U'}{U} &= \frac{T'}{T} \\ \Rightarrow \gamma &= \frac{\alpha^2}{\beta^2}. \end{aligned}$$

Using Eq. (2.8), we get

$$\begin{aligned} \gamma &= \frac{1}{\beta^2} \\ \Rightarrow \beta &= \sqrt{\frac{1}{\gamma}}. \end{aligned}$$

The ratio of the times is then

Answer to Problem 2.

$$\frac{t'}{t} = \sqrt{\frac{U}{U'}}$$

Chapter 3.

Integration of the Equations of Motion

§11 Motion in one dimension

Problem 1.

Determine the period of oscillations of a simple pendulum (a particle of mass m suspended by a string of length l in a gravitational field) as a function of the amplitude of the oscillations.

Solution: The potential energy of a simple pendulum is

$$U(\phi) = -mgl\cos\phi,$$

see §5 Prob. 1. If we define ϕ_0 as the maximum value of ϕ, the potential energy is equal to the total energy at this point, that is

$$U(\phi_0) = E = -mgl\cos\phi_0.$$

The energy of the pendulum could also be write using the kinetic energy,

$$E = \frac{1}{2}ml^2\dot{\phi}^2 + U(\phi).$$

This first order differential equation can be integrate to give the period of oscillation

$$\begin{aligned} T &= 2\sqrt{\frac{ml^2}{2}}\int_{-\phi_0}^{\phi_0}\frac{d\phi}{\sqrt{E-U(\phi)}} \\ &= 4\sqrt{\frac{ml^2}{2}}\int_{0}^{\phi_0}\frac{d\phi}{\sqrt{mgl\cos\phi - mgl\cos\phi_0}} \\ &= 4\sqrt{\frac{l}{2g}}\int_{0}^{\phi_0}\frac{d\phi}{\sqrt{\cos\phi-\cos\phi_0}}. \end{aligned}$$

To solve this integral, we need first to use a trigonometric identity (I'll let you find or derive this identity), giving us

$$T = 2\sqrt{\frac{l}{g}}\int_0^{\phi_0}\frac{d\phi}{\sqrt{\sin^2\frac{\phi_0}{2}-\sin^2\frac{\phi}{2}}}.$$

Next, we use the substitution

$$\sin\xi = \frac{\sin\frac{\phi}{2}}{\sin\frac{\phi_0}{2}}.$$

It's differential form is

$$\begin{aligned} \frac{d\phi}{d\xi} &= \frac{d}{d\xi}\left(2\arcsin\left(\sin\frac{\phi_0}{2}\sin\xi\right)\right) \\ &= \frac{2\sin\frac{\phi_0}{2}\cos\xi}{\sqrt{1-\sin^2\frac{\phi_0}{2}\sin^2\xi}} \end{aligned}$$

The substitution, finally, give

$$
\begin{aligned}
T &= 2\sqrt{\frac{l}{g}}\int_0^{\phi_0}\frac{d\phi}{\sqrt{\sin^2\frac{\phi_0}{2}-\sin^2\xi\sin^2\frac{\phi_0}{2}}}\\
&= 2\sqrt{\frac{l}{g}}\int_0^{\phi_0}\frac{d\phi}{\sin\frac{\phi_0}{2}\sqrt{1-\sin^2\xi}}\\
&= 2\sqrt{\frac{l}{g}}\int_0^{\phi_0}\frac{d\phi}{\sin\frac{\phi_0}{2}\cos\xi}\\
&= 2\sqrt{\frac{l}{g}}\int_{\arcsin(0)}^{\arcsin(1)}\frac{2\sin\frac{\phi_0}{2}\cos\xi d\xi}{\sin\frac{\phi_0}{2}\cos\xi\sqrt{1-\sin^2\frac{\phi_0}{2}\sin^2\xi}}\\
&= 4\sqrt{\frac{l}{g}}\int_0^{\frac{\pi}{2}}\frac{d\xi}{\sqrt{1-\sin^2\frac{\phi_0}{2}\sin^2\xi}}.
\end{aligned}
$$

We can also write

$$T = 4\sqrt{\frac{l}{g}}K\left(\sin\frac{\phi_0}{2}\right),$$

where

$$K(k) = \int_0^{\frac{\pi}{2}}\frac{d\xi}{\sqrt{1-k^2\sin^2\xi}}. \tag{3.1}$$

The integral K is known as the complete elliptic integral of the first kind. To solve this integral, we first need to see that the integrand is of the form

$$f(x) = (1-x)^{-\frac{1}{2}}.$$

The n derivative of this function is

$$f^{(n)}(x) = \frac{(2n-1)!!}{2^n}(1-x)^{-\frac{2n+1}{2}},$$

thus the Maclaurin Series is

$$f(x) = \sum_{n\geq 0}\frac{(2n-1)!!}{2^n\cdot n!}x^n.$$

Using this result in Eq. (3.1), we get

$$
\begin{aligned}
K(k) &= \int_0^{\frac{\pi}{2}}\sum_{n\geq 0}\frac{(2n-1)!!}{2^n\cdot n!}k^{2n}\sin^{2n}\xi\, d\xi\\
&= \sum_{n\geq 0}\frac{(2n-1)!!}{2^n\cdot n!}k^{2n}\int_0^{\frac{\pi}{2}}\sin^{2n}\xi\, d\xi.
\end{aligned}
$$

This last integral can be solve using the beta function

$$\mathcal{B}(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = 2\int_0^{\frac{\pi}{2}}\sin^{2x-1}t\cos^{2y-1}t\, dt,$$

with $x=\frac{2n+1}{2}$ and $y=\frac{1}{2}$. Thus, using $\Gamma\left(n+\frac{1}{2}\right)=\frac{(2n-1)!!}{2^n}\sqrt{\pi}$, the beta function become

$$
\begin{aligned}
\frac{1}{2}\mathcal{B}\left(n+\frac{1}{2},\frac{1}{2}\right) &= \frac{\Gamma(n+\frac{1}{2})\Gamma(\frac{1}{2})}{2\Gamma(n+1)}\\
&= \frac{\sqrt{\pi}}{2}\frac{\Gamma(\frac{2n+1}{2})}{n!}\\
&= \frac{\pi}{2}\frac{(2n-1)!!}{2^n\cdot n!}
\end{aligned}
$$

The elliptic integral is thereby

$$K(k) = \frac{\pi}{2} \sum_{n \geq 0} \left[\frac{(2n-1)!!}{2^n \cdot n!} k^n \right]^2$$
$$= \frac{\pi}{2} \sum_{n \geq 0} \left[\frac{(2n-1)!!}{(2n)!!} k^n \right]^2 .$$

The period of a simple pendulum is

$$T = 2\pi \sqrt{\frac{l}{g}} \sum_{n \geq 0} \left[\frac{(2n-1)!!}{(2n)!!} \sin^n \frac{\phi_0}{2} \right]^2$$

By expanding the sum, we get

$$T = 2\pi \sqrt{\frac{l}{g}} \left(1 + \left(\frac{1}{2} \right)^2 \sin^2 \frac{\phi_0}{2} + \left(\frac{1 \cdot 3}{2 \cdot 4} \right)^2 \sin^4 \frac{\phi_0}{2} + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \right)^2 \sin^6 \frac{\phi_0}{2} + \dots \right).$$

Using the Maclaurin series

$$\sin \frac{\phi_0}{2} = \frac{1}{2}\phi_0 - \frac{1}{48}\phi_0^3 + \frac{1}{3840}\phi_0^5 - \frac{1}{645120}\phi_0^7 + \dots,$$

we finally get

Answer to Problem 1.

$$T = 2\pi \sqrt{\frac{l}{g}} \left(1 + \frac{1}{16}\phi_0^2 + \frac{11}{3072}\phi_0^4 + \dots \right)$$

Problem 2.

Determine the period of oscillation, as a function of the energy, when a particle of mass m moves in fields for which the potential energy is

(a)

$$U = A|x|^n$$

Solution: For the motion to be an oscillation, we assume $A > 0$, $n > 0$ and $E > 0$. The total energy of the particle is

$$E = \frac{1}{2}m\dot{x}^2 + U(x).$$

At the turning points x_1 of the motion, the kinetic energy vanishes, so they satisfy $E = U(x_1)$, that is

$$\begin{aligned} E = U(x_1) &= A|x_1|^n \\ \Rightarrow |x_1| &= \left(\frac{E}{A}\right)^{1/n}, \\ \Rightarrow x_1 &= \pm\left(\frac{E}{A}\right)^{1/n}, \end{aligned}$$

we get the period of oscillation,

$$\begin{aligned} T &= 2\sqrt{\frac{m}{2}}\int_{-\left(\frac{E}{A}\right)^{1/n}}^{\left(\frac{E}{A}\right)^{1/n}} \frac{dx}{\sqrt{E - A|x|^n}} \\ &= 2\sqrt{2m}\int_0^{\left(\frac{E}{A}\right)^{1/n}} \frac{dx}{\sqrt{E - A|x|^n}}. \end{aligned}$$

It is possible to remove the absolute value, because the integral is over positive x, thus

$$\begin{aligned} T &= 2\sqrt{2m}\int_0^{\left(\frac{E}{A}\right)^{1/n}} \frac{dx}{\sqrt{E - Ax^n}} \\ &= 2\sqrt{\frac{2m}{E}}\int_0^{\left(\frac{E}{A}\right)^{1/n}} \frac{dx}{\sqrt{1 - \frac{A}{E}x^n}}. \end{aligned}$$

Using the substitution $y = \left(\frac{A}{E}\right)^{\frac{1}{n}} x$, for which $dx = \left(\frac{E}{A}\right)^{\frac{1}{n}} dy$, we get

$$\begin{aligned} T &= 2\sqrt{\frac{2m}{E}}\int_0^1 \frac{\left(\frac{E}{A}\right)^{\frac{1}{n}} dy}{\sqrt{1 - \frac{A}{E}\left(\left(\frac{A}{E}\right)^{\frac{-1}{n}} y\right)^n}} \\ &= 2\sqrt{\frac{2m}{E}}\left(\frac{E}{A}\right)^{\frac{1}{n}}\int_0^1 \frac{dy}{\sqrt{1 - y^n}}. \end{aligned}$$

Using another substitution $u = y^n$, we get

$$T = 2\sqrt{\frac{2m}{E}}\left(\frac{E}{A}\right)^{\frac{1}{n}}\int_0^1 \frac{u^{\frac{1}{n}}\,du}{nu\sqrt{1-u}}.$$

It is possible to express this integral in term of the beta function

$$\mathcal{B}(z_1, z_2) = \int_0^1 t^{z_1 - 1}(1-t)^{z_2 - 1}\,dt = \frac{\Gamma(z_1)\,\Gamma(z_2)}{\Gamma(z_1 + z_2)},$$

with $z_1 = \frac{1}{n}$ and $z_2 = \frac{1}{2}$. This give us

$$T = \frac{2}{n}\sqrt{\frac{2m}{E}}\left(\frac{E}{A}\right)^{\frac{1}{n}} \frac{\Gamma\left(\frac{1}{n}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}+\frac{1}{n}\right)}.$$

Knowing that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, we finally have

Answer to Problem 2 (a).

$$T = \frac{2}{n}\sqrt{\frac{2\pi m}{E}}\left(\frac{E}{A}\right)^{\frac{1}{n}} \frac{\Gamma\left(\frac{1}{n}\right)}{\Gamma\left(\frac{1}{2}+\frac{1}{n}\right)}$$

(b)

$$U = \frac{-U_0}{\cosh^2 \alpha x}, \qquad -U_0 < E < 0$$

Solution: Since $E < 0$, we write $E = -|E|$. The motion is symmetric about $x = 0$, where the potential has its minimum $-U_0$, and the turning points $\pm x_1$ are where the kinetic energy vanishes, $E = U(x_1)$, that is

$$\cosh^2(\alpha x_1) = \frac{U_0}{|E|},$$

which has a real solution $x_1 > 0$ because $|E| < U_0$. The period of oscillation is thus

$$\begin{aligned} T &= 2\sqrt{2m}\int_0^{x_1} \frac{dx}{\sqrt{E-U}} \\ &= 2\sqrt{2m}\int_0^{x_1} \frac{dx}{\sqrt{\frac{U_0}{\cosh^2(\alpha x)} - |E|}} \\ &= 2\sqrt{2m}\int_0^{x_1} \frac{\cosh(\alpha x)\,dx}{\sqrt{U_0 - |E|\cosh^2(\alpha x)}}, \end{aligned}$$

where we multiplied the numerator and the denominator by $\cosh(\alpha x) > 0$. Using $\cosh^2(\alpha x) = 1 + \sinh^2(\alpha x)$, we get

$$T = 2\sqrt{2m}\int_0^{x_1} \frac{\cosh(\alpha x)\,dx}{\sqrt{U_0 - |E| - |E|\sinh^2(\alpha x)}}.$$

The substitution $y = \sinh(\alpha x)$ gives us $dy = \alpha\cosh(\alpha x)\,dx$ and, since $x_1 > 0$,

$$y(x_1) = \sinh(\alpha x_1) = \sqrt{\cosh^2(\alpha x_1) - 1} = \sqrt{\frac{U_0}{|E|} - 1} \equiv a, \qquad a^2 = \frac{U_0}{|E|} - 1.$$

Using this substitution and factoring $|E|$ out of the square root, we get

$$\begin{aligned} T &= \frac{2}{\alpha}\sqrt{2m}\int_0^a \frac{dy}{\sqrt{U_0 - |E| - |E|y^2}} \\ &= \frac{2}{\alpha}\sqrt{\frac{2m}{|E|}}\int_0^a \frac{dy}{\sqrt{a^2 - y^2}} \\ &= \frac{2}{\alpha}\sqrt{\frac{2m}{|E|}}\left[\sin^{-1}\left(\frac{y}{a}\right)\right]_0^a \\ &= \frac{2}{\alpha}\sqrt{\frac{2m}{|E|}}\,\frac{\pi}{2}. \end{aligned}$$

Thus, the period of oscillation is

Answer to Problem 2 (b).

$$T = \frac{\pi}{\alpha}\sqrt{\frac{2m}{|E|}}$$

(c)

$$U = U_0 \tan^2 \alpha x$$

Solution: We assume $U_0 > 0$ and $E > 0$; the particle oscillates in the well $|\alpha x| < \pi/2$, symmetrically about $x = 0$, between the turning points $\pm x_1$ where $E = U(x_1)$, that is

$$x_1 = \frac{1}{\alpha}\tan^{-1}\sqrt{\frac{E}{U_0}}.$$

The period of oscillation is

$$\begin{aligned} T &= 2\sqrt{2m}\int_0^{x_1} \frac{dx}{\sqrt{E - U}} \\ &= 2\sqrt{2m}\int_0^{x_1} \frac{dx}{\sqrt{E - U_0\tan^2(\alpha x)}} \\ &= 2\sqrt{2m}\int_0^{x_1} \frac{\cos(\alpha x)\,dx}{\sqrt{E\cos^2(\alpha x) - U_0\sin^2(\alpha x)}} \\ &= 2\sqrt{2m}\int_0^{x_1} \frac{\cos(\alpha x)\,dx}{\sqrt{E - (E + U_0)\sin^2(\alpha x)}}, \end{aligned}$$

where we multiplied the numerator and the denominator by $\cos(\alpha x) > 0$. We use the real substitution

$$y = \sin(\alpha x), \qquad dy = \alpha\cos(\alpha x)\,dx,$$

whose upper limit is

$$y(x_1) = \sin\left(\tan^{-1}\sqrt{\frac{E}{U_0}}\right) = \frac{\sqrt{E/U_0}}{\sqrt{1 + E/U_0}} = \sqrt{\frac{E}{E + U_0}} \equiv a.$$

Factoring $E + U_0$ out of the square root, we get

$$\begin{aligned} T &= \frac{2\sqrt{2m}}{\alpha}\int_0^a \frac{dy}{\sqrt{E-(E+U_0)\,y^2}} \\ &= \frac{2}{\alpha}\sqrt{\frac{2m}{E+U_0}}\int_0^a \frac{dy}{\sqrt{a^2-y^2}} \\ &= \frac{2}{\alpha}\sqrt{\frac{2m}{E+U_0}}\left[\sin^{-1}\left(\frac{y}{a}\right)\right]_0^a \\ &= \frac{2}{\alpha}\sqrt{\frac{2m}{E+U_0}}\,\frac{\pi}{2}. \end{aligned}$$

Thus, the period of oscillation is

Answer to Problem 2 (c).

$$T = \frac{\pi}{\alpha}\sqrt{\frac{2m}{E+U_0}}$$

§13 The reduced mass

Problem 1.

A system consists of one particle of mass M and n particles with equal masses m. Eliminate the motion of the centre of mass and so reduce the problem to one involving n particles.

Solution: Let $\mathbf{R}$ be the radius vector of the particle of mass M, and $\mathbf{R}_a\,(a = 1, 2, \ldots, n)$ those of the particles of mass m. We put $\mathbf{r}_a \equiv \mathbf{R}_a - \mathbf{R}$ and take the origin to be at the centre of mass, namely

$$M\mathbf{R} + m\sum_a \mathbf{R}_a = 0.$$

Thus, we got

$$\mathbf{R} = -\frac{m}{M}\left(\sum_a \mathbf{r}_a + n\mathbf{R}\right) = -\frac{m}{\mu}\sum_a \mathbf{r}_a, \tag{3.2}$$

where $\mu \equiv M + nm$. The Lagrangian of the system is

$$L = \frac{1}{2}M\dot{\mathbf{R}}^2 + \frac{1}{2}m\sum_a \dot{\mathbf{R}}_a^2 - U.$$

If we substitute Eq. (3.2) and $\mathbf{R}_a = \mathbf{R} + \mathbf{r}_a$, we get

$$\begin{aligned}
L &= \frac{1}{2}\frac{Mm^2}{\mu^2}\left(\sum_a \dot{\mathbf{r}}_a\right)^2 + \frac{1}{2}nm\dot{\mathbf{R}}^2 + \frac{1}{2}m\sum_a \dot{\mathbf{r}}_a^2 + m\dot{\mathbf{R}}\sum_a \dot{\mathbf{r}}_a - U \\
&= \frac{1}{2}\frac{Mm^2}{\mu^2}\left(\sum_a \dot{\mathbf{r}}_a\right)^2 + \frac{1}{2}\frac{nm^3}{\mu^2}\left(\sum_a \dot{\mathbf{r}}_a\right)^2 + \frac{1}{2}m\sum_a \dot{\mathbf{r}}_a^2 - \frac{m^2}{\mu}\left(\sum_a \dot{\mathbf{r}}_a\right)^2 - U \\
&= \frac{1}{2}m\sum_a \dot{\mathbf{r}}_a^2 - \frac{1}{2}\frac{m^2}{\mu}\left(\frac{2\mu - M - nm}{\mu}\right)\left(\sum_a \dot{\mathbf{r}}_a\right)^2 - U \\
&= \frac{1}{2}m\sum_a \dot{\mathbf{r}}_a^2 - \frac{1}{2}\frac{m^2}{\mu}\left(\sum_a \dot{\mathbf{r}}_a\right)^2 - U.
\end{aligned}$$

The potential energy depends only on the relative positions of the particles, so $U = U(\mathbf{r}_1, \ldots, \mathbf{r}_n)$. The Lagrangian is thus only a function of $\mathbf{r}_a$ and their time derivatives, which is

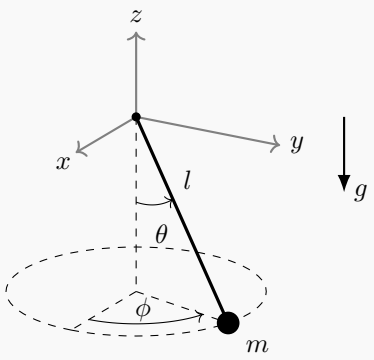
Answer to Problem 1.

$$L = \frac{1}{2}m\sum_a \dot{\mathbf{r}}_a^2 - \frac{1}{2}\frac{m^2}{\mu}\left(\sum_a \dot{\mathbf{r}}_a\right)^2 - U(\mathbf{r}_1, \ldots, \mathbf{r}_n), \qquad \mu = M + nm$$

§14 Motion in a central field

Problem 1.

Integrate the equations of motion for a spherical pendulum (a particle of mass m moving on the surface of a sphere of radius l in a gravitational field).



Solution: The Cartesian position of the particle m is

$$\begin{aligned} x &= l\sin\theta\cos\phi \\ y &= l\sin\theta\sin\phi \\ z &= -l\cos\theta. \end{aligned}$$

The time derivatives of these coordinates are

$$\begin{aligned} \dot{x} &= l\dot{\theta}\cos\theta\cos\phi - l\dot{\phi}\sin\theta\sin\phi \\ \dot{y} &= l\dot{\theta}\cos\theta\sin\phi + l\dot{\phi}\sin\theta\cos\phi \\ \dot{z} &= l\dot{\theta}\sin\theta. \end{aligned}$$

Substituting these expressions in the Lagrangian, we obtain the desired Lagrangian form, that is

$$\begin{aligned} L &= \frac{1}{2}m\left(\dot{x}^2+\dot{y}^2+\dot{z}^2\right) - mgz \\ &= \frac{1}{2}ml^2\left(\dot{\theta}^2+\dot{\phi}^2\sin^2\theta\right) + mgl\cos\theta. \end{aligned}$$

The Lagrangian does not involve ϕ explicitly, thus this co-ordinate is cyclic and the generalized momentum p_ϕ is conserved. This momentum is the same as the z-component of angular momentum M_z (see §9 Prob. 2 with $r = l$) and is written

$$M_z = p_\phi = \frac{\partial L}{\partial\dot{\phi}} = ml^2\dot{\phi}\sin^2\theta. \tag{3.3}$$

The energy of the pendulum is

$$E = \frac{1}{2}ml^2\left(\dot{\theta}^2+\dot{\phi}^2\sin^2\theta\right) - mgl\cos\theta.$$

Substituting $\dot{\phi}$ with Eq. (3.3) in the energy we get

$$\begin{aligned} E &= \frac{1}{2}ml^2\dot{\theta}^2 + \frac{M_z^2}{2ml^2\sin^2\theta} - mgl\cos\theta \\ &= \frac{1}{2}ml^2\dot{\theta}^2 + U_{\text{eff}}(\theta), \end{aligned}$$

where $U_{\text{eff}}(\theta)$ is the effective potential energy. Notice that the energy was re-written as a function of only one co-ordinate (*i.e.* θ). This is equivalent to the one-dimensional problem we have already seen. On a

branch of the motion where θ increases, $\dot{\theta}$ is the positive square root, thus

$$\begin{aligned}\frac{d\theta}{dt} &= \sqrt{\frac{2\left(E - U_{\text{eff}}(\theta)\right)}{ml^2}} \\ \Rightarrow t - t_0 &= l\sqrt{\frac{m}{2}}\int_{\theta_0}^{\theta}\frac{d\vartheta}{\sqrt{E - U_{\text{eff}}(\vartheta)}},\end{aligned}$$

where $\theta_0 = \theta(t_0)$ and ϑ is the integration variable. On a branch where θ decreases, $\dot{\theta}$ is the negative square root, and the right-hand side changes sign. This integral lead to an elliptic integral of the first kind see §11 Prob. 1. We also need to find the solution to the angle ϕ. To do so, we can use the Eq. (3.3) with the chain rule from calculus, that is, on an increasing-θ branch,

$$\begin{aligned}\frac{M_z}{ml^2\sin^2\theta} &= \frac{d\phi}{dt} \\ &= \frac{d\phi}{d\theta}\frac{d\theta}{dt} \\ &= \frac{d\phi}{d\theta}\sqrt{\frac{2\left(E - U_{\text{eff}}(\theta)\right)}{ml^2}}.\end{aligned}$$

Finally, we get

$$\begin{aligned}\frac{d\phi}{d\theta} &= \frac{M_z}{l\sin^2\theta}\sqrt{\frac{1}{2m\left(E - U_{\text{eff}}(\theta)\right)}} \\ \Rightarrow \phi - \phi_0 &= \frac{M_z}{l\sqrt{2m}}\int_{\theta_0}^{\theta}\frac{d\vartheta}{\sin^2\vartheta\sqrt{E - U_{\text{eff}}(\vartheta)}},\end{aligned}$$

with $\phi_0 = \phi(t_0)$. Again, the sign of the right-hand side is reversed on a decreasing-θ branch, so that ϕ always changes in the direction given by the sign of M_z, as Eq. (3.3) requires. This integral lead to an elliptic integral of the third kind (I really don't want to solve this). When θ reaches a turning point, $\dot{\theta}$ changes sign: the motion is continued by restarting both integrals from that turning point with the opposite sign, so that t keeps increasing and ϕ keeps changing monotonically. It is possible to find the range of angle θ. At the maximum and minimum value of θ, $\dot{\theta} = 0$, while the azimuthal kinetic energy $M_z^2/(2ml^2\sin^2\theta)$ can remain nonzero, thus

$$\begin{aligned}E &= U_{\text{eff}} \\ &= \frac{M_z^2}{2ml^2\sin^2\theta} - mgl\cos\theta \\ &= \frac{M_z^2}{2ml^2\left(1 - \cos^2\theta\right)} - mgl\cos\theta\end{aligned}$$

We can find a cubic algebraic equation for $\cos\theta$, that is

$$\begin{aligned}0 &= \left(E + mgl\cos\theta\right)\left(2ml^2\left(1 - \cos^2\theta\right)\right) - M_z^2 \\ &= mgl\left(\cos^3\theta - \cos\theta\right) + E\left(\cos^2\theta - 1\right) + \frac{M_z^2}{2ml^2}.\end{aligned}$$

The centrifugal part of the effective potential U_{eff}, namely

$$\frac{M_z^2}{2ml^2\sin^2\theta},$$

is never negative. If $M_z \neq 0$, it is positive and, as we can see, diverges at $\theta \to 0$ and $\theta \to \pi$, thus the angle θ is bound away from the poles. This can be write as

$$\theta \in [\theta_p, \theta_a] \qquad\qquad 0 < \theta_p \leq \theta_a < \pi \qquad (M_z \neq 0),$$

where the subscripts of the angles θ_p and θ_a are for the perigee and apogee. This mean that the two roots of the cubic algebraic equation for $\cos\theta$ giving the turning points are bound between -1 and $+1$ (the cubic is positive at $\cos\theta = \pm 1$, so its third root is smaller than -1). Except for the solution with

constant θ at the minimum of U_{eff} (when E equals this minimum, the two roots merge and $\theta_p = \theta_a$), the turning points are distinct, $\theta_p < \theta_a$, and θ oscillates between them. If $M_z = 0$, there is no centrifugal barrier: the motion is planar (a pendulum swinging, or revolving, in a fixed vertical plane) and it can reach a pole. To summarize, the solution of the equation of motion of the spherical pendulum is

Answer to Problem 1.

$$t - t_0 = l\sqrt{\frac{m}{2}} \int_{\theta_0}^{\theta} \frac{d\vartheta}{\sqrt{E - U_{\text{eff}}(\vartheta)}}$$

$$\phi - \phi_0 = \frac{M_z}{l\sqrt{2m}} \int_{\theta_0}^{\theta} \frac{d\vartheta}{\sin^2\vartheta \sqrt{E - U_{\text{eff}}(\vartheta)}}$$

$$U_{\text{eff}}(\theta) = \frac{M_z^2}{2ml^2 \sin^2\theta} - mgl\cos\theta$$

on a branch where θ increases; both right-hand sides change sign on a branch where θ decreases, and the integrals are restarted from each turning point θ_p, θ_a (roots of $E = U_{\text{eff}}(\theta)$) with the opposite sign.

Problem 2.

Integrate the equations of motion for a particle moving on the surface of a cone (of vertical angle 2α) placed vertically and with vertex downwards in a gravitational field.

Solution: We proceed as in §14 Prob. 1. By using the spherical co-ordinate, we can write the position of the particle as

$$\begin{aligned} x &= r\cos\phi\sin\alpha \\ y &= r\sin\phi\sin\alpha \\ z &= r\cos\alpha. \end{aligned}$$

By taking the time derivative of those, we obtain

$$\begin{aligned} \dot{x} &= \dot{r}\cos\phi\sin\alpha - r\dot{\phi}\sin\phi\sin\alpha \\ \dot{y} &= \dot{r}\sin\phi\sin\alpha + r\dot{\phi}\cos\phi\sin\alpha \\ \dot{z} &= \dot{r}\cos\alpha. \end{aligned}$$

The Lagrangian is thus

$$\begin{aligned} L &= \frac{1}{2}m\left(\dot{x}^2+\dot{y}^2+\dot{z}^2\right) - mgz \\ &= \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\phi}^2\sin^2\alpha\right) - mgr\cos\alpha. \end{aligned}$$

The co-ordinate ϕ is cyclic, thus

$$M_z = p_\phi = \frac{\partial L}{\partial\dot{\phi}} = mr^2\dot{\phi}\sin^2\alpha \tag{3.4}$$

is conserved. This is the z-component of the angular momentum found in §9 Prob. 1, with the distance $r\sin\alpha$ from the axis. The energy of the particle is

$$E = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\phi}^2\sin^2\alpha\right) + mgr\cos\alpha. \tag{3.5}$$

We can rewrite Eq. (3.4) as

$$\dot{\phi}^2 = \frac{M_z^2}{m^2r^4\sin^4\alpha}$$

and substituting in Eq. (3.5), we get

$$\begin{aligned} E &= \frac{1}{2}m\dot{r}^2 + \frac{M_z^2}{2mr^2\sin^2\alpha} + mgr\cos\alpha \\ &= \frac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r). \end{aligned}$$

Hence, on a branch where r increases,

$$t = \sqrt{\frac{m}{2}}\int\frac{dr}{\sqrt{E-U_{\text{eff}}(r)}}.$$

From Eq. (3.4), it is possible to write, on the same branch,

$$\begin{aligned} &\frac{d\phi}{dt} = \frac{M_z}{mr^2\sin^2\alpha} \\ \Rightarrow &\frac{d\phi}{dr}\frac{dr}{dt} = \frac{M_z}{mr^2\sin^2\alpha} \\ \Rightarrow &\frac{d\phi}{dr}\sqrt{\frac{2\left(E-U_{\text{eff}}(r)\right)}{m}} = \frac{M_z}{mr^2\sin^2\alpha} \\ \Rightarrow &\phi = \frac{M_z}{\sqrt{2m}\sin^2\alpha}\int\frac{dr}{r^2\sqrt{E-U_{\text{eff}}(r)}}. \end{aligned}$$

Written with explicit limits, the co-ordinates evolve as

$$t - t_0 = \sqrt{\frac{m}{2}} \int_{r_0}^{r} \frac{ds}{\sqrt{E - U_{\text{eff}}(s)}},$$
$$\phi - \phi_0 = \frac{M_z}{\sqrt{2m}\sin^2\alpha} \int_{r_0}^{r} \frac{ds}{s^2\sqrt{E - U_{\text{eff}}(s)}},$$

where $r_0 = r(t_0)$, $\phi_0 = \phi(t_0)$ and s is the integration variable. On a branch where r decreases, $\dot{r}$ is the negative square root and both right-hand sides change sign, so that t always increases and ϕ always changes in the direction given by the sign of M_z. At a turning point, $\dot{r}$ changes sign and the integrals are restarted from it with the opposite sign. It is possible to find the range of r. At the maximum and minimum value of r, the radial velocity is zero (the azimuthal velocity is not), thus

$$\begin{aligned} E &= U_{\text{eff}} \\ &= \frac{M_z^2}{2mr^2\sin^2\alpha} + mgr\cos\alpha \end{aligned}$$

We can find a cubic algebraic equation for r, that is

$$\begin{aligned} 0 &= r^2\left(E - mgr\cos\alpha\right) - \frac{M_z^2}{2m\sin^2\alpha} \\ &= mgr^3\cos\alpha - Er^2 + \frac{M_z^2}{2m\sin^2\alpha}. \end{aligned}$$

When $M_z \neq 0$ and E is above the minimum of $U_{\text{eff}}(r)$, this equation has two distinct positive roots, $r_p < r_a$, which are the turning points of the motion (the third root is negative). When E equals the minimum of U_{eff}, the two roots merge and the orbit is a circle, $r =$ constant. When $M_z = 0$, there is no centrifugal barrier at the vertex: the particle moves along a generator of the cone and reaches the vertex $r = 0$. To summarize, the solution of the equation of motion for a particle moving on the surface of a cone is

Answer to Problem 2.

$$t - t_0 = \sqrt{\frac{m}{2}} \int_{r_0}^{r} \frac{ds}{\sqrt{E - U_{\text{eff}}(s)}}$$
$$\phi - \phi_0 = \frac{M_z}{\sqrt{2m}\sin^2\alpha} \int_{r_0}^{r} \frac{ds}{s^2\sqrt{E - U_{\text{eff}}(s)}}$$
$$U_{\text{eff}}(r) = \frac{M_z^2}{2mr^2\sin^2\alpha} + mgr\cos\alpha$$

on a branch where r increases; both right-hand sides change sign on a branch where r decreases, and the integrals are restarted from each turning point r_p, r_a (roots of $E = U_{\text{eff}}(r)$) with the opposite sign.

Problem 3.

Integrate the equations of motion for a pendulum of mass m_2, with a mass m_1 at the point of support which can move on a horizontal line lying in the plane in which m_2 moves (see the figure in §5 Prob. 2).

Solution: From §5 Prob. 2, the Lagrangian of the system is

$$L = \frac{1}{2}(m_1+m_2)\dot{x}^2 + \frac{1}{2}m_2(l^2\dot{\phi}^2 + 2l\dot{x}\dot{\phi}\cos\phi) + m_2 gl\cos\phi.$$

The co-ordinate x is cyclic, thus

$$p_x = \frac{\partial L}{\partial \dot{x}} = (m_1+m_2)\dot{x} + m_2 l\dot{\phi}\cos\phi = \text{constant} \tag{3.6}$$

is conserved. It is always possible to find an inertial frame of reference where $p_x = 0$, using this frame we get

$$\begin{aligned}
&(m_1+m_2)\dot{x} + m_2 l\dot{\phi}\cos\phi = 0 \\
\Rightarrow &\int \Big((m_1+m_2)\dot{x} + m_2 l\dot{\phi}\cos\phi\Big)\,dt = \text{constant} \\
\Rightarrow &(m_1+m_2)x + m_2 l\sin\phi = (m_1+m_2)R_x = \text{constant}.
\end{aligned} \tag{3.7}$$

This express the fact that the centre of mass $\mathbf{R}$ of the system does not move horizontally. It is also possible to rewrite Eq. (3.6) as

$$\begin{aligned}
(m_1+m_2)\dot{x} + m_2 l\dot{\phi}\cos\phi &= 0 \\
\Rightarrow \dot{x} &= \frac{-m_2 l\dot{\phi}\cos\phi}{m_1+m_2}
\end{aligned}$$

Plugging this into the energy of the system we get

$$\begin{aligned}
E &= \frac{1}{2}(m_1+m_2)\dot{x}^2 + \frac{1}{2}m_2(l^2\dot{\phi}^2 + 2l\dot{x}\dot{\phi}\cos\phi) - m_2 gl\cos\phi \\
&= \frac{m_2^2 l^2\dot{\phi}^2\cos^2\phi}{2(m_1+m_2)} + \frac{1}{2}m_2 l^2\dot{\phi}^2 - \frac{m_2^2 l^2\dot{\phi}^2\cos^2\phi}{m_1+m_2} - m_2 gl\cos\phi \\
&= \frac{1}{2}m_2 l^2\dot{\phi}^2\left(1 - \frac{m_2}{m_1+m_2}\cos^2\phi\right) - m_2 gl\cos\phi.
\end{aligned}$$

Hence,

$$\begin{aligned}
t &= l\sqrt{\frac{m_2}{2}}\int\sqrt{\frac{1-\frac{m_2}{m_1+m_2}\cos^2\phi}{E+m_2 gl\cos\phi}}\,d\phi \\
&= l\sqrt{\frac{m_2}{2(m_1+m_2)}}\int\sqrt{\frac{m_1+m_2\sin^2\phi}{E+m_2 gl\cos\phi}}\,d\phi.
\end{aligned}$$

Using Eq. (3.7), we can express the position of the mass m_2 as

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x + l\sin\phi \\ l\cos\phi \end{bmatrix} = \begin{bmatrix} R_x + l\sin\phi\left(1 - \frac{m_2}{m_1+m_2}\right) \\ l\cos\phi \end{bmatrix} = \begin{bmatrix} R_x + \frac{m_1}{m_1+m_2}l\sin\phi \\ l\cos\phi \end{bmatrix},$$

since Eq. (3.7) gives $x = R_x - \frac{m_2}{m_1+m_2}l\sin\phi$. The path of the particle of mass m_2 is thus an arc of an ellipse center at $(R_x, 0)$ with horizontal semi-axis $lm_1/(m_1+m_2)$ and vertical semi-axis l. It is possible to see that when $m_1 \to \infty$, the path return to the simple pendulum. The solution of the equations of

motion is thus

Answer to Problem 3.

$$t = l\sqrt{\frac{m_2}{2}} \int \sqrt{\frac{1 - \frac{m_2}{m_1+m_2}\cos^2\phi}{E + m_2 g l \cos\phi}} d\phi$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} R_x + l\sin\phi\left(1 - \frac{m_2}{m_1+m_2}\right) \\ l\cos\phi \end{bmatrix} = \begin{bmatrix} R_x + \frac{m_1}{m_1+m_2} l\sin\phi \\ l\cos\phi \end{bmatrix}$$

§15 Kepler's problem

Problem 1.

Find the time dependence of the co-ordinates of a particle with energy $E = 0$ moving in a parabola in a field $U = -\alpha/r$.

Solution: From formula (14.6) of the book, we have the integral

$$\begin{aligned} t &= \int \frac{dr}{\sqrt{\frac{2}{m}\left(E - U(r)\right) - \frac{M^2}{m^2 r^2}}} \\ &= \int \frac{dr}{\sqrt{\frac{2\alpha}{mr} - \frac{M^2}{m^2 r^2}}} \\ &= \int \frac{r dr}{\sqrt{\frac{2\alpha}{m} r - \frac{M^2}{m^2}}}, \end{aligned}$$

written here for the outgoing branch ($\dot{r} > 0$), and taking $M > 0$. We define the parameter of the parabola

$$p = \frac{M^2}{m\alpha}$$

and use the substitution

$$\begin{aligned} r &= \frac{M^2}{2m\alpha}\left(1 + \eta^2\right) = \frac{1}{2} p \left(1 + \eta^2\right), \\ \Rightarrow \sqrt{\frac{2\alpha}{m} r - \frac{M^2}{m^2}} &= \sqrt{\frac{M^2}{m^2}\eta^2} = \frac{M}{m}|\eta|, \end{aligned} \tag{3.8}$$

with the differential form

$$dr = \frac{M^2}{m\alpha}\eta d\eta = p\,\eta\, d\eta.$$

On the outgoing branch we take $\eta \geq 0$, so that $|\eta| = \eta$, and the integral become

$$\begin{aligned} t - t_0 &= \frac{m}{M}\int_0^\eta \frac{1}{2} p \left(1 + \eta'^2\right) p\, d\eta' \\ &= \frac{M^3}{2m\alpha^2}\int_0^\eta \left(1 + \eta'^2\right) d\eta' \\ &= \frac{M^3}{2m\alpha^2}\left(\eta + \frac{1}{3}\eta^3\right) \\ &= \frac{1}{2}\sqrt{\frac{mp^3}{\alpha}}\left(\eta + \frac{1}{3}\eta^3\right), \end{aligned}$$

where t_0 is the time of passage at the perihelion $r = p/2$ ($\eta = 0$). On the incoming branch ($\dot{r} < 0$), the integrand of t changes sign; taking there $\eta \leq 0$, so that $|\eta| = -\eta$, the two sign changes cancel and the same formula holds. It is thus important to specify that the parameter η is signed and varies from $-\infty$ to ∞: $\eta < 0$ before the perihelion and $\eta > 0$ after it. Using Eq. (3.8) and the equation of the parabola (eccentricity $e = 1$)

$$\cos\phi = \frac{p}{r} - 1 = \frac{1 - \eta^2}{1 + \eta^2},$$

it is possible to find the Cartesian co-ordinates

$$x = r\cos\phi = \frac{1}{2} p \left(1 - \eta^2\right)$$

and

$$y^2 = r^2 - x^2 = \frac{p^2}{4}\left[\left(1 + \eta^2\right)^2 - \left(1 - \eta^2\right)^2\right] = p^2\eta^2,$$

whose branch is selected by the sign of η: for $M > 0$ the particle moves counterclockwise, so $y = p\eta$

(negative before the perihelion, positive after it). The parametric form of the required dependence are thus

Answer to Problem 1.

$$r = \frac{1}{2}p\left(1+\eta^2\right) \qquad t - t_0 = \frac{1}{2}\sqrt{\frac{mp^3}{\alpha}}\left(\eta + \frac{1}{3}\eta^3\right)$$
$$x = \frac{1}{2}p\left(1-\eta^2\right) \qquad y = p\eta$$

with $p = M^2/(m\alpha)$ $(M > 0)$, the signed parameter $-\infty < \eta < \infty$, and t_0 the time of passage at the perihelion.

Problem 2.

Integrate the equations of motion for a particle in a central field

$$U = -\frac{\alpha}{r^2} \qquad (\alpha > 0).$$

Solution: The energy of the particle, written with the effective potential energy, is

$$E = \frac{1}{2}m\dot{r}^2 + \frac{M^2}{2mr^2} - \frac{\alpha}{r^2} = \frac{1}{2}m\dot{r}^2 - \frac{C}{2mr^2}, \qquad C \equiv 2m\alpha - M^2.$$

The real constant C compares the attraction with the centrifugal barrier: the effective potential energy $-C/(2mr^2)$ is repulsive for $C < 0$, attractive for $C > 0$, and vanishes identically for the critical angular momentum $M^2 = 2m\alpha$ ($C = 0$). From formula (14.6) of the book, we have

$$\begin{aligned} dt &= \pm \frac{dr}{\sqrt{\frac{2}{m}\left(E - U(r)\right) - \frac{M^2}{m^2r^2}}} \\ &= \pm \frac{dr}{\sqrt{\frac{2E}{m} + \frac{C}{m^2r^2}}} \\ &= \pm \frac{mr\,dr}{\sqrt{2mEr^2 + C}}, \end{aligned}$$

with the upper sign on a branch where r increases and the lower sign on a branch where r decreases.

Time dependence of r. Rather than integrating this case by case, which would require dividing by E, we use the radial equation of motion, obtained by differentiating the energy with respect to time,

$$m\ddot{r} = -\frac{C}{mr^3}.$$

With the energy equation $\dot{r}^2 = 2E/m + C/(m^2r^2)$, it gives

$$\frac{d^2}{dt^2}\left(r^2\right) = 2\dot{r}^2 + 2r\ddot{r} = 2\left(\frac{2E}{m} + \frac{C}{m^2r^2}\right) - \frac{2C}{m^2r^2} = \frac{4E}{m},$$

a constant. Integrating twice from a reference time t_{ref}, at which $r = r_0$ and $\dot{r} = \dot{r}_0$, we get

$$r^2(t) = r_0^2 + 2r_0\dot{r}_0\left(t - t_{\text{ref}}\right) + \frac{2E}{m}\left(t - t_{\text{ref}}\right)^2, \tag{3.9}$$

where the initial data must satisfy the energy equation, $\dot{r}_0^2 = 2E/m + C/(m^2r_0^2)$. This law is valid on each physical segment of the motion, *i.e.* as long as $r > 0$: if r^2 reaches zero, the particle falls into the centre, where the field is singular. It covers the case $E = 0$, where r^2 is a linear function of time, without any division by zero. When $E \neq 0$, completing the square and using the energy equation give

$$r^2 = \frac{2E}{m}\left(t - t_c\right)^2 - \frac{C}{2mE}, \qquad t_c = t_{\text{ref}} - \frac{mr_0\dot{r}_0}{2E},$$

which is also what the direct integration of dt gives (with the substitution $w = 2mEr^2 + C$, $dw = 4mEr\,dr$):

$$t - t_c = \pm\frac{1}{2E}\sqrt{2mEr^2 + C}.$$

Equation of the path. Formula (14.7) of the book gives, on the same branches,

$$\begin{aligned} d\phi &= \pm \frac{M\,dr}{r^2\sqrt{2m\left(E - U(r)\right) - \frac{M^2}{r^2}}} \\ &= \pm \frac{M\,dr}{r^2\sqrt{2mE + \frac{C}{r^2}}}. \end{aligned}$$

Using the substitution

$$u = \frac{1}{r}$$

with the differential form

$$du = -\frac{1}{r^2}dr,$$

it becomes

$$d\phi = -\frac{M\,du}{\sqrt{2mE + Cu^2}} \tag{3.10}$$

on a branch where r increases (u decreases), with the opposite sign on a branch where r decreases. In both cases ϕ changes in the direction of $\dot\phi = M/(mr^2)$. Rather than integrating Eq. (3.10) with a different substitution for each sign of C and E, we take $M > 0$ (the case $M < 0$ follows by changing ϕ into $-\phi$) and consider u as a function of ϕ, primes meaning differentiation with respect to ϕ. Since $\dot r = -u'\dot\phi/u^2 = -Mu'/m$, the energy becomes

$$E = \frac{M^2}{2m}\left(u'^2 + u^2\right) - \alpha u^2,$$

i.e.

$$(u')^2 - \frac{C}{M^2}u^2 = \frac{2mE}{M^2}, \tag{3.11}$$

which is also Eq. (3.10) squared. Differentiating with respect to ϕ (or using the Binet equation $u'' + u = -\frac{m}{M^2}\frac{dU}{du}$ with $U = -\alpha u^2$) gives the linear equation

$$u'' - \frac{C}{M^2}u = 0.$$

Its real solutions must also satisfy the first integral Eq. (3.11), which fixes their amplitude, and they describe a motion only where $u > 0$. This gives the following orbit families.

- $C < 0$, $E > 0$: $u = \sqrt{\frac{2mE}{-C}}\cos\left(\frac{\sqrt{-C}}{M}(\phi - \phi_0)\right)$, with $|\phi - \phi_0| < \frac{\pi M}{2\sqrt{-C}}$. The particle comes from infinity, reaches the minimum distance $r_{\min} = \sqrt{-C/(2mE)}$ at $\phi = \phi_0$ and goes back to infinity, after turning through the angle $\pi M/\sqrt{-C} > \pi$.
- $C > 0$, $E > 0$: $u = \sqrt{\frac{2mE}{C}}\sinh\left(\frac{\sqrt{C}}{M}(\phi - \phi_0)\right)$, with $\phi > \phi_0$. The particle comes from infinity ($\phi \to \phi_0$) and spirals into the centre ($\phi \to \infty$). The reflected branch, $u = \sqrt{\frac{2mE}{C}}\sinh\left(\frac{\sqrt{C}}{M}(\phi_0 - \phi)\right)$ with $\phi < \phi_0$, describes a particle spiralling out of the centre to infinity.
- $C > 0$, $E < 0$: $u = \sqrt{\frac{2m|E|}{C}}\cosh\left(\frac{\sqrt{C}}{M}(\phi - \phi_0)\right)$. The particle spirals out of the centre, reaches the maximum distance $r_{\max} = \sqrt{C/(2m|E|)}$ at $\phi = \phi_0$, and spirals back into the centre.
- $C > 0$, $E = 0$: $u = A\exp\left[\pm\frac{\sqrt{C}}{M}(\phi - \phi_0)\right]$, $A > 0$, a logarithmic spiral, traversed inwards (upper sign) or outwards (lower sign).
- $C = 0$, $E > 0$: $u = \pm\frac{\sqrt{2mE}}{M}(\phi - \phi_0)$, with the sign making $u > 0$: a hyperbolic spiral, traversed inwards for $\phi > \phi_0$ (upper sign) or outwards for $\phi < \phi_0$ (lower sign).
- $C = 0$, $E = 0$: $u = A > 0$, a uniform circular motion of arbitrary positive radius $1/A$, the centrifugal barrier exactly balancing the attraction.

One verifies each of them by substitution in the linear equation and in Eq. (3.11); for instance, for $C < 0$, $u'^2 + \frac{|C|}{M^2}u^2 = \frac{2mE}{-C}\frac{|C|}{M^2} = \frac{2mE}{M^2}$. The time dependence along each path is given by Eq. (3.9). For $E \neq 0$,

since $\frac{d}{dt}\left(r^2\right) = 2r\dot{r} = -\frac{2M}{m}\frac{u'}{u}$ and $\frac{d}{dt}\left(r^2\right) = \frac{4E}{m}(t-t_c)$, we have the equivalent parameter relation

$$t - t_c = -\frac{M}{2E}\frac{u'}{u},$$

which fixes the orientation consistently with $\dot{\phi} = M/(mr^2) > 0$. For example,

$$t - t_c = \frac{\sqrt{-C}}{2E}\tan\left(\frac{\sqrt{-C}}{M}(\phi-\phi_0)\right) \qquad (C<0,\ E>0),$$
$$t - t_c = \frac{\sqrt{C}}{2|E|}\tanh\left(\frac{\sqrt{C}}{M}(\phi-\phi_0)\right) \qquad (C>0,\ E<0),$$

both increasing with ϕ. For $C > 0$ (and for $C = 0$, $E > 0$), the particle falls into (or emerges from) the centre at a finite time, while turning an infinite number of times around it.

Radial motion, $M = 0$**.** Then $C = 2m\alpha > 0$, the direction ϕ is constant and the particle moves on a straight line through the centre. Its distance follows the same law Eq. (3.9), for any energy, until it reaches the singular centre.

No motion. Since $\dot{r}^2 = 2E/m + C/(m^2r^2)$ cannot be negative, there is no motion for $C < 0$ and $E \le 0$, nor for $C = 0$ and $E < 0$.

It is also interesting to know that these paths are Cotes's spirals[1]. To summarize, the equations of motion for a particle in a central inverse-cube law force are

Answer to Problem 2.

With $C = 2m\alpha - M^2$, $u = 1/r$ and $M > 0$ (change ϕ into $-\phi$ for $M < 0$), the paths are, where $u > 0$,

$$C<0,\ E>0: \quad u = \sqrt{\frac{2mE}{-C}}\cos\left(\frac{\sqrt{-C}}{M}(\phi-\phi_0)\right)$$
$$C>0,\ E>0: \quad u = \sqrt{\frac{2mE}{C}}\sinh\left(\frac{\sqrt{C}}{M}(\phi-\phi_0)\right) \quad \text{(or its reflection } \phi-\phi_0 \to \phi_0-\phi)$$
$$C>0,\ E<0: \quad u = \sqrt{\frac{2m|E|}{C}}\cosh\left(\frac{\sqrt{C}}{M}(\phi-\phi_0)\right)$$
$$C>0,\ E=0: \quad u = A\exp\left[\pm\frac{\sqrt{C}}{M}(\phi-\phi_0)\right], \qquad A>0$$
$$C=0,\ E>0: \quad u = \pm\frac{\sqrt{2mE}}{M}(\phi-\phi_0)$$
$$C=0,\ E=0: \quad u = A>0 \quad \text{(circular motion of arbitrary radius)}$$

On each segment of the motion (until r reaches the centre),

$$r^2(t) = r_0^2 + 2r_0\dot{r}_0(t-t_{\text{ref}}) + \frac{2E}{m}(t-t_{\text{ref}})^2, \qquad \dot{r}_0^2 = \frac{2E}{m} + \frac{C}{m^2r_0^2},$$

and, for $E \neq 0$, $t - t_c = -\frac{M}{2E}\frac{u'}{u}$. For $M = 0$, the motion is radial, with constant ϕ and the same $r^2(t)$. There is no motion for $C < 0$, $E \le 0$, nor for $C = 0$, $E < 0$.

[1] Whittaker ET (1937). A Treatise on the Analytical Dynamics of Particles and Rigid Bodies, with an Introduction to the Problem of Three Bodies (4th ed.). New York: Dover Publications. pp. 80–83. ISBN 978-0-521-35883-5.

Problem 3.

When a small correction $\delta U(r)$ is added to the potential energy $U = -\alpha/r$, the paths of finite motion are no longer closed, and at each revolution the perihelion is displaced through a small angle $\delta\phi$. Find $\delta\phi$ when

(a)

$$\delta U = \frac{\beta}{r^2}$$

Solution: From formula (14.10) of the book, we have

$$\begin{aligned}\Delta\phi &= 2\int_{r_{\min}}^{r_{\max}} \frac{M dr}{r^2\sqrt{2m(E-U) - \frac{M^2}{r^2}}} \\ &= -2\frac{\partial}{\partial M}\int_{r_{\min}}^{r_{\max}} \sqrt{2m(E-U) - \frac{M^2}{r^2}}\, dr \\ &= -2\frac{\partial}{\partial M}\int_{r_{\min}}^{r_{\max}} \sqrt{2m\left(E + \frac{\alpha}{r} - \delta U\right) - \frac{M^2}{r^2}}\, dr.\end{aligned}$$

Expanding the integrand in powers of δU involves using a Taylor series expansion. Let's denote the integrand as F:

$$F = \sqrt{2m\left(E + \frac{\alpha}{r} - \delta U\right) - \frac{M^2}{r^2}} = \sqrt{F_0^2 - 2m\,\delta U},$$

where

$$F_0 = \sqrt{2m\left(E + \frac{\alpha}{r}\right) - \frac{M^2}{r^2}}$$

is the value of F for the unperturbed motion. We want to expand F around $\delta U = 0$. The expansion will look like:

$$F = F_0 + F_1\delta U + F_2(\delta U)^2 + \dots$$

Here, the Taylor coefficients are $F_k = \frac{1}{k!}\frac{\partial^k F}{\partial(\delta U)^k}\big|_{\delta U=0}$: F_0 is the value of F at $\delta U = 0$, F_1 is the first derivative with respect to δU at $\delta U = 0$, F_2 is half the second derivative, and so on. Let's find the derivatives:

$$\begin{aligned}\frac{\partial F}{\partial(\delta U)} &= -\frac{m}{\sqrt{F_0^2 - 2m\,\delta U}}, & \frac{\partial F}{\partial(\delta U)}\Big|_{\delta U=0} &= -\frac{m}{F_0}, \\ \frac{\partial^2 F}{\partial(\delta U)^2} &= -\frac{m^2}{(F_0^2 - 2m\,\delta U)^{3/2}}, & \frac{\partial^2 F}{\partial(\delta U)^2}\Big|_{\delta U=0} &= -\frac{m^2}{F_0^3},\end{aligned}$$

so that $F_1 = -m/F_0$ and $F_2 = -m^2/(2F_0^3)$. Therefore, the expanded expression in powers of δU is:

$$F = F_0 - \frac{m\,\delta U}{F_0} - \frac{m^2(\delta U)^2}{2F_0^3} + \dots$$

After plugging the expended expression of F in the integral, we see that the zero-order term gives 2π. The first-order term gives the required change $\delta\phi$:

$$\delta\phi = \frac{\partial}{\partial M}\int_{r_{\min}}^{r_{\max}} \frac{2m\,\delta U}{F_0}\, dr. \tag{3.12}$$

We can change the integration over r to one over ϕ, along the path of the unperturbed motion. On its outward branch, from the perihelion ($r = r_{\min}$, $\phi = 0$) to the aphelion ($r = r_{\max}$, $\phi = \pi$), formula (14.7)

of the book gives

$$d\phi = \frac{M\,dr}{r^2 F_0}, \qquad i.e. \qquad \frac{dr}{F_0} = \frac{r^2}{M}\,d\phi.$$

Hence, the integral Eq. (3.12) become

$$\begin{aligned}\delta\phi &= \frac{\partial}{\partial M}\left(\frac{2m}{M}\int_0^\pi r^2 \delta U d\phi\right) \\ &= \frac{\partial}{\partial M}\left(\frac{2m}{M}\int_0^\pi r^2 \frac{\beta}{r^2} d\phi\right) \\ &= \frac{\partial}{\partial M}\left(\frac{2\pi m\beta}{M}\right) \\ &= -\frac{2\pi m\beta}{M^2}.\end{aligned} \tag{3.13}$$

Using formula (15.4) of the book we can finally write

Answer to Problem 3 (a).

$$\delta\phi = -\frac{2\pi\beta}{\alpha p}$$

(b)

$$\delta U = \frac{\gamma}{r^3}$$

Solution: From Eq. (3.13), we have

$$\delta\phi = \frac{\partial}{\partial M}\left(\frac{2m}{M}\int_0^\pi \frac{\gamma}{r} d\phi\right).$$

Using formula (15.5) of the book, the integral become

$$\begin{aligned}\delta\phi &= \frac{\partial}{\partial M}\left(\frac{2m}{M}\int_0^\pi \frac{\gamma}{p}\left(1 + e\cos\phi\right) d\phi\right) \\ &= \frac{\partial}{\partial M}\left(\frac{2\pi\gamma m}{Mp}\right) \\ &= -\frac{6\pi\alpha\gamma m^2}{M^4}.\end{aligned}$$

The last expression can also be written as

Answer to Problem 3 (b).

$$\delta\phi = -\frac{6\pi\gamma}{\alpha p^2}$$

Chapter 4.

Collisions Between Particles

§16 Disintegration of particles

Problem 1.

Find the relation between the angles θ_1, θ_2 (in the L system) after a disintegration into two particles.

Solution: The angles of the particles, in the C system, are related by

$$\theta_0 = \theta_{10} = \pi - \theta_{20},$$

where θ_0 as been defined to simplify the notation. From formula (16.5) of the book, we have

$$\begin{aligned} V + v_{10}\cos\theta_0 &= v_{10}\sin\theta_0\cot\theta_1 \\ V - v_{20}\cos\theta_0 &= v_{20}\sin\theta_0\cot\theta_2. \end{aligned}$$

We must eliminate θ_0 from the two equations above. To do so, we can first solve for $\cos\theta_0$ and $\sin\theta_0$ which give us

$$\sin\theta_0 = \frac{V + v_{10}\cos\theta_0}{v_{10}\cot\theta_1} = \frac{V - v_{20}\cos\theta_0}{v_{20}\cot\theta_2} \tag{4.1}$$

$$\Rightarrow V v_{20}\cot\theta_2 + v_{10}v_{20}\cos\theta_0\cot\theta_2 = V v_{10}\cot\theta_1 - v_{10}v_{20}\cos\theta_0\cot\theta_1$$

$$\Rightarrow \cos\theta_0 = \frac{V\left(v_{10}\cot\theta_1 - v_{20}\cot\theta_2\right)}{v_{10}v_{20}\left(\cot\theta_1 + \cot\theta_2\right)}. \tag{4.2}$$

Using the sum of square of Eq. (4.1) and Eq. (4.2), we get

$$\left(\frac{V + v_{10}\cos\theta_0}{v_{10}\cot\theta_1}\right)^2 + \left(\frac{V\left(v_{10}\cot\theta_1 - v_{20}\cot\theta_2\right)}{v_{10}v_{20}\left(\cot\theta_1 + \cot\theta_2\right)}\right)^2 = 1$$

$$\Rightarrow \left(v_{10} + v_{20}\right)^2 + \left(v_{10}\cot\theta_1 - v_{20}\cot\theta_2\right)^2 = \frac{v_{10}^2 v_{20}^2}{V^2}\left(\cot\theta_1 + \cot\theta_2\right)^2$$

$$\Rightarrow v_{10}^2\csc\theta_1{}^2 + v_{20}^2\csc\theta_2{}^2 + 2v_{10}v_{20} - 2v_{10}v_{20}\cot\theta_1\cot\theta_2 = \frac{v_{10}^2 v_{20}^2\sin^2\left(\theta_1+\theta_2\right)}{V^2\sin^2\theta_1\sin^2\theta_2}$$

$$\Rightarrow v_{20}^2\sin^2\theta_1 + v_{10}^2\sin^2\theta_2 + 2v_{10}v_{20}\sin\theta_1\sin\theta_2\left(\sin\theta_1\sin\theta_2 - \cos\theta_1\cos\theta_2\right) = \frac{v_{10}^2 v_{20}^2}{V^2}\sin^2\left(\theta_1+\theta_2\right)$$

$$\Rightarrow v_{20}^2\sin^2\theta_1 + v_{10}^2\sin^2\theta_2 - 2v_{10}v_{20}\sin\theta_1\sin\theta_2\cos\left(\theta_1+\theta_2\right) = \frac{v_{10}^2 v_{20}^2}{V^2}\sin^2\left(\theta_1+\theta_2\right),$$

where we used $(v_{10}+v_{20})^2 = v_{10}^2 + 2v_{10}v_{20} + v_{20}^2$ and $1 + \cot^2\theta = \csc^2\theta$. Dividing by $v_{10}v_{20}$, and using formula (16.2) of the book, which gives $v_{10}v_{20} = p_0^2/(m_1m_2) = 2\epsilon/(m_1+m_2)$, and the relation $v_{10}/v_{20} = m_2/m_1$, we obtain

Answer to Problem 1.

$$\frac{m_2}{m_1}\sin^2\theta_2 + \frac{m_1}{m_2}\sin^2\theta_1 - 2\sin\theta_1\sin\theta_2\cos\left(\theta_1+\theta_2\right) = \frac{2\epsilon}{(m_1+m_2)V^2}\sin^2\left(\theta_1+\theta_2\right)$$

Problem 2.

Find the angular distribution of the resulting particles in the L system.

Solution: The distribution of the resulting particles with respect to the angle θ_0 is given by formula (16.7) of the book, *i.e.* , for increasing θ_0,

$$\mathrm{d}P = \frac{1}{2}\sin\theta_0 \mathrm{d}\theta_0 = -\frac{1}{2}\mathrm{d}(\cos\theta_0).$$

To transform it into a probability density in θ, we must use the absolute value of the Jacobian, $\mathrm{d}P/\mathrm{d}\theta = \frac{1}{2}\left|\mathrm{d}(\cos\theta_0)/\mathrm{d}\theta\right|$, summed over all the values of θ_0 that give the same θ. Let

$$q = \frac{V}{v_0}, \qquad A_q(\theta) = \frac{1+q^2\cos 2\theta}{\sqrt{1-q^2\sin^2\theta}}.$$

Formula (16.6) of the book reads $\cos\theta_0 = -q\sin^2\theta \pm \cos\theta\sqrt{1-q^2\sin^2\theta}$, so that

$$\begin{aligned}\frac{\mathrm{d}(\cos\theta_0)}{\mathrm{d}\theta} &= -2q\sin\theta\cos\theta \mp \left(\sin\theta\sqrt{1-q^2\sin^2\theta} + \frac{q^2\sin\theta\cos^2\theta}{\sqrt{1-q^2\sin^2\theta}}\right)\\ &= -\sin\theta\left[2q\cos\theta \pm A_q(\theta)\right],\end{aligned}$$

since $1-q^2\sin^2\theta+q^2\cos^2\theta = 1+q^2\cos 2\theta$. Where $1-q^2\sin^2\theta > 0$, the numerator of A_q is positive (for $q>1$, $\sin^2\theta < 1/q^2$ gives $1+q^2\cos 2\theta > q^2-1$), and one verifies that

$$A_q^2 - 4q^2\cos^2\theta = \frac{(1-q^2)^2}{1-q^2\sin^2\theta} \geq 0,$$

so that $A_q \geq 2q|\cos\theta|$ and both $A_q \pm 2q\cos\theta$ are nonnegative. Three cases must be distinguished.

- $0 \leq q < 1$: only the upper sign of (16.6) is valid, and it maps $0<\theta_0<\pi$ one-to-one onto $0<\theta<\pi$. The density is $\frac{1}{2}\sin\theta\left[2q\cos\theta + A_q(\theta)\right]$.
- $q=1$: $\sqrt{1-\sin^2\theta} = |\cos\theta|$, and the upper sign gives $\theta_0 = 2\theta$, $0<\theta<\pi/2$, while the lower sign gives $\theta_0=\pi$, for which the particle is at rest in the L system and has no direction. The density is $\frac{1}{2}\left|\mathrm{d}(\cos 2\theta)/\mathrm{d}\theta\right| = 2\sin\theta\cos\theta$, which is also the formula of the first case with $A_1 = 2\cos\theta$.
- $q>1$: the angle is limited by $\sin\theta < 1/q$, and both signs are valid: each θ is reached from two values of θ_0. The two nonnegative branch densities are $\frac{1}{2}\sin\theta\left[A_q(\theta)\pm 2q\cos\theta\right]$, and they must be added, giving $\sin\theta\, A_q(\theta)$.

The density is zero outside these supports, and its values at the endpoints are defined by limits (for $q>1$ it diverges, integrably, at $\sin\theta = 1/q$). In each case, it is normalized to 1. The angular distribution is thus

Answer to Problem 2.

$$\frac{\mathrm{d}P}{\mathrm{d}\theta} = \begin{cases} \frac{1}{2}\sin\theta\left[2q\cos\theta + A_q(\theta)\right], & 0\leq q<1, \quad 0<\theta<\pi,\\ 2\sin\theta\cos\theta, & q=1, \quad 0<\theta<\pi/2,\\ \sin\theta\, A_q(\theta), & q>1, \quad 0<\theta<\sin^{-1}(1/q),\end{cases}$$

and zero outside these intervals, with

$$q = \frac{V}{v_0}, \qquad A_q(\theta) = \frac{1+q^2\cos 2\theta}{\sqrt{1-q^2\sin^2\theta}}.$$

Problem 3.

Determine the range of possible values of the angle θ between the directions of motion of the two resulting particles in the L system.

Solution: The tangent of the angle $\theta = \theta_1 + \theta_2$ is

$$\tan\theta = \frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1 \tan\theta_2}.$$

Using formula (16.5) of the book to get $\tan\theta_1$ and $\tan\theta_2$, and using $\theta_0 = \theta_{10} = \pi - \theta_{20}$, the last equation become

$$\tan\theta = \frac{(v_{10} + v_{20})V \sin\theta_0}{V^2 + (v_{10} - v_{20})V\cos\theta_0 - v_{10}v_{20}}$$

or equivalently

$$\cot\theta = \frac{V^2 + (v_{10} - v_{20})V\cos\theta_0 - v_{10}v_{20}}{(v_{10} + v_{20})V\sin\theta_0}.$$

Exchanging the two particles ($v_{10} \leftrightarrow v_{20}$, $\theta_0 \to \pi - \theta_0$) leaves this expression unchanged, so only the smaller and the larger of the two (positive) speeds matter. Let

$$a = \min(v_{10}, v_{20}), \qquad b = \max(v_{10}, v_{20});$$

we may label the particles so that $v_{10} = a$ and $v_{20} = b$. The expression of $\cot\theta$ holds for $0 < \theta_0 < \pi$. At $\theta_0 = 0$ or π, the emission is collinear: both particles move along the line of $\mathbf{V}$, with the velocities $V \pm v_{10}$ and $V \mp v_{20}$, and θ is 0 if these have the same sign, or π if they have opposite signs. When one of them vanishes, the corresponding particle is at rest in the L system and its direction is undefined; such configurations are excluded. In order to find the extrema, we take the derivative with respect to θ_0,

$$\frac{\partial\cot\theta}{\partial\theta_0} = 0 = \frac{(v_{20} - v_{10})V^2 - V\cos\theta_0(V^2 - v_{10}v_{20})}{(v_{10} + v_{20})V^2\sin^2\theta_0}$$
$$\Rightarrow \cos\theta_0 = \frac{(v_{20} - v_{10})V}{V^2 - v_{10}v_{20}}.$$

We know that $-1 \le \cos\theta_0 \le 1$, thus the stationary point exists only if

$$(V^2 - v_{10}v_{20})^2 - (v_{20} - v_{10})^2V^2 = (V^2 - a^2)(V^2 - b^2) \ge 0,$$

i.e. if $V \le a$ or $V \ge b$: there is no stationary point in the middle regime $a < V < b$. It is an interior point, $0 < \theta_0 < \pi$, when $V < a$ or $V > b$. For $V = a$ or $V = b$, it reaches an end, $\cos\theta_0 = \pm 1$, where one particle is at rest in the L system: these are limiting cases, which are not attained. Moreover, we could calculate the derivative of the cotangent with respect to V and find the expression:

$$\frac{\partial\cot\theta}{\partial V} = \frac{V^2 + v_{10}v_{20}}{(v_{10} + v_{20})V^2\sin\theta_0}$$

which is positive for all values of V: at fixed θ_0, the angle θ decreases when V increases.

The range of θ follows from the behaviour of $\cot\theta$ at the ends. When $\theta_0 \to 0^+$ and $\theta_0 \to \pi^-$, its denominator tends to 0^+ and its numerator tends to

$$(V - b)(V + a) \qquad \text{and} \qquad (V + b)(V - a),$$

respectively. We have the following regimes.

Middle regime, $a < V < b$. We have $\cot\theta \to -\infty$ when $\theta_0 \to 0^+$ and $\cot\theta \to +\infty$ when $\theta_0 \to \pi^-$, with no stationary point in between. The cotangent therefore takes every real value, *i.e.* the angle θ takes every value between 0 and π. The two collinear emissions are attained at finite V: for $\theta_0 = 0$, the particles move in opposite directions ($V + a > 0$, $V - b < 0$), so $\theta = \pi$; for $\theta_0 = \pi$, they move in the same direction ($V - a > 0$, $V + b > 0$), so $\theta = 0$. Hence $0 \le \theta \le \pi$.

Slow regime, $0 < V < a$. We have $\cot\theta \to -\infty$ at both ends (both collinear emissions give opposite directions, $\theta = \pi$), so $\cot\theta$ has a maximum at the interior stationary point. The calculation of the

maximum value leads to:

$$\cot\theta = \frac{V^2 + (v_{10} - v_{20})V\cos\theta_0 - v_{10}v_{20}}{(v_{10} + v_{20})V\sin\theta_0} = -\frac{\sqrt{(V^2 - v_{10}v_{20})^2 - (v_{20} - v_{10})^2V^2}}{(v_{10} + v_{20})V}$$

Then, after transforming the cotangent into a sine, using $(a+b)^2V^2 + (V^2 - a^2)(V^2 - b^2) = (V^2 + ab)^2$, one gets:

$$\sin\theta = \frac{(a+b)V}{V^2 + ab}$$

If we put:

$$\alpha = \sin^{-1}\left(\frac{(a+b)V}{V^2 + ab}\right),$$

which is defined because $V^2 + ab - (a+b)V = (V-a)(V-b) \geq 0$ outside the middle regime, then, since $\cot\theta < 0$ at the maximum, the smallest angle is $\pi - \alpha$, and we have the range $\pi - \alpha \leq \theta \leq \pi$. For $V = 0$, the particles fly apart in opposite directions and $\theta = \pi$.

Fast regime, $V > b$. Similarly, $\cot\theta \to +\infty$ at both ends: both collinear emissions, $\theta_0 = 0$ and $\theta_0 = \pi$, give particles moving in the same direction, so $\theta = 0$ is attained at finite V. The cotangent has a minimum at the interior stationary point, where its value is the same as above with the opposite sign, and the largest angle is α. In this case, the range for θ is $0 \leq \theta \leq \alpha$.

Boundary cases. For $V = a < b$, the stationary point is at the excluded end $\theta_0 = \pi$, where the particle of speed a is at rest. The angle decreases monotonically from $\theta = \pi$ ($\theta_0 = 0$) and tends to $\pi/2$ (here $\alpha = \pi/2$) when $\theta_0 \to \pi^-$, without reaching it, so $\pi/2 < \theta \leq \pi$. For $V = b > a$, the end $\theta_0 = 0$ is excluded instead (the particle of speed b is at rest), $\theta \to \pi/2$ when $\theta_0 \to 0^+$, and $\theta = 0$ is attained at $\theta_0 = \pi$, so $0 \leq \theta < \pi/2$. For $V = a = b$, the numerator of $\cot\theta$ vanishes identically, so $\theta = \pi/2$ for every $0 < \theta_0 < \pi$, both ends being excluded. The final solution is thus

Answer to Problem 3.

With $a = \min(v_{10}, v_{20})$, $b = \max(v_{10}, v_{20})$ and

$$\alpha = \sin^{-1}\left(\frac{(a+b)V}{V^2 + ab}\right),$$

the attainable values of θ, excluding the configurations where a particle is at rest in the L system, are

$$\begin{array}{ll} V = 0: & \theta = \pi, \\ 0 < V < a: & \pi - \alpha \leq \theta \leq \pi, \\ a < V < b: & 0 \leq \theta \leq \pi, \\ V > b: & 0 \leq \theta \leq \alpha, \end{array} \qquad \begin{array}{ll} V = a < b: & \pi/2 < \theta \leq \pi, \\ V = b > a: & 0 \leq \theta < \pi/2, \\ V = a = b: & \theta = \pi/2. \end{array}$$

§17 Elastic collisions

Problem 1.

Express the velocity of each particle after a collision between a moving particle (m_1) and another at rest (m_2) in terms of their directions of motion in the L system.

Solution: To solve this problem, we need to use Fig. 16 of the book, where v is the initial speed of the particle m_1 and $m = m_1m_2/(m_1 + m_2)$ is the reduced mass, so that $\mathrm{AO} = m_1^2v/(m_1 + m_2)$ and $\mathrm{OB} = \mathrm{OC} = mv$. The triangle OBC is isosceles, so the direction of the second particle makes the angle $\theta_2 = \frac{1}{2}(\pi - \chi)$ with the incident direction. First, using the law of sines, the magnitude of the momentum of the second particle after the collision is

$$p_2' = \mathrm{OB}\frac{\sin\chi}{\sin\theta_2} = mv\frac{\sin\chi}{\cos(\chi/2)} = 2mv\sin\frac{\chi}{2} = 2mv\cos\theta_2.$$

We get formula (17.5) of the book if we isolate the velocity, and then express it in terms of the direction θ_2 of the second particle,

$$v_2' = \frac{2m_1v}{m_1 + m_2}\sin\frac{\chi}{2} = \frac{2m_1v}{m_1 + m_2}\cos\theta_2.$$

Second, using the law of cosines, we get

$$\begin{aligned}\mathrm{OC}^2 &= \mathrm{AO}^2 + p_1'^2 - 2\mathrm{AO}p_1'\cos\theta_1\\ \Rightarrow p_1' &= \mathrm{AO}\cos\theta_1 \pm \sqrt{\mathrm{AO}^2\cos^2\theta_1 + \mathrm{OC}^2 - \mathrm{AO}^2}\\ \Rightarrow v_1' &= \frac{m_1\cos\theta_1 \pm \sqrt{m_2^2 - m_1^2\sin^2\theta_1}}{m_1 + m_2}v.\end{aligned}$$

Only the roots giving a nonnegative speed are physical. For the case where $m_2 > m_1$, the radical is larger than $m_1|\cos\theta_1|$, so only the upper sign is physical and we also get formula (17.5) of the book, but when $m_1 > m_2$, the angle is limited by $\sin\theta_1 \le m_2/m_1$ and both roots are nonnegative: the radical may have either sign. To summarize, the answer is

Answer to Problem 1.

$$\begin{aligned}v_2' &= \frac{2m_1v}{m_1 + m_2}\cos\theta_2\\ v_1' &= \frac{m_1\cos\theta_1 \pm \sqrt{m_2^2 - m_1^2\sin^2\theta_1}}{m_1 + m_2}v,\end{aligned}$$

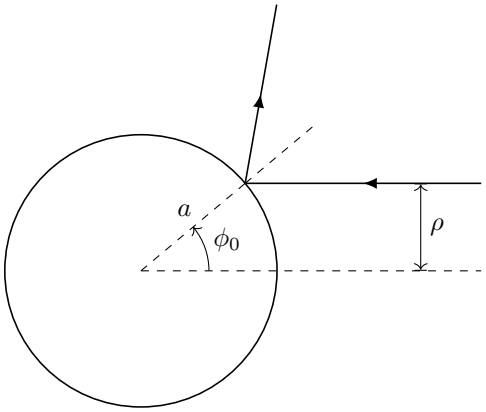
where v is the initial speed of m_1 and only the nonnegative roots for v_1' are physical (both signs for $m_1 > m_2$, the upper sign only for $m_1 < m_2$).

§18 Scattering

Problem 1.

Determine the effective cross-section for scattering of particles from a perfectly rigid sphere of radius a (*i.e.* when the interaction is such that $U = \infty$ for $r < a$ and $U = 0$ for $r > a$).

Solution: The particle is reflected from the sphere at the same angle ϕ_0 to the normal (*i.e.* the radius) as it is incident, as shown in the figure.



From the figure, we can see

$$\rho = a\sin\phi_0 = a\sin\frac{1}{2}(\pi-\chi) = a\cos\frac{1}{2}\chi. \tag{4.3}$$

Subsituing Eq. (4.3) in formula (18.7) or (18.8) of the book, we have

$$\mathrm{d}\sigma = \frac{1}{2}\pi a^2\sin\chi\mathrm{d}\chi = \frac{1}{4}a^2\mathrm{d}o. \tag{4.4}$$

If we integrate $\mathrm{d}\sigma$ over all angles, we find

$$\sigma = \frac{1}{2}\pi a^2\int_0^\pi \sin\chi\mathrm{d}\chi = \pi a^2,$$

which is the projection of the sphere perpendicular to the beam (*i.e.* the impact area). We need to express Eq. (4.4) in the L system using formula (17.4) of the book. If this equation is solve for $\cos\chi$, we get, as in §16 Prob. 2 with V/v_0 replaced by

$$q = \frac{m_1}{m_2}, \qquad A_q(\theta_1) = \frac{1+q^2\cos 2\theta_1}{\sqrt{1-q^2\sin^2\theta_1}},$$

and θ_0 by χ,

$$\begin{aligned}
&\tan\theta_1 = \frac{m_2\sin\chi}{m_1+m_2\cos\chi}\\
&\Rightarrow \tan^2\theta_1\,(q+\cos\chi)^2 = 1-\cos^2\chi\\
&\Rightarrow \cos\chi = -q\sin^2\theta_1 \pm \cos\theta_1\sqrt{1-q^2\sin^2\theta_1}\\
&\Rightarrow \frac{\mathrm{d}(\cos\chi)}{\mathrm{d}\theta_1} = -\sin\theta_1\left[2q\cos\theta_1 \pm A_q(\theta_1)\right].
\end{aligned}$$

Since $\mathrm{d}\sigma = \frac{1}{2}\pi a^2|\mathrm{d}(\cos\chi)|$ and $\mathrm{d}o_1 = 2\pi\sin\theta_1\mathrm{d}\theta_1$, each branch contributes $\frac{a^2}{4}\left|2q\cos\theta_1 \pm A_q(\theta_1)\right|$ to $\mathrm{d}\sigma_1/\mathrm{d}o_1$, the absolute value of the Jacobian being needed because $\cos\chi$ decreases when χ increases. As shown in §16 Prob. 2, $A_q \geq 2q|\cos\theta_1|$, so these contributions are $\frac{a^2}{4}\left[A_q(\theta_1) \pm 2q\cos\theta_1\right]$. For $q < 1$ ($m_1 < m_2$), only the upper sign is valid, and it maps $0 < \chi < \pi$ onto $0 < \theta_1 < \pi$. For $q = 1$, the upper

sign gives $\chi = 2\theta_1$, $0 < \theta_1 < \pi/2$, while the lower sign gives the head-on collision $\chi = \pi$, after which the particle is at rest and has no direction. For $q > 1$ ($m_1 > m_2$), the angle is limited by $\sin\theta_1 < 1/q$, and each θ_1 is reached from two values of χ: the two branch contributions must be added, giving $\frac{a^2}{2}A_q(\theta_1)$. In every case, the integral of $\mathrm{d}\sigma_1$ over the physical support is πa^2, and the cross-section is zero outside it.

For the recoiling sphere, $\theta_2 = \frac{1}{2}(\pi - \chi)$, so $\cos\chi = -\cos 2\theta_2$, and χ from 0 to π corresponds to θ_2 from $\pi/2$ to 0. Hence

$$\mathrm{d}\sigma = \frac{1}{2}\pi a^2 \left|\mathrm{d}(\cos 2\theta_2)\right| = 2\pi a^2 \sin\theta_2 \cos\theta_2 \mathrm{d}\theta_2 = a^2 \cos\theta_2 \, \mathrm{d}o_2,$$

whose integral over $0 < \theta_2 < \pi/2$ is again πa^2. We thus get

Answer to Problem 1.

$$\frac{\mathrm{d}\sigma_1}{\mathrm{d}o_1} = \begin{cases} \frac{a^2}{4}\left[2q\cos\theta_1 + A_q(\theta_1)\right], & q < 1, \quad 0 < \theta_1 < \pi, \\ a^2\cos\theta_1, & q = 1, \quad 0 < \theta_1 < \pi/2, \\ \frac{a^2}{2}A_q(\theta_1), & q > 1, \quad 0 < \theta_1 < \sin^{-1}(1/q), \end{cases}$$

and zero outside these intervals, with

$$q = \frac{m_1}{m_2}, \qquad A_q(\theta_1) = \frac{1 + q^2\cos 2\theta_1}{\sqrt{1 - q^2\sin^2\theta_1}}.$$

For the recoiling particle,

$$\frac{\mathrm{d}\sigma_2}{\mathrm{d}o_2} = a^2\cos\theta_2, \qquad 0 < \theta_2 < \pi/2.$$

Problem 2.

Express the effective cross-section (§18 Prob. 1) as a function of the energy ϵ lost by a scattered particle.

Solution: The energy loss by a particle of mass m_1 is equal to the energy gained by the sphere of mass m_2. Knowing that the sphere is initially at rest, and using v_2' from §17 Prob. 1, the energy gained is

$$\epsilon = E_2' = \frac{1}{2}m_2 v_2'^2 = \frac{2m_1^2 m_2}{(m_1+m_2)^2} v_\infty^2 \sin^2\frac{1}{2}\chi = \epsilon_{max}\sin^2\frac{1}{2}\chi.$$

Using the double angle identity, the variation of ϵ is

$$\mathrm{d}\epsilon = \epsilon_{max}\sin\frac{1}{2}\chi\cos\frac{1}{2}\chi\mathrm{d}\chi = \frac{1}{2}\epsilon_{max}\sin\chi\mathrm{d}\chi.$$

Using Eq. (4.4) from §18 Prob. 1, the effective cross section can thus be express as

Answer to Problem 2.

$$\mathrm{d}\sigma = \frac{\pi a^2 \mathrm{d}\epsilon}{\epsilon_{max}}$$

Problem 3.

Find the effective cross-section as a function of the velocity v_∞ for particles scattered in a field $U \sim r^{-n}$.

Solution: For a central potential field, the scattering angle is determined by the integral

$$\chi = \pi - 2\int_{r_{min}}^{\infty} \frac{\rho \mathrm{d}r}{r^2\sqrt{1-\frac{\rho^2}{r^2}-\frac{2U(r)}{mv_\infty^2}}}.$$

Subsituing $U(r) = \alpha r^{-n}$, with $n > 0$, and define the dimensionless variable $x = \frac{r}{\rho}$, then

$$\chi = \pi - 2\int_{x_{min}}^{\infty} \frac{\mathrm{d}x}{x^2\sqrt{1-\frac{1}{x^2}-\frac{2\alpha}{mv_\infty^2\rho^n x^n}}}.$$

Notice the dimensionless combination $\xi = \frac{2\alpha}{mv_\infty^2\rho^n}$, the integral simplifies to

$$\chi = \pi - 2\int_{x_{min}}^{\infty} \frac{\mathrm{d}x}{x^2\sqrt{1-\frac{1}{x^2}-\frac{\xi}{x^n}}}.$$

Thus, the scattering angle depends only on ξ,

$$\chi = f(\xi),$$

where $f(\xi)$ is a dimensionless function determined by the integral. We take $\alpha > 0$ (repulsive field), so that $\xi > 0$, and we assume that $\chi = f(\xi)$ can be inverted, *i.e.* that each scattering angle comes from a single impact parameter. (For an attractive field, $\alpha < 0$, the same argument holds with α and ξ replaced by $|\alpha|$ and $|\xi|$, in the range of ρ where the particle is scattered rather than captured and f is invertible.) To invert this relationship, solve for ρ in terms of χ and v_∞,

$$\xi = \frac{2\alpha}{mv_\infty^2\rho^n} \Rightarrow \rho = \left(\frac{2\alpha}{mv_\infty^2\xi}\right)^{1/n}.$$

Let $\xi = g(\chi)$, where $g(\chi)$ is a function encoding the angular dependence. Then,

$$\begin{aligned}\rho &= \left(\frac{2\alpha}{mv_\infty^2}\right)^{1/n}(g(\chi))^{-1/n}\\ \Rightarrow \rho &\sim v_\infty^{-2/n}(g(\chi))^{-1/n}.\end{aligned}$$

The differential cross-section is thus

$$\begin{aligned}\mathrm{d}\sigma &= \frac{\rho}{\sin\chi}\left|\frac{\mathrm{d}\rho}{\mathrm{d}\chi}\right|\mathrm{d}o\\ &= \left(\frac{2\alpha}{mv_\infty^2}\right)^{2/n}\frac{F(\chi)}{\sin\chi}\mathrm{d}o\\ \Rightarrow \mathrm{d}\sigma &\sim v_\infty^{-4/n}\frac{F(\chi)}{\sin\chi}\mathrm{d}o,\end{aligned}$$

where

$$F(\chi) = (g(\chi))^{-1/n}\left|\frac{\mathrm{d}}{\mathrm{d}\chi}(g(\chi))^{-1/n}\right|$$

depends on n and χ only. At a fixed scattering angle, the differential cross-section thus varies as $\mathrm{d}\sigma/\mathrm{d}o \propto v_\infty^{-4/n}$.

The total cross-section, however, is not finite. Because the field extends to infinity and does not vanish, a particle with any impact parameter is deflected, by an angle which tends to zero only as $\rho \to \infty$ ($\xi \to 0$). Integrating over all angles therefore amounts to integrating $2\pi\rho\,\mathrm{d}\rho$ over all $0 < \rho < \infty$, which

diverges at small angles. Only the cross-section for scattering into a fixed angular interval $\chi_1 < \chi < \chi_2$ that excludes $\chi = 0$ is finite:

$$\sigma(\chi_1, \chi_2) = 2\pi \left(\frac{2\alpha}{m v_\infty^2} \right)^{2/n} \int_{\chi_1}^{\chi_2} F(\chi) \mathrm{d}\chi.$$

Knowing that the integral depends only on n and the interval, not v_∞, we have

Answer to Problem 3.

$$\frac{\mathrm{d}\sigma}{\mathrm{d}o} \propto v_\infty^{-4/n} \quad \text{at fixed } \chi, \qquad \sigma(\chi_1 < \chi < \chi_2) \propto v_\infty^{-4/n} \quad (\chi_1 > 0).$$

The unrestricted total cross-section diverges because of the small deflections at large impact parameters.

Problem 4.

Determine the effective cross-section for a particle to "fall" to the centre of a field $U = -\alpha/r^2$.

Solution: Let the particle come from infinity with velocity v_∞ and impact parameter ρ, and take $\alpha > 0$. Its energy and angular momentum are

$$E = \frac{1}{2}mv_\infty^2, \qquad M = mv_\infty\rho.$$

The radial motion is governed by the energy equation

$$\frac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r) = E,$$

where the effective potential energy is

$$U_{\text{eff}}(r) = \frac{M^2}{2mr^2} - \frac{\alpha}{r^2} = \frac{1}{r^2}\left(\frac{M^2}{2m} - \alpha\right).$$

This is the field of §15 Prob. 2, where the same condition on M^2 separates the possible paths. Since both terms vary as r^{-2}, the sign of the bracket determines the behaviour of the particle near the centre. If $M^2 > 2m\alpha$, then U_{eff} is a repulsive centrifugal barrier which tends to $+\infty$ as $r \to 0$, so the particle reaches a turning point where $U_{\text{eff}}(r_{min}) = E$ and goes back to infinity. If $M^2 < 2m\alpha$, then $U_{\text{eff}} < 0$ for all r, so $E - U_{\text{eff}} > 0$ everywhere. The radial velocity never vanishes and an incoming particle has no radial turning point: it falls to the centre. In the limiting case $M^2 = 2m\alpha$, we have $U_{\text{eff}} = 0$, thus $\dot{r} = -v_\infty$ and the particle also reaches the centre. The critical trajectory falls to the centre because the centrifugal and attractive terms have the same dependence r^{-2}. At the critical value of M they cancel identically, so there is no barrier at all and nothing can stop the particle at a finite distance. This differs from §18 Prob. 5, where the two terms have different powers of r and U_{eff} has a maximum at a finite radius.

The particle therefore falls to the centre if $M^2 \leq 2m\alpha$, *i.e.* if

$$m^2v_\infty^2\rho^2 \leq 2m\alpha \Rightarrow \rho \leq \rho_{max}, \qquad \rho_{max}^2 = \frac{2\alpha}{mv_\infty^2}.$$

The effective cross-section is obtained by summing the areas of all the annuli of the incident beam that lead to a fall,

$$\sigma = \int_0^{\rho_{max}} 2\pi\rho\,\mathrm{d}\rho = \pi\rho_{max}^2.$$

We thus get

Answer to Problem 4.

$$\sigma = \frac{2\pi\alpha}{mv_\infty^2} = \frac{\pi\alpha}{E}$$

Problem 5.

The same as §18 Prob. 4, but for a field $U = -\alpha/r^n$ $(n > 2, \alpha > 0)$.

Solution: As in §18 Prob. 4, the particle comes from infinity with velocity v_∞ and impact parameter ρ, so that

$$E = \frac{1}{2}mv_\infty^2, \qquad M = mv_\infty\rho.$$

The effective potential energy is now

$$U_{\text{eff}}(r) = \frac{M^2}{2mr^2} - \frac{\alpha}{r^n}.$$

Since $n > 2$, the centrifugal term dominates at large r and the attractive term dominates at small r. Thus $U_{\text{eff}} \to 0^+$ as $r \to \infty$, $U_{\text{eff}} \to -\infty$ as $r \to 0$, and U_{eff} has a maximum at some r_b, given by

$$U'_{\text{eff}}(r_b) = -\frac{M^2}{mr_b^3} + \frac{n\alpha}{r_b^{n+1}} = 0 \Rightarrow \frac{M^2}{m} = \frac{n\alpha}{r_b^{n-2}}.$$

This barrier analysis assumes $M \neq 0$. A head-on particle ($\rho = 0$) has no centrifugal barrier: $U_{\text{eff}} = -\alpha/r^n$ is monotonic, and the particle falls to the centre. The height of this barrier is

$$U_{\text{eff}}(r_b) = \frac{n\alpha}{2r_b^n} - \frac{\alpha}{r_b^n} = \frac{(n-2)\alpha}{2r_b^n}.$$

If E is smaller than the barrier height, the particle reaches a turning point outside the barrier and goes back to infinity. If E exceeds the barrier height, then $E - U_{\text{eff}} > 0$ for all r, so the particle has no radial turning point and falls to the centre. Since the barrier height increases with M, the capture occurs for $\rho < \rho_{max}$, where ρ_{max} is the impact parameter for which E is equal to the barrier height. In this limiting case, the incoming particle approaches asymptotically the unstable circular orbit $r = r_b$. Indeed, at $r = r_b$ both $E - U_{\text{eff}}$ and U'_{eff} vanish, so r_b is a point of (unstable) equilibrium of the radial motion. Near it, $\frac{1}{2}m\dot{r}^2 = E - U_{\text{eff}} \propto (r - r_b)^2$, so the time $\int \mathrm{d}r/|\dot{r}| \propto \int \mathrm{d}r/|r - r_b|$ needed to reach r_b diverges logarithmically. In contrast with §18 Prob. 4, the critical particle therefore never reaches the centre. This single value of ρ does not affect the cross-section.

Denote by r_c the position of the maximum at the critical impact parameter. The condition $E = U_{\text{eff}}(r_c)$ gives

$$\frac{1}{2}mv_\infty^2 = \frac{(n-2)\alpha}{2r_c^n} \Rightarrow r_c^n = \frac{(n-2)\alpha}{mv_\infty^2}.$$

Using $M = mv_\infty\rho_{max}$ in the condition for the maximum, we find

$$\begin{aligned}\rho_{max}^2 &= \frac{M^2}{m^2v_\infty^2} = \frac{n\alpha}{mv_\infty^2 r_c^{n-2}} \\ &= \frac{n}{n-2}\frac{r_c^n}{r_c^{n-2}} = \frac{n}{n-2}r_c^2,\end{aligned}$$

where we used $\alpha/(mv_\infty^2) = r_c^n/(n-2)$ in the second line. The effective cross-section is obtained by summing the areas of all the annuli of the incident beam that lead to a fall,

$$\sigma = \int_0^{\rho_{max}} 2\pi\rho\,\mathrm{d}\rho = \pi\rho_{max}^2 = \frac{\pi n}{n-2}r_c^2.$$

We thus get

Answer to Problem 5.

$$\sigma = \frac{\pi n}{n-2}\left(\frac{(n-2)\alpha}{mv_\infty^2}\right)^{2/n} = \frac{\pi n}{n-2}\left(\frac{(n-2)\alpha}{2E}\right)^{2/n}$$

Problem 6.

Determine the effective cross-section for particles of mass m_1 to strike a sphere of mass m_2 and radius R to which they are attracted in accordance with Newton's law.

Solution: We consider the relative motion of the particle and the sphere, with r the distance between their centres. This is equivalent to the motion of a single particle of reduced mass

$$m = \frac{m_1 m_2}{m_1 + m_2}$$

in the field

$$U(r) = -\frac{\gamma m_1 m_2}{r},$$

where γ is the gravitational constant. Let v_∞ be the relative velocity at infinity and ρ the impact parameter. The energy and angular momentum of the relative motion are

$$E = \frac{1}{2} m v_\infty^2, \qquad M = m v_\infty \rho,$$

and the radial motion is governed by the energy equation

$$\frac{1}{2} m \dot{r}^2 + \frac{M^2}{2mr^2} - \frac{\gamma m_1 m_2}{r} = E.$$

The distance of closest approach decreases as ρ decreases, so the particle strikes the sphere if $\rho \leq \rho_{max}$, where ρ_{max} corresponds to the limiting trajectory which grazes the sphere. For this trajectory, $\dot{r} = 0$ at $r = R$. Substituting $M = m v_\infty \rho_{max}$, we have

$$\frac{m v_\infty^2 \rho_{max}^2}{2R^2} - \frac{\gamma m_1 m_2}{R} = \frac{1}{2} m v_\infty^2.$$

Multiplying by $2R^2/(m v_\infty^2)$ and using $m_1 m_2 / m = m_1 + m_2$,

$$\begin{aligned} \rho_{max}^2 &= R^2 \left(1 + \frac{2\gamma m_1 m_2}{m R v_\infty^2} \right) \\ &= R^2 \left(1 + \frac{2\gamma (m_1 + m_2)}{R v_\infty^2} \right). \end{aligned}$$

The effective cross-section is obtained by summing the areas of all the annuli of the incident beam that lead to a collision with the sphere,

$$\sigma = \int_0^{\rho_{max}} 2\pi \rho \, \mathrm{d}\rho = \pi \rho_{max}^2 = \pi R^2 \left(1 + \frac{2\gamma (m_1 + m_2)}{R v_\infty^2} \right).$$

The second term describes the gravitational focusing of the trajectories towards the sphere. In the high-speed limit $v_\infty^2 \gg \gamma (m_1 + m_2)/R$, it becomes negligible and we recover the geometric cross-section $\sigma = \pi R^2$. As in the book, we assume $m_2 \gg m_1$, so that $m_1 + m_2 \simeq m_2$. We thus get

Answer to Problem 6.

$$\sigma \simeq \pi R^2 \left(1 + \frac{2\gamma m_2}{R v_\infty^2} \right)$$

Problem 7.

Deduce the form of a scattering field $U(r)$, given the effective cross-section as a function of the angle of scattering for a given energy E. It is assumed that $U(r)$ decreases monotonically with r (a repulsive field), with $U(0) > E$ and $U(\infty) = 0$ (O. B. Firsov 1953).

Solution: Let $\mathrm{d}\sigma/\mathrm{d}\theta$ be the effective cross-section per unit scattering angle, *i.e.* already integrated over the azimuth. In terms of the cross-section per unit solid angle, since $\mathrm{d}o = 2\pi \sin\theta \, \mathrm{d}\theta$,

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\theta} = 2\pi \sin\theta \, \frac{\mathrm{d}\sigma}{\mathrm{d}o}.$$

As in the book, we assume that the relation between ρ and χ is one-to-one, with χ decreasing monotonically from π (head-on collision, $\rho = 0$) to 0 (no deflection, $\rho = \infty$). Note that this is an additional assumption: the monotonicity of U alone does not guarantee it. The particles scattered through angles between χ and π are then exactly those with impact parameters smaller than $\rho(\chi)$, so they fill a disc of area $\pi\rho^2$ in the incident beam, and

$$\pi\rho^2(\chi) = \int_\chi^\pi \frac{\mathrm{d}\sigma}{\mathrm{d}\theta} \, \mathrm{d}\theta.$$

Knowing the cross-section, this integral gives $\rho(\chi)$, which we invert to get the scattering angle $\chi(\rho)$ as a function of the impact parameter. The problem is thus reduced to finding $U(r)$ from $\chi(\rho)$.

The scattering angle in a central field is given by

$$\chi = \pi - 2\int_{r_{min}}^\infty \frac{\rho \, \mathrm{d}r}{r^2\sqrt{1 - \frac{\rho^2}{r^2} - \frac{U(r)}{E}}}.$$

We now introduce the variables

$$s = \frac{1}{r}, \qquad x = \frac{1}{\rho^2}, \qquad w(s) = \sqrt{1 - \frac{U(1/s)}{E}},$$

and write $\widehat{\chi}(x) = \chi(x^{-1/2})$ for the same scattering angle expressed in terms of x. Since $\mathrm{d}r = -\mathrm{d}s/s^2$, we have $\rho \, \mathrm{d}r/r^2 = -\rho \, \mathrm{d}s$. When r goes from r_{min} to ∞, s goes from $s_0 = 1/r_{min}$ to 0, and the minus sign is absorbed by reversing the limits. Thus

$$\frac{\pi - \widehat{\chi}(x)}{2} = \int_0^{s_0} \frac{\rho \, \mathrm{d}s}{\sqrt{w^2(s) - \rho^2 s^2}} = \int_0^{s_0(x)} \frac{\mathrm{d}s}{\sqrt{x w^2(s) - s^2}},$$

where we divided the numerator and the denominator by $\rho = x^{-1/2}$. The turning point $s_0(x)$ is the zero of the radicand,

$$x w^2(s_0) - s_0^2 = 0.$$

Since U decreases with r, it increases with s, so $w(s)$ decreases and $s/w(s)$ increases with s. The radicand is therefore positive for $0 \le s < s_0(x)$, and the condition $s < s_0(x)$ is equivalent to $x > s^2/w^2(s)$.

This is an integral equation of Abel's type, which we solve as follows. Let A be an auxiliary parameter (denoted α in the book). We divide both sides by $\sqrt{A - x}$ and integrate over x from 0 to A,

$$\frac{1}{2}\int_0^A \frac{\pi - \widehat{\chi}(x)}{\sqrt{A - x}} \, \mathrm{d}x = \int_0^A \int_0^{s_0(x)} \frac{\mathrm{d}s \, \mathrm{d}x}{\sqrt{(x w^2(s) - s^2)\,(A - x)}}.$$

The domain of integration on the right is $0 < x < A$, $0 < s < s_0(x)$. From the previous remark, it can equally be described by $0 < s < s_0(A)$ and $s^2/w^2(s) < x < A$. Reversing the order of integration and writing $x w^2 - s^2 = w^2(x - s^2/w^2)$, we get

$$\frac{1}{2}\int_0^A \frac{\pi - \widehat{\chi}(x)}{\sqrt{A - x}} \, \mathrm{d}x = \int_0^{s_0(A)} \frac{\mathrm{d}s}{w(s)} \int_{s^2/w^2(s)}^A \frac{\mathrm{d}x}{\sqrt{(x - s^2/w^2(s))\,(A - x)}}.$$

For the inner integral, let $a = s^2/w^2$ and substitute $x = a + (A - a)\sin^2\varphi$, so that $\mathrm{d}x = 2(A -$

a) $\sin\varphi\cos\varphi\,\mathrm{d}\varphi$ and $\sqrt{(x-a)(A-x)} = (A-a)\sin\varphi\cos\varphi$. Then

$$\int_a^A \frac{\mathrm{d}x}{\sqrt{(x-a)(A-x)}} = \int_0^{\pi/2} 2\,\mathrm{d}\varphi = \pi,$$

independently of a and A. Hence

$$\frac{1}{2}\int_0^A \frac{\pi - \widehat{\chi}(x)}{\sqrt{A-x}}\,\mathrm{d}x = \pi\int_0^{s_0(A)} \frac{\mathrm{d}s}{w(s)}.$$

On the left, the first term gives $\frac{1}{2}\pi\int_0^A (A-x)^{-1/2}\,\mathrm{d}x = \pi\sqrt{A}$. For the second term, we integrate by parts with $\mathrm{d}x/\sqrt{A-x} = -\mathrm{d}\left(2\sqrt{A-x}\right)$,

$$\begin{aligned}\int_0^A \frac{\widehat{\chi}(x)\,\mathrm{d}x}{\sqrt{A-x}} &= \left[-2\widehat{\chi}(x)\sqrt{A-x}\right]_0^A + 2\int_0^A \widehat{\chi}'(x)\sqrt{A-x}\,\mathrm{d}x \\ &= 2\int_0^A \widehat{\chi}'(x)\sqrt{A-x}\,\mathrm{d}x,\end{aligned}$$

since the boundary term vanishes at $x = A$ and $\widehat{\chi}(0) = 0$ (a particle with infinite impact parameter is not deflected). Therefore

$$\pi\sqrt{A} - \int_0^A \widehat{\chi}'(x)\sqrt{A-x}\,\mathrm{d}x = \pi\int_0^{s_0(A)} \frac{\mathrm{d}s}{w(s)}.$$

We differentiate this with respect to A. The integrand on the left vanishes at the upper limit $x = A$, so only the derivative of $\sqrt{A-x}$ contributes,

$$\frac{\pi}{2\sqrt{A}} - \frac{1}{2}\int_0^A \frac{\widehat{\chi}'(x)\,\mathrm{d}x}{\sqrt{A-x}} = \frac{\pi}{w(s_0)}\frac{\mathrm{d}s_0}{\mathrm{d}A}.$$

This holds for every A. Since A and s_0 are related by $Aw^2(s_0) = s_0^2$, we may instead take s_0 as the independent variable, rename it s, and put $A = s^2/w^2$ with $w = w(s)$. Multiplying by $\mathrm{d}A = 2(s/w)\,\mathrm{d}(s/w)$ and using $\sqrt{A} = s/w$, we get

$$\pi\,\mathrm{d}\left(\frac{s}{w}\right) - \frac{s}{w}\,\mathrm{d}\left(\frac{s}{w}\right)\int_0^{s^2/w^2} \frac{\widehat{\chi}'(x)\,\mathrm{d}x}{\sqrt{s^2/w^2 - x}} = \frac{\pi\,\mathrm{d}s}{w}.$$

Since

$$\pi\,\mathrm{d}\left(\frac{s}{w}\right) - \frac{\pi\,\mathrm{d}s}{w} = -\frac{\pi s\,\mathrm{d}w}{w^2} = \frac{s}{w}\left(-\pi\,\mathrm{d}\log w\right),$$

dividing by s/w gives

$$-\pi\,\mathrm{d}\log w = \mathrm{d}\left(\frac{s}{w}\right)\int_0^{s^2/w^2} \frac{\widehat{\chi}'(x)\,\mathrm{d}x}{\sqrt{s^2/w^2 - x}}.$$

Let $h = s/w$, which increases from 0 with s. At $s = 0$ (*i.e.* $r = \infty$), $U = 0$ and $w = 1$, so $\log w = 0$. Integrating from $s = 0$, and calling η the variable of integration for h,

$$-\pi\log w = \int_0^h \mathrm{d}\eta \int_0^{\eta^2} \frac{\widehat{\chi}'(x)\,\mathrm{d}x}{\sqrt{\eta^2 - x}}.$$

The domain $0 < \eta < h$, $0 < x < \eta^2$ is also $0 < x < h^2$, $\sqrt{x} < \eta < h$. Interchanging the integrations,

$$-\pi\log w = \int_0^{h^2} \widehat{\chi}'(x)\left[\int_{\sqrt{x}}^h \frac{\mathrm{d}\eta}{\sqrt{\eta^2 - x}}\right]\mathrm{d}x = \int_0^{h^2} \widehat{\chi}'(x)\,\operatorname{arcosh}\left(\frac{h}{\sqrt{x}}\right)\mathrm{d}x,$$

where we used $\int \mathrm{d}\eta/\sqrt{\eta^2 - x} = \operatorname{arcosh}(\eta/\sqrt{x})$ and $\operatorname{arcosh} 1 = 0$.

We now go back to the variables r and ρ. We have $h = 1/(rw)$ and $\sqrt{x} = 1/\rho$, so $h/\sqrt{x} = \rho/(rw)$, and $\widehat{\chi}'(x)\,\mathrm{d}x = \mathrm{d}\chi = (\mathrm{d}\chi/\mathrm{d}\rho)\,\mathrm{d}\rho$. The limit $x = 0$ corresponds to $\rho = \infty$ and $x = h^2$ to $\rho = rw$. Hence

$$-\pi \log w = \int_{\infty}^{rw} \operatorname{arcosh}\left(\frac{\rho}{rw}\right) \frac{\mathrm{d}\chi}{\mathrm{d}\rho}\,\mathrm{d}\rho = -\int_{rw}^{\infty} \operatorname{arcosh}\left(\frac{\rho}{rw}\right) \frac{\mathrm{d}\chi}{\mathrm{d}\rho}\,\mathrm{d}\rho,$$

i.e.

$$w(r) = \exp\left\{\frac{1}{\pi}\int_{rw(r)}^{\infty} \operatorname{arcosh}\left(\frac{\rho}{rw(r)}\right) \frac{\mathrm{d}\chi}{\mathrm{d}\rho}\,\mathrm{d}\rho\right\}.$$

With the assumption adopted at the start, that χ decreases monotonically with ρ, we have $\mathrm{d}\chi/\mathrm{d}\rho < 0$. The exponent is then negative and $w < 1$, *i.e.* $U > 0$, as expected for a repulsive field.

To obtain the second form, we integrate by parts in ρ, keeping r and $w(r)$ fixed. Since

$$\frac{\mathrm{d}}{\mathrm{d}\rho} \operatorname{arcosh}\left(\frac{\rho}{rw}\right) = \frac{1}{\sqrt{\rho^2 - r^2w^2}},$$

we have

$$\int_{rw}^{\infty} \operatorname{arcosh}\left(\frac{\rho}{rw}\right) \frac{\mathrm{d}\chi}{\mathrm{d}\rho}\,\mathrm{d}\rho = \left[\chi(\rho) \operatorname{arcosh}\left(\frac{\rho}{rw}\right)\right]_{rw}^{\infty} - \int_{rw}^{\infty} \frac{\chi(\rho)\,\mathrm{d}\rho}{\sqrt{\rho^2 - r^2w^2}}.$$

Two conditions on the behaviour of $\chi(\rho)$ at large ρ are needed here. **(a)** The boundary term vanishes at $\rho = rw$ because $\operatorname{arcosh} 1 = 0$. At $\rho \to \infty$, $\operatorname{arcosh}(\rho/rw) \simeq \log(2\rho/rw)$ grows logarithmically, so the boundary term vanishes there only if $\chi(\rho)\log\rho \to 0$. **(b)** The remaining integral must converge. Its integrand behaves as $\chi(\rho)/\rho$ at large ρ, so at infinity its convergence is equivalent to that of $\int^{\infty} \chi(\rho)\,\mathrm{d}\rho/\rho$. Near the lower limit, the singularity $1/\sqrt{\rho - rw}$ is integrable. Condition (a) alone does not guarantee (b): for example, $\chi = 1/(\log\rho\,\log\log\rho)$ satisfies $\chi\log\rho \to 0$, but $\int^{\infty} \chi\,\mathrm{d}\rho/\rho$ diverges. A sufficient condition for both is $\chi(\rho) = O(\rho^{-\varepsilon})$ with some $\varepsilon > 0$: then $\chi\log\rho = O(\rho^{-\varepsilon}\log\rho) \to 0$, and $\int^{\infty} \rho^{-1-\varepsilon}\,\mathrm{d}\rho$ converges. For instance, for a field decreasing as r^{-n} at large distances, $\chi \propto \rho^{-n}$ for large ρ (see §20 Prob. 2). Under these conditions,

$$w(r) = \exp\left\{-\frac{1}{\pi}\int_{rw(r)}^{\infty} \frac{\chi(\rho)\,\mathrm{d}\rho}{\sqrt{\rho^2 - r^2w^2(r)}}\right\}.$$

Both formulas give w only implicitly, since w also appears in the lower limit. However, the right-hand sides depend on r and w only through the product rw. They therefore give w as an explicit function of rw, from which $w(r)$ follows, and then

$$U(r) = E\left[1 - w^2(r)\right].$$

The derivation covered all s between 0 and the turning points s_0. The smallest distance reached is that of the head-on collision ($\rho = 0$), r_{min}, for which $U(r_{min}) = E$ and $w = 0$. The potential is thus determined in the accessible region $r > r_{min}$. The scattering data give no information on the interior region $r < r_{min}$, where $U > E$. In summary, we get

Answer to Problem 7.

$$w(r) = \exp\left\{\frac{1}{\pi}\int_{rw(r)}^{\infty} \operatorname{arcosh}\left(\frac{\rho}{rw(r)}\right) \frac{\mathrm{d}\chi}{\mathrm{d}\rho}\,\mathrm{d}\rho\right\} = \exp\left\{-\frac{1}{\pi}\int_{rw(r)}^{\infty} \frac{\chi(\rho)\,\mathrm{d}\rho}{\sqrt{\rho^2 - r^2w^2(r)}}\right\},$$

$$w(r) = \sqrt{1 - \frac{U(r)}{E}}, \qquad U(r) = E\left[1 - w^2(r)\right], \qquad r > r_{min}$$

§19 Rutherford's formula

Problem 1.

Find the effective cross-section for scattering in a field $U = \alpha/r^2$ ($\alpha > 0$).

Solution: Let the particle of mass m come from infinity with velocity v_∞ and impact parameter ρ. It is convenient to introduce the positive length a defined by

$$a^2 = \frac{2\alpha}{mv_\infty^2}.$$

With $U(r) = \alpha/r^2$, we have $2U/(mv_\infty^2) = a^2/r^2$, so the angle between the direction of closest approach and the asymptote of the trajectory is

$$\phi_0 = \int_{r_{min}}^{\infty} \frac{\rho\,\mathrm{d}r}{r^2\sqrt{1-\frac{\rho^2}{r^2}-\frac{a^2}{r^2}}} = \int_{r_{min}}^{\infty} \frac{\rho\,\mathrm{d}r}{r^2\sqrt{1-(\rho^2+a^2)/r^2}}.$$

The distance of closest approach is the zero of the radicand,

$$r_{min} = \sqrt{\rho^2+a^2}.$$

We substitute $u = 1/r$, so that $\mathrm{d}r/r^2 = -\mathrm{d}u$. When r goes from r_{min} to ∞, u goes from $1/\sqrt{\rho^2+a^2}$ to 0, and the minus sign is absorbed by reversing the limits,

$$\begin{aligned}\phi_0 &= \rho\int_0^{1/\sqrt{\rho^2+a^2}} \frac{\mathrm{d}u}{\sqrt{1-(\rho^2+a^2)u^2}} \\ &= \frac{\rho}{\sqrt{\rho^2+a^2}}\left[\arcsin\left(\sqrt{\rho^2+a^2}\,u\right)\right]_0^{1/\sqrt{\rho^2+a^2}} = \frac{\pi\rho}{2\sqrt{\rho^2+a^2}}.\end{aligned}$$

The scattering angle is therefore

$$\chi = \pi - 2\phi_0 = \pi\left(1-\frac{\rho}{\sqrt{\rho^2+a^2}}\right) = \pi\left[1-\left(1+\frac{2\alpha}{mv_\infty^2\rho^2}\right)^{-1/2}\right].$$

Since $\rho/\sqrt{\rho^2+a^2}$ increases from 0 to 1 as ρ goes from 0 to ∞, χ decreases monotonically from π for a head-on collision to 0 as $\rho\to\infty$. Thus each angle $0<\chi<\pi$ corresponds to a single impact parameter, and we can invert this relation. Writing it as

$$\frac{\rho}{\sqrt{\rho^2+a^2}} = \frac{\pi-\chi}{\pi},$$

and squaring, we get $\pi^2\rho^2 = (\pi-\chi)^2(\rho^2+a^2)$, *i.e.*

$$\rho^2\left[\pi^2-(\pi-\chi)^2\right] = a^2(\pi-\chi)^2 \Rightarrow \rho^2 = a^2\frac{(\pi-\chi)^2}{\chi(2\pi-\chi)},$$

where we used $\pi^2-(\pi-\chi)^2 = \chi(2\pi-\chi)$. To differentiate, let $y = \pi-\chi$, so that $\rho^2 = a^2y^2/(\pi^2-y^2)$ and

$$\frac{\mathrm{d}(\rho^2)}{\mathrm{d}y} = a^2\frac{2y(\pi^2-y^2)+2y^3}{(\pi^2-y^2)^2} = \frac{2\pi^2a^2y}{(\pi^2-y^2)^2}.$$

Since $\mathrm{d}y = -\mathrm{d}\chi$,

$$\frac{\mathrm{d}(\rho^2)}{\mathrm{d}\chi} = -\frac{2\pi^2a^2(\pi-\chi)}{\chi^2(2\pi-\chi)^2}.$$

The particles scattered into the range $\mathrm{d}\chi$ are those passing through an annulus of radius ρ and width $|\mathrm{d}\rho|$ in the incident beam, so

$$\mathrm{d}\sigma = 2\pi\rho\left|\frac{\mathrm{d}\rho}{\mathrm{d}\chi}\right|\mathrm{d}\chi = \pi\left|\frac{\mathrm{d}(\rho^2)}{\mathrm{d}\chi}\right|\mathrm{d}\chi.$$

In terms of the element of solid angle $\mathrm{d}o = 2\pi\sin\chi\,\mathrm{d}\chi$, this becomes

$$\mathrm{d}\sigma = \frac{1}{2\sin\chi}\left|\frac{\mathrm{d}(\rho^2)}{\mathrm{d}\chi}\right|\mathrm{d}o = \pi^2a^2\frac{\pi-\chi}{\chi^2(2\pi-\chi)^2}\frac{\mathrm{d}o}{\sin\chi}.$$

This is a differential cross-section. Its integral over all angles diverges at $\chi = 0$, since particles with

arbitrarily large impact parameters are still deflected, by arbitrarily small angles. As a check, for $\rho \gg a$ we have $\rho/\sqrt{\rho^2+a^2} \simeq 1 - a^2/(2\rho^2)$, hence

$$\chi \simeq \frac{\pi a^2}{2\rho^2} = \frac{\pi\alpha}{m v_\infty^2 \rho^2}, \qquad \frac{\mathrm{d}\sigma}{\mathrm{d}o} \simeq \pi^2 a^2 \frac{\pi}{4\pi^2\chi^3} = \frac{\pi\alpha}{2 m v_\infty^2 \chi^3}.$$

The same small-angle result follows directly from $\rho^2 \simeq \pi\alpha/(m v_\infty^2 \chi)$, which gives $\mathrm{d}\sigma = \pi|\mathrm{d}(\rho^2)/\mathrm{d}\chi|\,\mathrm{d}\chi = \pi^2\alpha\,\mathrm{d}\chi/(m v_\infty^2\chi^2)$ with $\mathrm{d}o \simeq 2\pi\chi\,\mathrm{d}\chi$. Substituting a^2, we thus get

Answer to Problem 1.

$$\chi = \pi\left[1 - \left(1 + \frac{2\alpha}{m v_\infty^2 \rho^2}\right)^{-1/2}\right],$$
$$\mathrm{d}\sigma = \frac{2\pi^2\alpha}{m v_\infty^2}\,\frac{\pi-\chi}{\chi^2(2\pi-\chi)^2}\,\frac{\mathrm{d}o}{\sin\chi}, \qquad 0 < \chi < \pi$$

Problem 2.

Find the effective cross-section for scattering by a spherical "potential well" of radius a and "depth" U_0 (*i.e.* a field with $U = 0$ for $r > a$ and $U = -U_0$ for $r < a$).

Solution: Let m be the mass of the particle and v_∞ its velocity at infinity. Since U is constant outside the sphere, the particle moves with the constant velocity v_∞ until it reaches $r = a$. Inside the well, U is again constant, so the particle moves with a constant velocity v_{in} given by the conservation of energy,

$$\frac{1}{2}mv_{\text{in}}^2 - U_0 = \frac{1}{2}mv_\infty^2.$$

As in the book, we define the ratio

$$n = \frac{v_{\text{in}}}{v_\infty} = \sqrt{1 + \frac{2U_0}{mv_\infty^2}} > 1.$$

The trajectory is therefore made of straight segments, and its direction changes only at the boundary $r = a$, where the particle receives an impulse directed along the radius. This is analogous to the refraction of a light ray by a sphere with refractive index n.

The geometry of the path is shown in Fig. 21 of the book. Let α be the angle of incidence and β the angle of refraction, both measured from the normal to the sphere (*i.e.* the radius) at the point of entry. At $r = a$, the angular momentum is conserved (the force is radial), so the component of the velocity perpendicular to the radius is the same on both sides,

$$mv_\infty a \sin\alpha = mv_{\text{in}} a \sin\beta \Rightarrow \frac{\sin\alpha}{\sin\beta} = n.$$

Particles with $\rho > a$ do not touch the sphere and are not deflected. For $\rho < a$, the impact parameter is the distance from the centre to the incident line, so

$$\rho = a\sin\alpha.$$

At the entry, the direction of motion turns towards the centre by an angle $\alpha - \beta$. Inside the sphere, the particle moves along a chord, which makes the same angle β with the radii at both of its ends. By the symmetry of the path with respect to the radius bisecting this chord, the particle leaves the sphere at the angle α with the normal and is again deflected by $\alpha - \beta$ in the same sense. The total angle of deflection is thus

$$\chi = 2(\alpha - \beta).$$

To express ρ in terms of χ, let $q = \chi/2$, so that $\beta = \alpha - q$. The law of refraction becomes

$$\sin(\alpha - q) = \sin\alpha\cos q - \cos\alpha\sin q = \frac{\sin\alpha}{n}.$$

Dividing by $\sin\alpha$,

$$\cos q - \cot\alpha\sin q = \frac{1}{n} \Rightarrow \cot\alpha = \frac{n\cos q - 1}{n\sin q}.$$

Using $\sin^2\alpha = 1/(1 + \cot^2\alpha)$, we get

$$\sin^2\alpha = \frac{n^2\sin^2 q}{n^2\sin^2 q + (n\cos q - 1)^2} = \frac{n^2\sin^2 q}{n^2 + 1 - 2n\cos q},$$

since $n^2\sin^2 q + n^2\cos^2 q - 2n\cos q + 1 = n^2 + 1 - 2n\cos q$. Therefore

$$\rho^2 = a^2\sin^2\alpha = \frac{a^2n^2\sin^2(\chi/2)}{n^2 + 1 - 2n\cos(\chi/2)}.$$

Let $D = n^2 + 1 - 2n\cos q$, so that $\mathrm{d}D/\mathrm{d}q = 2n\sin q$. Since $\mathrm{d}q/\mathrm{d}\chi = 1/2$, the quotient rule gives

$$\frac{\mathrm{d}(\rho^2)}{\mathrm{d}\chi} = \frac{a^2n^2}{2}\frac{2\sin q\cos q\, D - \sin^2 q\cdot 2n\sin q}{D^2} = \frac{a^2n^2\sin q\left[\cos q\, D - n\sin^2 q\right]}{D^2}.$$

The bracket factorizes as

$$\begin{aligned}\cos q\, D - n\sin^2 q &= n^2\cos q + \cos q - 2n\cos^2 q - n + n\cos^2 q\\ &= n^2\cos q - n\cos^2 q - n + \cos q\\ &= (n\cos q - 1)(n - \cos q).\end{aligned}$$

We now check that χ increases with ρ. As ρ increases from 0 to a, α increases from 0 to $\pi/2$, and since $\sin\beta = \sin\alpha/n$,

$$\frac{\mathrm{d}\chi}{\mathrm{d}\alpha} = 2\left(1 - \frac{\mathrm{d}\beta}{\mathrm{d}\alpha}\right) = 2\left(1 - \frac{\cos\alpha}{n\cos\beta}\right) = 2\left(1 - \frac{\cos\alpha}{\sqrt{n^2 - \sin^2\alpha}}\right) > 0,$$

because $n^2 - \sin^2\alpha > 1 - \sin^2\alpha = \cos^2\alpha$. The relation between ρ and χ is thus one-to-one, and $\mathrm{d}(\rho^2)/\mathrm{d}\chi > 0$. The particles deflected into the range $\mathrm{d}\chi$ pass through an annulus of area

$$\mathrm{d}\sigma = \pi\frac{\mathrm{d}(\rho^2)}{\mathrm{d}\chi}\,\mathrm{d}\chi = \frac{1}{2\sin\chi}\frac{\mathrm{d}(\rho^2)}{\mathrm{d}\chi}\,\mathrm{d}o,$$

where we used $\mathrm{d}o = 2\pi\sin\chi\,\mathrm{d}\chi$. With $\sin\chi = 2\sin q\cos q$, the factor $\sin q$ cancels, and

$$\mathrm{d}\sigma = \frac{a^2n^2(n\cos q - 1)(n - \cos q)}{4\cos q\, D^2}\,\mathrm{d}o.$$

Let us find the range of χ. For $\rho = 0$, the particle crosses the sphere along a diameter, so $\alpha = \beta = 0$ and $\chi = 0$. As $\rho \to a$ from below, the particle grazes the sphere, with

$$\alpha \to \frac{\pi}{2}, \qquad \beta \to \arcsin\frac{1}{n}.$$

Hence

$$\chi_{max} = 2\left(\frac{\pi}{2} - \arcsin\frac{1}{n}\right) = 2\arccos\frac{1}{n}, \qquad \cos\frac{\chi_{max}}{2} = \frac{1}{n}.$$

The formula for $\mathrm{d}\sigma$ applies for $0 < \chi < \chi_{max}$. Accordingly, the factor $n\cos(\chi/2) - 1$ is positive in this range and vanishes at $\chi = \chi_{max}$. Particles with $\rho > a$ are not deflected and do not contribute to the scattering.

Finally, the total cross-section for scattering is obtained directly from the area of the annuli,

$$\sigma = \int_0^{\chi_{max}} \mathrm{d}\sigma = \pi\left[\rho^2(\chi_{max}) - \rho^2(0)\right].$$

At $\chi = \chi_{max}$, $\cos q = 1/n$, so $\sin^2 q = (n^2 - 1)/n^2$ and $D = n^2 - 1$, which gives $\rho^2(\chi_{max}) = a^2$. Since $\rho^2(0) = 0$, $\sigma = \pi a^2$. This is simply the cross-sectional area of the sphere: every particle which hits the sphere is deflected. We thus get

Answer to Problem 2.

$$\mathrm{d}\sigma = \frac{a^2n^2\left[n\cos(\chi/2) - 1\right]\left[n - \cos(\chi/2)\right]}{4\cos(\chi/2)\left[n^2 + 1 - 2n\cos(\chi/2)\right]^2}\,\mathrm{d}o, \qquad n = \sqrt{1 + \frac{2U_0}{mv_\infty^2}},$$
$$0 < \chi < \chi_{max} = 2\arccos\frac{1}{n}, \qquad \sigma = \pi a^2$$

§20 Small-angle scattering

Problem 1.

Derive formula (20.3) from (18.4).

Solution: Consider the relative motion, with reduced mass m and relative velocity at infinity v_∞. Formula (18.4) of the book gives

$$\phi_0 = \int_{r_{min}}^{\infty} \frac{\rho\,\mathrm{d}r}{r^2\sqrt{1-\frac{\rho^2}{r^2}-\frac{2U(r)}{mv_\infty^2}}}, \qquad \chi = \pi - 2\phi_0.$$

For a weak field, it is tempting to expand the integrand in powers of U. However, the integrand behaves as $1/\sqrt{r-r_{min}}$ near the turning point. Expanding it gives terms like $U/(1-\rho^2/r^2)^{3/2}$, which are not integrable at $r=\rho$. These divergences are spurious: they only come from expanding near the point where the radicand vanishes.

To avoid this, we write ϕ_0 as the derivative of an integral whose integrand vanishes at the turning point. Since this integral does not converge at infinity, we first introduce a finite upper limit R,

$$J_R(\rho) = \int_{r_{min}}^{R} \sqrt{1-\frac{\rho^2}{r^2}-\frac{2U(r)}{mv_\infty^2}}\,\mathrm{d}r.$$

We differentiate with respect to ρ at fixed R. The lower limit r_{min} depends on ρ, but its contribution is the integrand at $r=r_{min}$ times $\mathrm{d}r_{min}/\mathrm{d}\rho$, which vanishes since the square root is zero there. Hence

$$\frac{\partial J_R}{\partial \rho} = -\int_{r_{min}}^{R} \frac{\rho\,\mathrm{d}r}{r^2\sqrt{1-\frac{\rho^2}{r^2}-\frac{2U(r)}{mv_\infty^2}}} \Rightarrow \phi_{0,R} = -\frac{\partial J_R}{\partial \rho},$$

where $\phi_{0,R}$ is the integral (18.4) cut off at $r=R$. We keep R finite until after the differentiation, since J_R itself grows like R and has no limit as $R\to\infty$.

We now expand J_R to first order in U. Pointwise,

$$\sqrt{1-\frac{\rho^2}{r^2}-\frac{2U}{mv_\infty^2}} \simeq \sqrt{1-\frac{\rho^2}{r^2}} - \frac{U}{mv_\infty^2\sqrt{1-\rho^2/r^2}} = \sqrt{1-\frac{\rho^2}{r^2}} - \frac{1}{mv_\infty^2}\frac{U(r)\,r}{\sqrt{r^2-\rho^2}}.$$

This expansion is not uniform near the turning point, but it can be integrated. To see why, regard J_R as a function of the strength of the field. Its first-order variation contains a term from the shift of the lower limit, from ρ (the turning point when $U=0$) to r_{min}. This term is proportional to the unperturbed integrand at its lower endpoint, which vanishes. The other term is the integral of the first-order variation of the integrand, which has only an integrable singularity $1/\sqrt{r-\rho}$. To first order, we can therefore take the lower limit at ρ,

$$J_R \simeq \int_\rho^R \sqrt{1-\frac{\rho^2}{r^2}}\,\mathrm{d}r - \frac{1}{mv_\infty^2}\int_\rho^R \frac{U(r)\,r\,\mathrm{d}r}{\sqrt{r^2-\rho^2}} = \int_\rho^R \sqrt{1-\frac{\rho^2}{r^2}}\,\mathrm{d}r - \frac{K_R(\rho)}{mv_\infty^2},$$

where

$$K_R(\rho) = \int_\rho^R \frac{U(r)\,r\,\mathrm{d}r}{\sqrt{r^2-\rho^2}}.$$

The free term can be integrated directly,

$$\int_\rho^R \sqrt{1-\frac{\rho^2}{r^2}}\,\mathrm{d}r = \int_\rho^R \frac{\sqrt{r^2-\rho^2}}{r}\,\mathrm{d}r = \left[\sqrt{r^2-\rho^2} - \rho\arccos\frac{\rho}{r}\right]_\rho^R = \sqrt{R^2-\rho^2} - \rho\arccos\frac{\rho}{R},$$

as can be checked by differentiating the bracket with respect to r,

$$\frac{r}{\sqrt{r^2-\rho^2}} - \frac{\rho^2}{r\sqrt{r^2-\rho^2}} = \frac{\sqrt{r^2-\rho^2}}{r}.$$

Its derivative with respect to ρ is

$$\frac{\partial}{\partial\rho}\left(\sqrt{R^2-\rho^2}-\rho\arccos\frac{\rho}{R}\right)=-\frac{\rho}{\sqrt{R^2-\rho^2}}-\arccos\frac{\rho}{R}+\frac{\rho}{\sqrt{R^2-\rho^2}}=-\arccos\frac{\rho}{R}.$$

Consequently,

$$\phi_{0,R}\simeq\arccos\frac{\rho}{R}+\frac{1}{mv_\infty^2}\frac{\partial K_R}{\partial\rho}.$$

We cannot differentiate K_R directly under the integral sign, because its integrand is singular at the lower limit $r=\rho$. We therefore integrate by parts first, using

$$\frac{r\,\mathrm{d}r}{\sqrt{r^2-\rho^2}}=\mathrm{d}\sqrt{r^2-\rho^2},$$

which gives

$$K_R=\Big[U(r)\sqrt{r^2-\rho^2}\Big]_\rho^R-\int_\rho^R U'(r)\sqrt{r^2-\rho^2}\,\mathrm{d}r=U(R)\sqrt{R^2-\rho^2}-\int_\rho^R U'(r)\sqrt{r^2-\rho^2}\,\mathrm{d}r.$$

The boundary term at $r=\rho$ vanishes since $\sqrt{r^2-\rho^2}=0$ there. The new integrand is finite at $r=\rho$, and it vanishes there. Hence, when we differentiate with respect to ρ, the variation of the lower limit again gives no contribution, and

$$\frac{\partial K_R}{\partial\rho}=-\frac{\rho U(R)}{\sqrt{R^2-\rho^2}}+\rho\int_\rho^R\frac{U'(r)\,\mathrm{d}r}{\sqrt{r^2-\rho^2}}.$$

Now we take $R\to\infty$. Then $\arccos(\rho/R)\to\pi/2$, and the first term vanishes since $U(R)\to0$ and $\rho/\sqrt{R^2-\rho^2}\to0$. Assuming the remaining integral converges (*i.e.* the force decreases fast enough), we get

$$\phi_0\simeq\frac{\pi}{2}+\frac{\rho}{mv_\infty^2}\int_\rho^\infty\frac{U'(r)\,\mathrm{d}r}{\sqrt{r^2-\rho^2}},$$

and therefore

$$\chi=\pi-2\phi_0\simeq-\frac{2\rho}{mv_\infty^2}\int_\rho^\infty\frac{U'(r)\,\mathrm{d}r}{\sqrt{r^2-\rho^2}}.$$

For a repulsive field, $U'<0$ and $\chi>0$, as it should be. This result is valid only if the deflection is small, *i.e.* if the field is weak along the whole path ($|U|\ll\frac12 mv_\infty^2$ for $r\gtrsim\rho$). Under this condition, the first-order expansion is justified.

The angle χ is the scattering angle in the centre-of-mass system. Formula (20.3) gives instead θ_1, the deflection of the incident particle in the laboratory system, where the particle m_2 is initially at rest. From formula (17.4) of the book, for small angles

$$\tan\theta_1=\frac{m_2\sin\chi}{m_1+m_2\cos\chi}\Rightarrow\theta_1\simeq\frac{m_2}{m_1+m_2}\chi=\frac{m}{m_1}\chi,\qquad m=\frac{m_1m_2}{m_1+m_2}.$$

Multiplying χ by m/m_1, the reduced mass cancels. We thus get

Answer to Problem 1.

$$\theta_1\simeq-\frac{2\rho}{m_1v_\infty^2}\int_\rho^\infty\frac{\mathrm{d}U}{\mathrm{d}r}\frac{\mathrm{d}r}{\sqrt{r^2-\rho^2}},$$
$$\chi\simeq-\frac{2\rho}{mv_\infty^2}\int_\rho^\infty\frac{\mathrm{d}U}{\mathrm{d}r}\frac{\mathrm{d}r}{\sqrt{r^2-\rho^2}}$$

Problem 2.

Determine the effective cross-section for small-angle scattering in a field $U = \alpha/r^n$ $(n > 0)$.

Solution: We take $\alpha > 0$ (a repulsive field). Let m_2 be the mass of the scattering particle, initially at rest, and

$$m = \frac{m_1 m_2}{m_1 + m_2}$$

the reduced mass. As shown in §20 Prob. 1, when the scattering angle χ in the centre-of-mass system is small ($\chi \ll 1$), the deflection of the incident particle of mass m_1 is $\theta_1 \simeq (m/m_1)\chi$, and it is given by formula (20.3) of the book,

$$\theta_1 \simeq -\frac{2\rho}{m_1 v_\infty^2} \int_\rho^\infty \frac{\mathrm{d}U}{\mathrm{d}r} \frac{\mathrm{d}r}{\sqrt{r^2 - \rho^2}}.$$

Since

$$\frac{\mathrm{d}U}{\mathrm{d}r} = -\frac{n\alpha}{r^{n+1}},$$

the deflection is

$$\theta_1 \simeq \frac{2n\alpha\rho}{m_1 v_\infty^2} \int_\rho^\infty \frac{\mathrm{d}r}{r^{n+1}\sqrt{r^2 - \rho^2}}.$$

To evaluate the integral, we substitute

$$u = \frac{\rho^2}{r^2}, \qquad r = \rho u^{-1/2}, \qquad \mathrm{d}r = -\frac{\rho}{2} u^{-3/2}\,\mathrm{d}u.$$

When r goes from ρ to ∞, u goes from 1 to 0, and the minus sign of $\mathrm{d}r$ is absorbed by reversing the limits. We also have

$$r^{n+1} = \rho^{n+1} u^{-(n+1)/2}, \qquad \sqrt{r^2 - \rho^2} = \rho\sqrt{\frac{1}{u} - 1} = \rho u^{-1/2}(1-u)^{1/2}.$$

Collecting the powers of u, $-\frac{3}{2} + \frac{n+1}{2} + \frac{1}{2} = \frac{n-1}{2}$, so

$$\int_\rho^\infty \frac{\mathrm{d}r}{r^{n+1}\sqrt{r^2 - \rho^2}} = \frac{1}{2\rho^{n+1}} \int_0^1 u^{(n-1)/2}(1-u)^{-1/2}\,\mathrm{d}u.$$

As in §11 Prob. 2, this is the Beta function $B(p,q) = \int_0^1 u^{p-1}(1-u)^{q-1}\,\mathrm{d}u = \Gamma(p)\Gamma(q)/\Gamma(p+q)$ with $p = (n+1)/2$ and $q = 1/2$. Since $\Gamma(1/2) = \sqrt{\pi}$,

$$\int_\rho^\infty \frac{\mathrm{d}r}{r^{n+1}\sqrt{r^2 - \rho^2}} = \frac{1}{2\rho^{n+1}} B\left(\frac{n+1}{2}, \frac{1}{2}\right) = \frac{\sqrt{\pi}}{2\rho^{n+1}} \frac{\Gamma\left((n+1)/2\right)}{\Gamma\left((n+2)/2\right)}.$$

Using $\Gamma\left((n+2)/2\right) = \frac{n}{2}\Gamma(n/2)$, the factor n cancels and

$$\theta_1 \simeq \frac{2\sqrt{\pi}\,\alpha}{m_1 v_\infty^2 \rho^n} \frac{\Gamma\left((n+1)/2\right)}{\Gamma(n/2)}.$$

Let us define

$$C_n = \frac{2\sqrt{\pi}\,\alpha}{m_1 v_\infty^2} \frac{\Gamma\left((n+1)/2\right)}{\Gamma(n/2)},$$

so that $\theta_1 \simeq C_n/\rho^n$. The deflection decreases monotonically with ρ, and the relation can be inverted,

$$\rho \simeq \left(\frac{C_n}{\theta_1}\right)^{1/n}, \qquad \left|\frac{\mathrm{d}\rho}{\mathrm{d}\theta_1}\right| = \frac{1}{n} C_n^{1/n} \theta_1^{-1/n-1} = \frac{\rho}{n\theta_1}.$$

The particles deflected into the range $\mathrm{d}\theta_1$ pass through an annulus of area $2\pi\rho|\mathrm{d}\rho|$ in the incident beam. For small angles, the element of solid angle is

$$\mathrm{d}o_1 = 2\pi \sin\theta_1\,\mathrm{d}\theta_1 \simeq 2\pi\theta_1\,\mathrm{d}\theta_1,$$

so

$$\frac{\mathrm{d}\sigma}{\mathrm{d}o_1} \simeq \frac{\rho}{\theta_1}\left|\frac{\mathrm{d}\rho}{\mathrm{d}\theta_1}\right| = \frac{\rho^2}{n\theta_1^2} = \frac{1}{n}\frac{C_n^{2/n}}{\theta_1^{2+2/n}}.$$

This result requires weak scattering in the centre-of-mass system, $\chi \simeq (m_1/m)\theta_1 \ll 1$. Multiplying

$\theta_1 \simeq C_n/\rho^n$ by m_1/m, the corresponding centre-of-mass deflection is

$$\chi \simeq \frac{2\sqrt{\pi}\,\Gamma\left((n+1)/2\right)}{\Gamma(n/2)}\frac{\alpha/\rho^n}{mv_\infty^2},$$

whose numerical coefficient is of order unity for n of order unity. The condition $\chi \ll 1$ is thus the weak-field condition

$$\frac{\alpha}{\rho^n} \ll \frac{1}{2}mv_\infty^2,$$

i.e. the potential energy at the distance of closest approach $r \simeq \rho$ must be small compared with the energy of the relative motion. Since $m \leq m_1$, it implies $\theta_1 \ll 1$, which justifies $\mathrm{d}o_1 \simeq 2\pi\theta_1\,\mathrm{d}\theta_1$. However, the converse does not hold for arbitrary masses. For instance, if $m_1 > m_2$, the laboratory angle never exceeds $\arcsin(m_2/m_1)$, so it can be small even when χ is not. The two conditions coincide only when $m_2 \gg m_1$, where $m \simeq m_1$ and $\theta_1 \simeq \chi$. The result must not be extended to arbitrary angles. Note also that $\mathrm{d}\sigma$ diverges as $\theta_1 \to 0$, so the total cross-section is infinite unless the angles are cut off.

As a check, for $n = 1$ we have $\Gamma(1)/\Gamma(1/2) = 1/\sqrt{\pi}$, so $C_1 = 2\alpha/(m_1v_\infty^2)$ and

$$\frac{\mathrm{d}\sigma}{\mathrm{d}o_1} \simeq \frac{4\alpha^2}{m_1^2v_\infty^4\theta_1^4},$$

which is the small-angle limit of the Rutherford formula. For $n = 2$, $\Gamma(3/2)/\Gamma(1) = \sqrt{\pi}/2$, so $C_2 = \pi\alpha/(m_1v_\infty^2)$ and

$$\frac{\mathrm{d}\sigma}{\mathrm{d}o_1} \simeq \frac{\pi\alpha}{2m_1v_\infty^2\theta_1^3},$$

in agreement with the small-angle limit found in §19 Prob. 1 for a fixed centre ($m_1 = m$, $\theta_1 = \chi$). We thus get

Answer to Problem 2.

$$\theta_1 \simeq \frac{2\sqrt{\pi}\,\Gamma\left((n+1)/2\right)}{\Gamma(n/2)}\frac{\alpha}{m_1v_\infty^2\rho^n},$$

$$\mathrm{d}\sigma \simeq \frac{1}{n}\left[\frac{2\sqrt{\pi}\,\Gamma\left((n+1)/2\right)}{\Gamma(n/2)}\frac{\alpha}{m_1v_\infty^2}\right]^{2/n}\frac{\mathrm{d}o_1}{\theta_1^{2+2/n}},$$

$$\text{valid for } \frac{\alpha}{\rho^n} \ll \frac{1}{2}mv_\infty^2, \qquad m = \frac{m_1m_2}{m_1+m_2}$$

Chapter 5.

Small Oscillations

§21 Free oscillations in one dimension

Problem 1.

Express the amplitude and initial phase of the oscillations in terms of the initial co-ordinate x_0 and velocity v_0.

Solution: The general solution of $\ddot{x} + \omega^2 x = 0$, with angular frequency $\omega > 0$, that satisfies $x(0) = x_0$ and $\dot{x}(0) = v_0$ is

$$x(t) = x_0 \cos \omega t + \frac{v_0}{\omega} \sin \omega t.$$

We write the same harmonic motion as

$$x(t) = a \cos(\omega t + \alpha) = a \cos\alpha \cos \omega t - a \sin\alpha \sin \omega t, \qquad a \geq 0.$$

Comparing the coefficients of $\cos \omega t$ and $\sin \omega t$, we get

$$a \cos\alpha = x_0, \qquad a \sin\alpha = -\frac{v_0}{\omega}.$$

These are the relations $x_0 = a\cos\alpha$ and $v_0 = -a\omega \sin\alpha$ obtained by evaluating x and $\dot{x}$ directly at $t = 0$. Squaring them and adding gives

$$x_0^2 + \frac{v_0^2}{\omega^2} = a^2 \left(\cos^2\alpha + \sin^2\alpha\right) = a^2.$$

For a nonzero amplitude, the phase is fixed modulo 2π by

$$\cos\alpha = \frac{x_0}{a}, \qquad \sin\alpha = -\frac{v_0}{a\omega}.$$

Thus, $\tan\alpha = -v_0/(\omega x_0)$ when $x_0 \neq 0$, with the quadrant determined by the signs of the sine and cosine. The two-argument arctangent $\operatorname{atan2}(Y, X)$ returns the angle whose cosine and sine have the signs of X and Y respectively, and so selects the correct quadrant. If $x_0 = v_0 = 0$, the particle remains at equilibrium and the phase is arbitrary.

Answer to Problem 1.

$$a = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}, \qquad \alpha = \operatorname{atan2}\left(-\frac{v_0}{\omega}, x_0\right) \pmod{2\pi}$$

Problem 2.

Find the ratio of frequencies ω and ω' of the oscillations of two diatomic molecules consisting of atoms of different isotopes, the masses of the atoms being m_1, m_2 and m_1', m_2'.

Solution: Within the harmonic approximation, isotopic substitution changes the masses while leaving the interatomic potential, and hence its curvature k at equilibrium, unchanged.

Let u_1 and u_2 be the displacements of the atoms from their equilibrium positions along the molecular axis, with atom 2 on the positive side of atom 1. We define the centre-of-mass displacement X and the change q in interatomic separation from equilibrium by

$$X = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}, \qquad q = u_2 - u_1.$$

Inverting these relations gives

$$u_1 = X - \frac{m_2}{m_1 + m_2} q, \qquad u_2 = X + \frac{m_1}{m_1 + m_2} q.$$

Substituting them into the kinetic energy, the cross terms in $\dot{X}\dot{q}$ cancel, and we get

$$T = \frac{1}{2} m_1 \dot{u}_1^2 + \frac{1}{2} m_2 \dot{u}_2^2 = \frac{1}{2}(m_1 + m_2)\dot{X}^2 + \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} \dot{q}^2.$$

The first term describes the free translation of the molecule as a whole, and the second describes its vibration. The potential energy depends only on the interatomic separation, and hence only on q; in the harmonic approximation, it is $\frac{1}{2}kq^2$. Dropping the translational term, the vibrational Lagrangian is

$$L_{\text{vib}} = \frac{1}{2}\mu \dot{q}^2 - \frac{1}{2} k q^2, \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}.$$

The equation of motion is

$$\mu \ddot{q} + kq = 0,$$

so that

$$\omega = \sqrt{\frac{k}{\mu}}.$$

For the isotopically substituted molecule,

$$\mu' = \frac{m_1' m_2'}{m_1' + m_2'}, \qquad \omega' = \sqrt{\frac{k}{\mu'}}.$$

Taking the ratio eliminates the common force constant.

Answer to Problem 2.

$$\frac{\omega'}{\omega} = \sqrt{\frac{\mu}{\mu'}} = \sqrt{\frac{m_1 m_2 \,(m_1' + m_2')}{m_1' m_2' \,(m_1 + m_2)}}$$

Problem 3.

Find the frequency of oscillations of a particle of mass m which is free to move along a line and is attached to a spring whose other end is fixed at a point A (Fig. 22) at a distance l from the line. A force F is required to extend the spring to length l.

Solution: Let x be the displacement along the line from the point nearest the fixed end A. The length of the spring is

$$d(x) = \sqrt{l^2 + x^2}.$$

Here, l is the length of the spring in the constrained equilibrium configuration $x = 0$, where the spring is already stretched and carries a tension $F > 0$; it is not the natural length of the spring. For $|x| \ll l$, the additional extension from this equilibrium length is

$$\delta l = d(x) - l = l\left(\sqrt{1 + \frac{x^2}{l^2}} - 1\right) \simeq \frac{x^2}{2l}.$$

Let $V(d)$ denote the potential energy of the spring. The force required to hold it at length l is $F = V'(l) > 0$. Expanding about that length and dropping the constant $V(l)$, we get

$$U(x) = V(l + \delta l) - V(l) = F\,\delta l + \frac{1}{2}V''(l)\,(\delta l)^2 + \cdots .$$

Since δl is already quadratic in x, its square contributes only at fourth order. Thus, the harmonic approximation gives

$$U(x) \simeq \frac{F}{2l}x^2.$$

The restoring potential therefore comes from the first-order term $F\,\delta l$, *i.e.* from the tension already present at equilibrium, and not from the stiffness $V''(l)$ of the spring.

The kinetic energy is

$$T = \frac{1}{2}m\dot{x}^2,$$

and the equation of motion becomes

$$m\ddot{x} + \frac{F}{l}x = 0.$$

As an independent check, the tension $V'(d)$ pulls the particle towards A along the spring, and its component along the line is

$$F_x = -V'(d)\frac{x}{d} \simeq -\frac{F}{l}x$$

to first order in x, which reproduces this equation of motion. Comparison with $\ddot{x} + \omega^2 x = 0$ gives the angular frequency.

Answer to Problem 3.

$$\omega = \sqrt{\frac{F}{ml}}$$

Problem 4.

The same as §21 Prob. 3, but for a particle of mass m moving on a circle of radius r (Fig. 23).

Solution: Let O be the centre of the circle of radius r. The fixed end A lies outside the circle, with $OA = l + r$, so that the spring has length l when the particle is at the point nearest A. (If A were inside the circle, with $OA = r - l$, the same steps would give $\omega^2 = F(r-l)/(mrl)$.) Let ϕ be the angle between OA and the radius to the particle.

By the cosine rule, the length of the spring is

$$\begin{aligned} d(\phi) &= \sqrt{r^2 + (l+r)^2 - 2r(l+r)\cos\phi} \\ &= \sqrt{l^2 + 2r(l+r)(1-\cos\phi)}. \end{aligned}$$

For small oscillations, $1 - \cos\phi \simeq \phi^2/2$. Factoring out l and using $\sqrt{1+\epsilon} \simeq 1 + \epsilon/2$, we get

$$d = l\sqrt{1 + \frac{2r(l+r)}{l^2}(1-\cos\phi)} \simeq l + \frac{r(l+r)}{2l}\phi^2,$$

and therefore

$$\delta l = d(\phi) - l \simeq \frac{r(l+r)}{2l}\phi^2.$$

The amplitude must be small enough for both expansions to hold: $\phi \ll 1$ for the cosine, and $r(l+r)\phi^2/l^2 \ll 1$ for the square root. As in §21 Prob. 3, the leading change in the energy of the spring is $F\,\delta l$. Hence,

$$U(\phi) \simeq \frac{Fr(l+r)}{2l}\phi^2.$$

The speed of the particle is $r\dot{\phi}$, giving

$$T = \frac{1}{2}mr^2\dot{\phi}^2.$$

The linearized equation of motion is

$$mr^2\ddot{\phi} + \frac{Fr(l+r)}{l}\phi = 0,$$

or

$$\ddot{\phi} + \frac{F(l+r)}{mrl}\phi = 0.$$

As a consistency check, at fixed l the circle becomes a straight line at distance l from A as $r \to \infty$, and

$$\omega^2 = \frac{F}{ml}\left(1 + \frac{l}{r}\right) \longrightarrow \frac{F}{ml} \quad \text{as } r \to \infty,$$

which recovers the result of §21 Prob. 3.

Answer to Problem 4.

$$\omega = \sqrt{\frac{F(l+r)}{mrl}}$$

Problem 5.

Find the frequency of oscillations of the pendulum shown in the figure in §5 Prob. 2, whose point of support carries a mass m_1 and is free to move horizontally (its motion is integrated in §14 Prob. 3).

Solution: Let x denote the horizontal position of the support and ϕ the angle of the pendulum from the downward vertical. The support has mass m_1, the bob has mass m_2, and the length of the pendulum is l.

From §5 Prob. 2, the Lagrangian is

$$L = \frac{1}{2}(m_1 + m_2)\dot{x}^2 + \frac{1}{2}m_2 l^2\dot{\phi}^2 + m_2 l\dot{x}\dot{\phi}\cos\phi + m_2 gl\cos\phi.$$

Because x is cyclic, the total horizontal momentum is conserved:

$$p_x = \frac{\partial L}{\partial \dot{x}} = (m_1 + m_2)\dot{x} + m_2 l\dot{\phi}\cos\phi.$$

A constant horizontal translation of the whole system does not affect its oscillation frequency. We may therefore work in the inertial frame in which $p_x = 0$, as in §14 Prob. 3. In this frame,

$$\dot{x} = -\frac{m_2 l}{m_1 + m_2}\dot{\phi}\cos\phi.$$

Substituting this expression into the kinetic energy, we get

$$\begin{aligned} T &= \frac{1}{2}(m_1 + m_2)\dot{x}^2 + \frac{1}{2}m_2 l^2\dot{\phi}^2 + m_2 l\dot{x}\dot{\phi}\cos\phi \\ &= \frac{m_2^2 l^2}{2(m_1 + m_2)}\dot{\phi}^2\cos^2\phi + \frac{1}{2}m_2 l^2\dot{\phi}^2 - \frac{m_2^2 l^2}{m_1 + m_2}\dot{\phi}^2\cos^2\phi \\ &= \frac{1}{2}m_2 l^2\left(1 - \frac{m_2}{m_1 + m_2}\cos^2\phi\right)\dot{\phi}^2. \end{aligned}$$

For small oscillations, we retain terms of second order in the displacement and velocity. Since $\dot{\phi}^2$ is already of second order, replacing $\cos^2\phi$ by 1 changes T only at fourth order. Hence,

$$T \simeq \frac{1}{2}\frac{m_1 m_2}{m_1 + m_2}l^2\dot{\phi}^2.$$

Taking the gravitational potential energy to vanish at $\phi = 0$, we have

$$U = m_2 gl(1 - \cos\phi) \simeq \frac{1}{2}m_2 gl\phi^2.$$

Eliminating a cyclic velocity directly in the Lagrangian is not legitimate in general. The correct Lagrangian for the remaining co-ordinate is $L - p_x\dot{x}$ (minus the Routhian), with $\dot{x}$ expressed in terms of p_x and ϕ. Here, $p_x = 0$, so the extra term vanishes and the Lagrangian for ϕ is simply $T - U$ with $\dot{x}$ eliminated. Equivalently, the energy $E = T + U$ then takes the one-dimensional form $\frac{1}{2}M(\phi)\dot{\phi}^2 + U(\phi)$, which determines the frequency.

The quadratic Lagrangian for the oscillation is therefore

$$L^{(2)} = \frac{1}{2}\frac{m_1 m_2}{m_1 + m_2}l^2\dot{\phi}^2 - \frac{1}{2}m_2 gl\phi^2.$$

Its Euler–Lagrange equation gives

$$\frac{m_1 m_2}{m_1 + m_2}l^2\ddot{\phi} + m_2 gl\phi = 0,$$

or

$$\ddot{\phi} + \frac{g(m_1 + m_2)}{m_1 l}\phi = 0.$$

The freedom of the support to recoil reduces the effective inertia of the oscillatory co-ordinate. Consequently, the frequency exceeds that of a pendulum with a fixed support. In the limit $m_1 \to \infty$, the

fixed-support result $\sqrt{g/l}$ is recovered.

Answer to Problem 5.

$$\omega = \sqrt{\frac{g(m_1 + m_2)}{m_1 l}}$$

Problem 6.

Determine the form of a curve such that the frequency of oscillations of a particle on it under the force of gravity is independent of the amplitude.

Solution: Consider a particle constrained to slide without friction along a plane curve in a uniform gravitational field. We set the lowest point at $x = y = 0$, with y measured vertically upwards, and let s be the signed arc length along the curve, measured from that point.

The arc length is a useful generalized co-ordinate because the speed is exactly $|\dot{s}|$. Thus,

$$T = \frac{1}{2}m\dot{s}^2, \qquad U = mgy(s),$$

and the equation of motion is

$$\ddot{s} + g\frac{\mathrm{d}y}{\mathrm{d}s} = 0.$$

To construct the required curve, we require the motion in s to be exactly harmonic with a prescribed constant angular frequency ω:

$$\ddot{s} + \omega^2 s = 0.$$

This is sufficient to make the frequency independent of the amplitude. Comparing the two equations and integrating, with $y(0) = 0$, gives

$$g\frac{\mathrm{d}y}{\mathrm{d}s} = \omega^2 s, \qquad y = \frac{\omega^2 s^2}{2g}.$$

No small-amplitude expansion has been used.

The required relation between height and arc length now determines the geometry. On the right-hand branch, $s > 0$, we have

$$s = \frac{\sqrt{2gy}}{\omega}, \qquad \left(\frac{\mathrm{d}s}{\mathrm{d}y}\right)^2 = \frac{g}{2\omega^2 y}.$$

Since $\mathrm{d}s^2 = \mathrm{d}x^2 + \mathrm{d}y^2$,

$$\left(\frac{\mathrm{d}x}{\mathrm{d}y}\right)^2 = \left(\frac{\mathrm{d}s}{\mathrm{d}y}\right)^2 - 1 = \frac{g}{2\omega^2 y} - 1.$$

We introduce the length $a = g/(4\omega^2)$. Then,

$$\frac{\mathrm{d}x}{\mathrm{d}y} = \sqrt{\frac{2a - y}{y}}.$$

We use the substitution

$$y = a(1 - \cos\theta), \qquad 0 \le \theta < \pi,$$

for which

$$\mathrm{d}y = a\sin\theta\,\mathrm{d}\theta, \qquad \sqrt{\frac{2a - y}{y}} = \cot\frac{\theta}{2}.$$

Therefore,

$$\mathrm{d}x = a\sin\theta\cot\frac{\theta}{2}\,\mathrm{d}\theta = a(1 + \cos\theta)\,\mathrm{d}\theta.$$

Integrating from the lowest point yields

$$x = a(\theta + \sin\theta), \qquad y = a(1 - \cos\theta).$$

Reflection gives the left-hand branch, represented by the same formulae with negative θ. This is a cycloidal trough, with its lowest point at $\theta = 0$ and adjacent cusps at $\theta = \pm\pi$.

We now verify the exact independence of the amplitude explicitly. On the smooth arc $-\pi < \theta < \pi$,

$$\begin{aligned}\left(\frac{\mathrm{d}s}{\mathrm{d}\theta}\right)^2 &= \left(\frac{\mathrm{d}x}{\mathrm{d}\theta}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)^2 \\ &= a^2(1+\cos\theta)^2 + a^2\sin^2\theta \\ &= 4a^2\cos^2\frac{\theta}{2}.\end{aligned}$$

Choosing s to increase with θ, we get

$$s = 4a\sin\frac{\theta}{2}.$$

It follows that

$$y = 2a\sin^2\frac{\theta}{2} = \frac{s^2}{8a}.$$

Thus, the exact Lagrangian becomes

$$L = \frac{1}{2}m\dot{s}^2 - \frac{mg}{8a}s^2,$$

and the exact equation of motion is

$$\ddot{s} + \frac{g}{4a}s = 0.$$

Hence, every oscillation with amplitude $|s| < 4a$, *i.e.* whose turning points lie below the height $2a$ of the cusps, has the same angular frequency $\omega = \sqrt{g/4a}$ and period $\tau = 2\pi/\omega$. This applies to finite amplitudes, not merely to infinitesimal oscillations.

Answer to Problem 6.

The curve is a cycloidal trough, where $a > 0$ is the radius of its generating circle, with angular frequency ω and period τ:

$$\begin{aligned}x &= a(\theta + \sin\theta), \\ y &= a(1-\cos\theta), \qquad -\pi < \theta < \pi, \\ \omega &= \sqrt{\frac{g}{4a}}, \qquad \tau = 4\pi\sqrt{\frac{a}{g}}.\end{aligned}$$

§22 Forced oscillations

Problem 1. *Not done*

Determine the forced oscillations of a system under a force $F(t)$ of the following forms, if at time $t = 0$ the system is at rest in equilibrium ($x = \mathrm{d}x/\mathrm{d}t = 0$):

(a) *Not done*

$F = F_0$, a constant.

(b) *Not done*

$F = at$.

(c) *Not done*

$F = F_0 e^{-\alpha t}$.

(d) *Not done*

$F = F_0 e^{-\alpha t} \cos \beta t$.

Problem 2. *Not done*

Determine the final amplitude for the oscillations of a system under a force which is zero for $t < 0$, $F_0 t/T$ for $0 < t < T$, and F_0 for $t > T$ (Fig. 24), if up to time $t = 0$ the system is at rest in equilibrium.

Problem 3. *Not done*

The same as §22 Prob. 2, but for a constant force F_0 which acts for a finite time T (Fig. 25).

Problem 4. *Not done*

The same as §22 Prob. 2, but for a force $F_0 t/T$ which acts between $t = 0$ and $t = T$ (Fig. 26).

Problem 5. *Not done*

The same as §22 Prob. 2, but for a force $F_0 \sin \omega t$ which acts between $t = 0$ and $t = T = 2\pi/\omega$ (Fig. 27).

§23 Oscillations of systems with more than one degree of freedom

Problem 1. *Not done*

Determine the oscillations of a system with two degrees of freedom whose Lagrangian is

$$L = \frac{1}{2}\left(\dot{x}^2 + \dot{y}^2\right) - \frac{1}{2}\omega_0^2\left(x^2 + y^2\right) + \alpha xy$$

(two identical one-dimensional systems of eigenfrequency ω_0 coupled by an interaction $-\alpha xy$).

Problem 2. *Not done*

Determine the small oscillations of a coplanar double pendulum (see the figure in §5 Prob. 1).

Problem 3. *Not done*

Find the path of a particle in a central field $U = \frac{1}{2}kr^2$ (called a space oscillator).

§24 Vibrations of molecules

Problem 1. *Not done*

Determine the frequencies of vibrations of a symmetrical linear triatomic molecule ABA (Fig. 28). It is assumed that the potential energy of the molecule depends only on the distances AB and BA and the angle ABA.

Problem 2. *Not done*

The same as §24 Prob. 1, but for a triangular molecule ABA (Fig. 29).

Problem 3. *Not done*

The same as §24 Prob. 1, but for an unsymmetrical linear molecule ABC (Fig. 30).

§26 Forced oscillations under friction

Problem 1. *Not done*

Determine the forced oscillations due to an external force $f = f_0 e^{\alpha t} \cos \gamma t$ in the presence of friction.

§27 Parametric resonance

Problem 1. *Not done*

Obtain an expression correct as far as the term in h^2 for the limits of the region of instability for resonance near $\gamma = 2\omega_0$.

Problem 2. *Not done*

Determine the limits of the region of instability in resonance near $\gamma = \omega_0$.

Problem 3. *Not done*

Find the conditions for parametric resonance in small oscillations of a simple pendulum whose point of support oscillates vertically (see the Lagrangian in §5 Prob. 3(c)).

§29 Resonance in non-linear oscillations

Problem 1. *Not done*

Determine the function $b(\epsilon)$ for resonance at frequencies $\gamma \approx 3\omega_0$.

§30 Motion in a rapidly oscillating field

Problem 1. *Not done*

Determine the positions of stable equilibrium of a pendulum whose point of support oscillates vertically with a high frequency $\gamma \gg \sqrt{g/l}$ (see the Lagrangian in §5 Prob. 3(c)).

Problem 2. *Not done*

The same as §30 Prob. 1, but for a pendulum whose point of support oscillates horizontally (see the Lagrangian in §5 Prob. 3(b)).

Chapter 6.

Motion of a Rigid Body

§32 The inertia tensor

Problem 1.

Determine the principal moments of inertia for the following types of molecule, regarded as systems of particles at fixed distances apart:

(a)

A molecule of collinear atoms.

Solution: Choose the x_3-axis along the line containing all atoms, and place the origin at the centre of mass. Then each particle has coordinates $(0, 0, x_a)$, so the inertia tensor is diagonal in the basis (x_1, x_2, x_3), with

$$I_3 = 0, \qquad I_1 = I_2.$$

Indeed, since the distance to the x_1-axis (or x_2-axis) is $|x_a - x_{CM}|$ for each particle,

$$I_1 = I_2 = \sum_a m_a \, (x_a - x_{CM})^2, \qquad x_{CM} = \frac{1}{\mu} \sum_a m_a x_a, \quad \mu = \sum_a m_a.$$

Expanding and using the definition of x_{CM} gives

$$\begin{aligned} I_1 = I_2 &= \sum_a m_a x_a^2 - 2 x_{CM} \sum_a m_a x_a + \mu x_{CM}^2 \\ &= \sum_a m_a x_a^2 - \frac{1}{\mu} \left(\sum_a m_a x_a \right)^2 . \end{aligned}$$

To rewrite this in terms of pairwise separations, expand the square:

$$\left(\sum_a m_a x_a \right)^2 = \sum_a m_a^2 x_a^2 + 2 \sum_{a<b} m_a m_b x_a x_b .$$

Hence

$$\begin{aligned} I_1 = I_2 &= \sum_a m_a x_a^2 - \frac{1}{\mu} \sum_a m_a^2 x_a^2 - \frac{2}{\mu} \sum_{a<b} m_a m_b x_a x_b \\ &= \frac{1}{\mu} \sum_a m_a (\mu - m_a) x_a^2 - \frac{2}{\mu} \sum_{a<b} m_a m_b x_a x_b . \end{aligned}$$

Now note that $\mu - m_a = \sum_{b \neq a} m_b$, so

$$\sum_a m_a (\mu - m_a) x_a^2 = \sum_a \sum_{b \neq a} m_a m_b x_a^2 = \sum_{a<b} m_a m_b (x_a^2 + x_b^2).$$

Therefore

$$\begin{aligned} I_1 = I_2 &= \frac{1}{\mu} \sum_{a<b} m_a m_b (x_a^2 + x_b^2) - \frac{2}{\mu} \sum_{a<b} m_a m_b x_a x_b \\ &= \frac{1}{\mu} \sum_{a<b} m_a m_b (x_a - x_b)^2 . \end{aligned}$$

Since for collinear atoms the pair distance is $l_{ab} = |x_a - x_b|$, we finally obtain

Answer to Problem 1 (a).

$$I_1 = I_2 = \frac{1}{\mu} \sum_{a<b} m_a m_b \, l_{ab}^2$$

$$I_3 = 0$$

(b)

A triatomic molecule which is an isosceles triangle (Fig. 36).

Solution: Let the molecule lie in the (x_1, x_2)-plane, with x_1 along the base and x_2 along the axis of symmetry. The two base atoms have equal masses m_1 and are separated by a distance a; the apex atom has mass m_2 and is at height h above the base midpoint. Denote the total mass by

$$\mu = 2m_1 + m_2.$$

Choose coordinates (before shifting to the centre of mass):

$$(x_1, x_2)_{(1)} = \left(-\frac{a}{2}, 0\right), \qquad (x_1, x_2)_{(3)} = \left(+\frac{a}{2}, 0\right), \qquad (x_1, x_2)_{(2)} = (0, h),$$

where atoms (1) and (3) have mass m_1 and atom (2) has mass m_2. By symmetry $x_{1,CM} = 0$, and

$$x_{2,CM} = \frac{m_2 h}{\mu}.$$

The symmetry implies that the axes through the centre of mass parallel to x_1, x_2, and x_3 are principal. Hence,

$$I_1 = \sum_a m_a (x_{2,a} - x_{2,CM})^2, \qquad I_2 = \sum_a m_a (x_{1,a} - x_{1,CM})^2, \qquad I_3 = I_1 + I_2,$$

since the molecule is planar ($x_3 \equiv 0$). Using the pair-distance identity in one dimension,

$$\sum_a m_a (x_{2,a} - x_{2,CM})^2 = \frac{1}{\mu} \sum_{a<b} m_a m_b (x_{2,a} - x_{2,b})^2,$$

and noting that the only non-zero vertical separations are between the apex and each base atom, equal to h, we obtain

$$I_1 = \frac{1}{\mu}\left(m_1 m_2 h^2 + m_1 m_2 h^2\right) = \frac{2m_1 m_2}{\mu} h^2.$$

Similarly,

$$I_2 = \sum_a m_a (x_{1,a} - x_{1,CM})^2 = \frac{1}{\mu} \sum_{a<b} m_a m_b (x_{1,a} - x_{1,b})^2.$$

The horizontal separations are between the two base atoms, and between the apex and either base atom. Hence

$$I_2 = \frac{1}{\mu}\left(m_1^2 a^2 + 2m_1 m_2 \left(\frac{a}{2}\right)^2\right) = \frac{1}{\mu}\left(m_1^2 a^2 + \frac{1}{2} m_1 m_2 a^2\right) = \frac{1}{2} m_1 a^2.$$

Therefore

$$I_3 = I_1 + I_2 = \frac{2m_1 m_2}{\mu} h^2 + \frac{1}{2} m_1 a^2.$$

Answer to Problem 1 (b).

$$I_1 = \frac{2m_1 m_2}{\mu} h^2$$
$$I_2 = \frac{1}{2} m_1 a^2$$
$$I_3 = I_1 + I_2.$$

(c)

A tetratomic molecule which is an equilateral-based tetrahedron (Fig. 37).

Solution: The base is an equilateral triangle of side a in the plane $x_3 = 0$, with three equal masses m_1 at its vertices. A fourth atom of mass m_2 lies on the symmetry axis through the base centroid at height h above the base. Let

$$\mu = 3m_1 + m_2.$$

Let ρ be the distance from the base centroid to a base vertex. For an equilateral triangle,

$$\rho = \frac{a}{\sqrt{3}}, \qquad \rho^2 = \frac{a^2}{3}.$$

By symmetry the centre of mass lies on the symmetry axis. Measuring x_3 from the base plane,

$$x_{3,CM} = \frac{m_2 h}{\mu}.$$

The symmetry axis is a principal axis; any two orthogonal axes through the centre of mass perpendicular to it are also principal and have equal moments. Thus

$$I_1 = I_2, \qquad I_3 \text{ about the symmetry axis } (x_3).$$

The apex mass lies on the axis and contributes nothing. Each base mass is at distance ρ from the axis, hence

$$I_3 = 3m_1\rho^2 = 3m_1\frac{a^2}{3} = m_1 a^2.$$

Let x_1 be any axis through the centre of mass perpendicular to x_3. Then

$$I_1 = \sum_a m_a\,(y_a^2 + z_a^2),$$

where y is a coordinate orthogonal to x_1 in the plane perpendicular to the axis, and z is measured along x_3 from the centre of mass. The vertical coordinates relative to the centre of mass are

$$z_{\text{base}} = -x_{3,CM} = -\frac{m_2 h}{\mu}, \qquad z_{\text{apex}} = h - x_{3,CM} = h\left(1 - \frac{m_2}{\mu}\right) = \frac{3m_1}{\mu}h.$$

For the three vertices of an equilateral triangle, for any in-plane axis through the centroid,

$$\sum_{\text{base}} y^2 = \frac{3}{2}\rho^2 = \frac{3}{2}\cdot\frac{a^2}{3} = \frac{1}{2}a^2.$$

Therefore,

$$\begin{aligned} I_1 &= m_1\left(\sum_{\text{base}} y^2\right) + 3m_1 z_{\text{base}}^2 + m_2 z_{\text{apex}}^2 \\ &= \frac{1}{2}m_1 a^2 + 3m_1\left(\frac{m_2^2 h^2}{\mu^2}\right) + m_2\left(\frac{9m_1^2 h^2}{\mu^2}\right) \\ &= \frac{1}{2}m_1 a^2 + \frac{3m_1 m_2}{\mu}\,h^2. \end{aligned}$$

Hence $I_2 = I_1$.

Answer to Problem 1 (c).

$$I_1 = I_2 = \frac{3m_1 m_2}{\mu}\,h^2 + \frac{1}{2}m_1 a^2$$
$$I_3 = m_1 a^2$$

Problem 2. *Not done*

Determine the principal moments of inertia for the following homogeneous bodies:

(a) *Not done*

A thin rod of length l.

(b) *Not done*

A sphere of radius R.

(c) *Not done*

A circular cylinder of radius R and height h.

(d) *Not done*

A rectangular parallelepiped of sides a, b, and c.

(e) *Not done*

A circular cone of height h and base radius R.

(f) *Not done*

An ellipsoid of semiaxes a, b, c.

Problem 3. *Not done*

Determine the frequency of small oscillations of a compound pendulum (a rigid body swinging about a fixed horizontal axis in a gravitational field).

Problem 4. *Not done*

Find the kinetic energy of the system shown in Fig. 39: OA and AB are thin uniform rods of length l hinged together at A. The rod OA rotates (in the plane of the diagram) about O, while the end B of the rod AB slides along Ox.

Problem 5. *Not done*

Find the kinetic energy of a cylinder of radius R rolling on a plane, if the mass of the cylinder is so distributed that one of the principal axes of inertia is parallel to the axis of the cylinder and at a distance a from it, and the moment of inertia about that principal axis is I.

Problem 6. *Not done*

Find the kinetic energy of a homogeneous cylinder of radius a rolling inside a cylindrical surface of radius R (Fig. 41).

Problem 7. *Not done*

Find the kinetic energy of a homogeneous cone rolling on a plane (its principal moments of inertia are found in §32 Prob. 2(e)).

Problem 8. *Not done*

Find the kinetic energy of a homogeneous cone whose base rolls on a plane and whose vertex is fixed at a height above the plane equal to the radius of the base, so that the axis of the cone is parallel to the plane (see §32 Prob. 2(e) and §32 Prob. 7).

Problem 9. *Not done*

Find the kinetic energy of a homogeneous ellipsoid which rotates about one of its axes (AB in Fig. 44) while that axis itself rotates about a line CD perpendicular to it and passing through the centre of the ellipsoid.

Problem 10. *Not done*

The same as §32 Prob. 9, but for the case where the axis AB is not perpendicular to CD and is an axis of symmetry of the ellipsoid (Fig. 45).

§35 Eulerian angles

Problem 1. *Not done*

Reduce to quadratures the problem of the motion of a heavy symmetrical top whose lowest point is fixed (Fig. 48).

Problem 2. *Not done*

Find the condition for the rotation of a top about a vertical axis to be stable (see §35 Prob. 1).

Problem 3. *Not done*

Determine the motion of a top when the kinetic energy of its rotation about its axis is large compared with its energy in the gravitational field (called a "fast" top; see §35 Prob. 1).

§37 The asymmetrical top

Problem 1. *Not done*

Determine the free rotation of a top about an axis near the x_3-axis or the x_1-axis.

Problem 2. *Not done*

Determine the free rotation of a top for which $M^2 = 2EI_2$.

§38 Rigid bodies in contact

Problem 1. *Not done*

Using d'Alembert's principle, find the equations of motion of a homogeneous sphere rolling on a plane under an external force **F** and torque **K**.

Problem 2. *Not done*

A uniform rod BD of weight P and length l rests against a wall as shown in Fig. 52 and its lower end B is held by a string AB. Find the reaction of the wall and the tension in the string.

Problem 3. *Not done*

A rod of weight P has one end A on a vertical plane and the other end B on a horizontal plane (Fig. 53), and is held in position by two horizontal strings AD and BC, the latter being in the same vertical plane as AB. Determine the reactions of the planes and the tensions in the strings.

Problem 4. *Not done*

Two rods of length l and negligible weight are hinged together, and their ends are connected by a string AB (Fig. 54). They stand on a plane, and a force F is applied at the midpoint of one rod. Determine the reactions.

§39 Motion in a non-inertial frame of reference

Problem 1. *Not done*

Find the deflection of a freely falling body from the vertical caused by the Earth's rotation, assuming the angular velocity of this rotation to be small.

Problem 2. *Not done*

Determine the deflection from coplanarity of the path of a particle thrown from the Earth's surface with velocity $\mathbf{v}_0$.

Problem 3. *Not done*

Determine the effect of the Earth's rotation on small oscillations of a pendulum (the problem of Foucault's pendulum).

Chapter 7.

The Canonical Equations

§40 Hamilton's equations

Problem 1.

Find the Hamiltonian for a single particle in Cartesian, cylindrical and spherical co-ordinates.

Solution: For a particle with a velocity-independent potential energy U, the generalized momenta and the Hamiltonian are

$$p_i = \frac{\partial L}{\partial \dot{q}_i}, \qquad H = \sum_i p_i \dot{q}_i - L.$$

The kinetic energy is quadratic in the velocities, so the Legendre transform gives $H = T + U$, with the velocities expressed in terms of the canonical momenta.

In Cartesian co-ordinates, we have

$$L = \frac{1}{2}m\left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\right) - U(x, y, z), \qquad p_x = m\dot{x}, \quad p_y = m\dot{y}, \quad p_z = m\dot{z}.$$

In cylindrical co-ordinates, the Lagrangian is

$$L = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\phi}^2 + \dot{z}^2\right) - U(r, \phi, z),$$

so that

$$p_r = m\dot{r}, \qquad p_\phi = mr^2\dot{\phi}, \qquad p_z = m\dot{z}.$$

In spherical co-ordinates, with θ the polar angle and ϕ the azimuth, the Lagrangian is

$$L = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\dot{\phi}^2\sin^2\theta\right) - U(r, \theta, \phi),$$

and hence

$$p_r = m\dot{r}, \qquad p_\theta = mr^2\dot{\theta}, \qquad p_\phi = mr^2\dot{\phi}\sin^2\theta.$$

Substituting these momenta in $H = T + U$ gives the three Hamiltonians.

Answer to Problem 1.

$$\text{Cartesian:} \quad H = \frac{1}{2m}\left(p_x^2 + p_y^2 + p_z^2\right) + U(x, y, z)$$

$$\text{Cylindrical:} \quad H = \frac{1}{2m}\left(p_r^2 + \frac{p_\phi^2}{r^2} + p_z^2\right) + U(r, \phi, z)$$

$$\text{Spherical:} \quad H = \frac{1}{2m}\left(p_r^2 + \frac{p_\theta^2}{r^2} + \frac{p_\phi^2}{r^2\sin^2\theta}\right) + U(r, \theta, \phi)$$

Problem 2.

Find the Hamiltonian for a particle in a uniformly rotating frame of reference.

Solution: Let $\mathbf{v} = \dot{\mathbf{r}}$ be the velocity relative to a frame rotating with constant angular velocity $\boldsymbol{\Omega}$ about a fixed origin. The inertial velocity, expressed along the rotating axes, is $\mathbf{v} + \boldsymbol{\Omega} \times \mathbf{r}$. Thus, the Lagrangian is

$$L = \frac{1}{2} m \left(\mathbf{v} + \boldsymbol{\Omega} \times \mathbf{r} \right)^2 - U,$$

and the canonical momentum is

$$\mathbf{p} = \frac{\partial L}{\partial \mathbf{v}} = m \left(\mathbf{v} + \boldsymbol{\Omega} \times \mathbf{r} \right).$$

Consequently,

$$\mathbf{v} = \frac{\mathbf{p}}{m} - \boldsymbol{\Omega} \times \mathbf{r}.$$

The Legendre transform gives

$$\begin{aligned} H &= \mathbf{p} \cdot \mathbf{v} - L \\ &= \mathbf{p} \cdot \left(\frac{\mathbf{p}}{m} - \boldsymbol{\Omega} \times \mathbf{r} \right) - \frac{\mathbf{p}^2}{2m} + U \\ &= \frac{\mathbf{p}^2}{2m} - \boldsymbol{\Omega} \cdot (\mathbf{r} \times \mathbf{p}) + U, \end{aligned}$$

where we used $\mathbf{p} \cdot (\boldsymbol{\Omega} \times \mathbf{r}) = \boldsymbol{\Omega} \cdot (\mathbf{r} \times \mathbf{p})$. Here, $\mathbf{p}$ is the canonical momentum, rather than $m\mathbf{v}$.

Answer to Problem 2.

$$H = \frac{\mathbf{p}^2}{2m} - \boldsymbol{\Omega} \cdot (\mathbf{r} \times \mathbf{p}) + U$$

Problem 3.

Find the Hamiltonian for a system comprising one particle of mass M and n particles each of mass m, excluding the motion of the centre of mass (see §13 Prob. 1).

Solution: We use the relative co-ordinates of §13 Prob. 1, namely $\mathbf{r}_a = \mathbf{R}_a - \mathbf{R}$ and $\mathbf{v}_a = \dot{\mathbf{r}}_a$, with $\mu \equiv M + nm$ the total mass. The reduced Lagrangian, including the potential energy, is

$$L = \frac{1}{2} m \sum_a \mathbf{v}_a^2 - \frac{1}{2} \frac{m^2}{\mu} \left(\sum_a \mathbf{v}_a \right)^2 - U.$$

The canonical momenta are

$$\mathbf{p}_a = \frac{\partial L}{\partial \mathbf{v}_a} = m\mathbf{v}_a - \frac{m^2}{\mu} \sum_b \mathbf{v}_b.$$

Summing over a, we get

$$\sum_a \mathbf{p}_a = \left(m - \frac{nm^2}{\mu} \right) \sum_a \mathbf{v}_a = \frac{mM}{\mu} \sum_a \mathbf{v}_a.$$

Substituting this relation back in the expression for $\mathbf{p}_a$ yields

$$\mathbf{v}_a = \frac{\mathbf{p}_a}{m} + \frac{1}{M} \sum_b \mathbf{p}_b.$$

Since the reduced kinetic energy is quadratic in the velocities, we have $\sum_a \mathbf{p}_a \cdot \mathbf{v}_a = 2T$ and

$$\begin{aligned} H = T + U &= \frac{1}{2} \sum_a \mathbf{p}_a \cdot \mathbf{v}_a + U \\ &= \frac{1}{2m} \sum_a \mathbf{p}_a^2 + \frac{1}{2M} \left(\sum_a \mathbf{p}_a \right)^2 + U. \end{aligned}$$

The momenta $\mathbf{p}_a$ are conjugate to the relative co-ordinates $\mathbf{r}_a$, and the kinetic energy of the centre of mass has already been removed.

Answer to Problem 3.

$$H = \frac{1}{2m} \sum_a \mathbf{p}_a^2 + \frac{1}{2M} \left(\sum_a \mathbf{p}_a \right)^2 + U(\mathbf{r}_1, \ldots, \mathbf{r}_n)$$

§41 The Routhian

Problem 1.

Find the Routhian for a symmetrical top in an external field $U(\phi, \theta)$, eliminating the cyclic coordinate ψ (where ψ, ϕ, θ are Eulerian angles).

Solution: Let $I_1' = I_2'$ be the transverse moments of inertia about the fixed point and I_3 the moment of inertia about the axis of symmetry. In Eulerian angles, the Lagrangian is

$$L = \frac{I_1'}{2}\left(\dot{\theta}^2 + \dot{\phi}^2 \sin^2\theta\right) + \frac{I_3}{2}\left(\dot{\psi} + \dot{\phi}\cos\theta\right)^2 - U(\phi, \theta).$$

Since L is independent of ψ, its conjugate momentum is conserved:

$$p_\psi = \frac{\partial L}{\partial \dot{\psi}} = I_3\left(\dot{\psi} + \dot{\phi}\cos\theta\right).$$

Solving for the velocity to be eliminated gives

$$\dot{\psi} = \frac{p_\psi}{I_3} - \dot{\phi}\cos\theta.$$

The Routhian is the partial Legendre transform $R = p_\psi\dot{\psi} - L$. Substituting the expression for $\dot{\psi}$, we get

$$R = p_\psi\left(\frac{p_\psi}{I_3} - \dot{\phi}\cos\theta\right) - \frac{I_1'}{2}\left(\dot{\theta}^2 + \dot{\phi}^2\sin^2\theta\right) - \frac{p_\psi^2}{2I_3} + U(\phi, \theta).$$

Thus, ψ and $\dot{\psi}$ are eliminated, while p_ψ is retained as a constant parameter.

Answer to Problem 1.

$$R(\phi, \theta, \dot{\phi}, \dot{\theta}; p_\psi) = \frac{p_\psi^2}{2I_3} - p_\psi\dot{\phi}\cos\theta - \frac{I_1'}{2}\left(\dot{\theta}^2 + \dot{\phi}^2\sin^2\theta\right) + U(\phi, \theta)$$

For a given value of p_ψ, the term $p_\psi^2/2I_3$ is a constant, so it may be omitted when deriving the reduced equations of motion for ϕ and θ.

§42 Poisson brackets

Problem 1.

Determine the Poisson brackets formed from the Cartesian components of the momentum $\mathbf{p}$ and the angular momentum $\mathbf{M} = \mathbf{r} \times \mathbf{p}$ of a particle.

Solution: We use Landau's definition of the Poisson bracket,

$$[F, G] = \sum_{k=x,y,z} \left(\frac{\partial F}{\partial p_k} \frac{\partial G}{\partial x_k} - \frac{\partial F}{\partial x_k} \frac{\partial G}{\partial p_k} \right).$$

Since the momentum components are independent canonical variables, we have

$$[p_i, p_j] = 0.$$

For a bracket involving one momentum component, the definition reduces to

$$[F, p_j] = -\frac{\partial F}{\partial x_j}.$$

Using $M_x = yp_z - zp_y$, we obtain

$$[M_x, p_x] = 0, \qquad [M_x, p_y] = -p_z, \qquad [M_x, p_z] = p_y.$$

The remaining brackets follow by cyclically permuting x, y, z. In index notation, with $\epsilon_{xyz} = 1$, these results take the form given below. Brackets with the arguments reversed follow from antisymmetry.

Answer to Problem 1.

$$[p_i, p_j] = 0, \qquad [M_i, p_j] = -\sum_k \epsilon_{ijk} p_k$$

Problem 2.

Determine the Poisson brackets formed from the components of $\mathbf{M}$.

Solution: The components of the angular momentum are

$$M_x = yp_z - zp_y, \qquad M_y = zp_x - xp_z, \qquad M_z = xp_y - yp_x.$$

For M_x and M_y, only the pair z, p_z contributes to the bracket defined in §42 Prob. 1:

$$\begin{aligned}[M_x, M_y] &= \frac{\partial M_x}{\partial p_z}\frac{\partial M_y}{\partial z} - \frac{\partial M_x}{\partial z}\frac{\partial M_y}{\partial p_z} \\ &= yp_x - (-p_y)(-x) \\ &= yp_x - xp_y = -M_z.\end{aligned}$$

Cyclic permutation gives the other two independent brackets. Brackets of a component with itself vanish by antisymmetry.

Answer to Problem 2.

$$[M_x, M_y] = -M_z, \qquad [M_y, M_z] = -M_x, \qquad [M_z, M_x] = -M_y$$

or, equivalently,

$$[M_i, M_j] = -\sum_k \epsilon_{ijk} M_k$$

Problem 3.

Show that $[\phi, M_z] = 0$, where ϕ is any function, spherically symmetrical about the origin, of the co-ordinates and momentum of a particle.

Solution: Since $M_z = xp_y - yp_x$, the bracket of §42 Prob. 1 with any smooth function $F(\mathbf{r}, \mathbf{p})$ is

$$[F, M_z] = y\frac{\partial F}{\partial x} - x\frac{\partial F}{\partial y} + p_y\frac{\partial F}{\partial p_x} - p_x\frac{\partial F}{\partial p_y}.$$

We denote this differential operator by $\mathcal{D}_z F$.

A scalar invariant under simultaneous rotations of $\mathbf{r}$ and $\mathbf{p}$ can depend on these vectors only through

$$u = \mathbf{r}^2, \qquad v = \mathbf{p}^2, \qquad w = \mathbf{r}\cdot\mathbf{p}.$$

Applying the operator to these combinations gives

$$\begin{aligned}
\mathcal{D}_z u &= 2xy - 2xy = 0,\\
\mathcal{D}_z v &= 2p_x p_y - 2p_x p_y = 0,\\
\mathcal{D}_z w &= yp_x - xp_y + xp_y - yp_x = 0.
\end{aligned}$$

For $\phi = \Phi(u, v, w)$, the chain rule therefore yields

$$[\phi, M_z] = \frac{\partial \Phi}{\partial u}\mathcal{D}_z u + \frac{\partial \Phi}{\partial v}\mathcal{D}_z v + \frac{\partial \Phi}{\partial w}\mathcal{D}_z w = 0.$$

Answer to Problem 3.

$$[\phi, M_z] = 0$$

Problem 4.

Show that $[\mathbf{f}, M_z] = \mathbf{f} \times \mathbf{n}$, where $\mathbf{f}$ is a vector function of the co-ordinates and momentum of a particle, and $\mathbf{n}$ is a unit vector parallel to the z-axis.

Solution: Here, a vector function means a function that transforms as a vector under simultaneous rotations of its arguments.

Under an infinitesimal active rotation through an angle $\delta\alpha$ about the z-axis, we have

$$\delta\mathbf{r} = \delta\alpha\,\mathbf{n} \times \mathbf{r}, \qquad \delta\mathbf{p} = \delta\alpha\,\mathbf{n} \times \mathbf{p}.$$

For each component f_i, the chain rule gives

$$\begin{aligned}\frac{\delta f_i}{\delta\alpha} &= (\mathbf{n} \times \mathbf{r}) \cdot \nabla_{\mathbf{r}} f_i + (\mathbf{n} \times \mathbf{p}) \cdot \nabla_{\mathbf{p}} f_i \\ &= -y\frac{\partial f_i}{\partial x} + x\frac{\partial f_i}{\partial y} - p_y\frac{\partial f_i}{\partial p_x} + p_x\frac{\partial f_i}{\partial p_y} \\ &= -[f_i, M_z],\end{aligned}$$

using the operator derived in §42 Prob. 3. Because $\mathbf{f}$ transforms as a vector, its change under the same rotation is

$$\frac{\delta\mathbf{f}}{\delta\alpha} = \mathbf{n} \times \mathbf{f}.$$

Comparing these expressions component by component gives

$$[\mathbf{f}, M_z] = -\mathbf{n} \times \mathbf{f} = \mathbf{f} \times \mathbf{n}.$$

Answer to Problem 4.

$$[\mathbf{f}, M_z] = \mathbf{f} \times \mathbf{n}$$

§44 Maupertuis' principle

Problem 1. *Not done*

Derive the differential equation of the path from the variational principle (44.10).

§48 Separation of the variables

Problem 1. *Not done*

Find a complete integral of the Hamilton–Jacobi equation for motion of a particle in a field $U = \alpha/r - Fz$ (a combination of a Coulomb field and a uniform field), and find a conserved function of the co-ordinates and momenta that is specific to this motion.

Problem 2. *Not done*

The same as §48 Prob. 1, but for a field $U = \alpha_1/r_1 + \alpha_2/r_2$ (the Coulomb field of two fixed points at a distance 2σ apart).

§50 Canonical variables

Problem 1. *Not done*

Write down the equations of motion in canonical variables for a harmonic oscillator (whose Hamiltonian is (49.11)) with time-dependent frequency.

§51 Accuracy of conservation of the adiabatic invariant

Problem 1. *Not done*

Estimate δI for a harmonic oscillator with a frequency that varies slowly according to

$$\omega^2 = \omega_0^2 \frac{1 + ae^{\alpha t}}{1 + e^{\alpha t}}$$

from $\omega_- = \omega_0$ for $t = -\infty$ to $\omega_+ = \sqrt{a}\,\omega_0$ for $t = +\infty$ ($a > 0$, $\alpha \ll \omega_0$).

Problem 2. *Not done*

A particle oscillates in a potential well. Determine how its energy varies under a frictional force $f_{\text{fr}} = -\alpha\dot{x}$ with a small coefficient α (x being a Cartesian co-ordinate).

§52 Conditionally periodic motion

Problem 1. *Not done*

Calculate the action variables for elliptic motion in a field $U = -\alpha/r$.